

Hadronic Shower Characterization in the Belle II Electromagnetic Calorimeter

Emanuele Zanusso

Dipartimento di Fisica di Torino
Master of Science in Physics Thesis

Supervisor: Professor Stefano Spataro

April 2026



UNIVERSITÀ
DI TORINO

Abstract

Abstract

Index

0	Chapters subdivision	8
1	Introduction	9
1.1	High Luminosity B Factory	9
1.2	Anti-neutron in HEP physics	10
1.2.1	Anti-neutron interactions in physics	11
1.2.2	The anti-nucleon–nucleon interaction	13
1.2.3	Electromagnetic and hadronic showers	15
1.3	The MANTRA Project	16
2	The Belle II experiment	18
2.1	The SuperKEKB collider	18
2.2	The Belle II detector	19
2.2.1	The vertex detector VXD	21
2.2.2	The Central Drift Chamber (CDC)	22
2.2.3	PID systems: TOP and ARICH	23
2.2.4	The Electromagnetic Calorimeter (ECL)	24
2.2.5	The solenoid and iron structure	27
2.2.6	The K_L and Muon Detector (KLM)	27
3	The Belle II Analysis Software Framework	28
3.1	Monte Carlo simulation	28
3.2	Reconstruction	29
3.2.1	The ECL reconstruction algorithm	29
3.2.2	ECL performance	31
3.3	basf2 modules and path	31
3.4	TopoAna: Topology Analysis	32
4	Study of the Monte Carlo signal channel	35
4.1	Kinematic variables comparison	37
4.1.1	ISR distributions	37
4.1.2	Event selections	39
4.1.3	Recoil mass distribution	40
4.1.4	The recoil kinematic	41
4.2	The $\bar{n}$ relatives analysis	48
4.2.1	The basf2 ancestor variables	49
4.2.2	Ancestor momentum	51
4.2.3	$\bar{n}$ production vertex	53
4.3	Kinematic Fit effects	55

4.4	Shower shape variables in the signal sample	56
4.4.1	The analyzed shower shape variables	56
4.4.2	Anti-neutrons ECL cluster variables	59
4.4.3	Ancestor cluster variables	62
4.4.4	Cluster variables in different ECL regions	67
4.5	Summary of Signal Channel Analysis	69
5	Study of the Monte Carlo cocktail production	70
5.1	Introduction to the Belle II Distributed Computing	70
5.1.1	Types of processes at the Belle experiment enviroment . .	72
5.2	Applied selections and cuts	73
5.3	Recoil mass in $u\bar{u}$ cocktail production (NO ISR case)	75
5.3.1	Recoil mass components analysis	76
5.3.2	Performance metrics study on α cut	79
5.4	$q\bar{q}$ cocktail production (NO ISR case)	82
6	Signal sample analysis scripts	84
6.1	The decay file and the generation steering file	84

List of Figures

1.1	Trends of S- and P-wave annihilation cross sections according to annihilation models: the full markers (circles for S-wave, squares for P-wave) are from the Dover–Richard model [1], while the open ones have been obtained by [2] using a version of the optical model.	12
1.2	Fit of $\sigma_{ann}(\bar{n}p)$ measured by OBELIX up to 400 MeV, including S- (solid line), P- (broken line) and D- (point line) wave contribution.	13
1.3	Electromagnetic shower vs hadronic shower comparison.	15
2.1	SuperKEKB design, where the Interaction Point (IP) is located in Tsukuba surrounded by the Belle II detector.	18
2.2	SuperKEKB operating parameters on March 18-19, 2026. From top to bottom: HER and LER ring currents (first two panels), instantaneous and integrated luminosity (bottom two panels).	20
2.3	An artistic view of the Belle II detector.	21
2.4	A scheme of the Belle II VXD.	21
2.5	A scheme of the Belle II CDC.	22
2.6	Scheme of the TOP counter module.	23
2.7	Representation of the side view of the TOP counter with internal reflection of the Cherenkov radiation.	24
2.8	(a) Working principle of the ARICH. (b) Ring produced by a cosmic-ray muon.	24
2.9	Typical ECL shower caused by a 7 GeV photon.	25
2.10	Properties of inorganic scintillators used in calorimetry [3].	25
2.11	A scheme of the Belle II ECL detector.	26
2.12	ECL coverage and segmentation for each ECL region.	26
3.1	Description of ECL clustering process: (a) Step 0; (b) Step 1; (c) Step 2; (d) Step 3.	30
3.2	Connected regions obtained in the example (threshold: 0.1 MeV).	30
3.3	The energy resolution as a function of incident photon energy for the (a) 3Y3 and (b) 5Y5 matrix with total energy sum. The error bars are not statistical but are the rms of four measurements taken with the crystals (3,3), (3,4), (4,3) and (4,4) into the photon beam [4].	31
3.4	Scheme of the processing chain in basf2.	32
3.5	Topology diagrams of (a) $e^+e^- \rightarrow J/\psi$ and (b) $e^+e^- \rightarrow \Upsilon(4S) \rightarrow B^0\bar{B}^0$, two typical processes at e^+e^- colliders, such as Belle II and BESIII.	33
3.6	Raw MC information example from 3.5 b.	33
3.7	Highly readable MC information example from 3.5 b, as presented by the TopoAna program.	34
4.1	Feynman Diagram of the channel of interest. The ISR gamma is a pre-interaction radiation emitted by the e^+ and/or e^- . The recoil system is made of $p\pi^-(\gamma_{ISR})$.	35

4.2	Feynman diagram of the channel of interest. <i>Phokhara</i> handles the initial hard photon emission (red), while <i>EvtGen</i> simulates the decay to final-state particles (blue).	36
4.3	Topology Analysis (TopoAna) output of the generated 100k events. The fourth column is the unambiguous decay tree index associated to the correspondent channel, the fifth the number of entries per channel, and the last the cumulative number of entries. . . .	37
4.4	(a) ISR energy distribution. (b) ISR θ distribution. Both distributions are shown on a logarithmic y -axis scale.	38
4.5	2D energy E and θ distribution of ISR radiation.	38
4.6	(a) α distribution. (b) α distribution after the rank selection. . .	40
4.7	Recoil mass distribution, in both ISR and NO ISR cases.	41
4.8	Reconstructed momentum and $\cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.	42
4.9	Generated momentum and $\cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.	43
4.10	2D distribution of proton momentum versus proton $\cos\theta$, in the NO ISR case.	43
4.11	Momentum distribution of the recoil system, in both ISR and NO ISR cases.	44
4.12	Reconstructed (left) and generated (right) momentum distribution of anti-neutron candidates, in both ISR and NO ISR cases. .	45
4.13	Recoil momentum and generated $\bar{n}$ list momentum. The blue distribution shows the recoil momentum, the red distribution corresponds to the generated $\bar{n}$ with MC PDG = -2112 (associated to $\bar{n}$ by MC matching), while the green distribution corresponds to the generated $\bar{n}$ with MC PDG $\neq$ -2112 (associated to no $\bar{n}$ by MC matching).	45
4.14	Obtained momentum and θ correlation distributions between recoil system and anti-neutron candidate, in both reconstructed and generated case, in both ISR and NO ISR cases.	47
4.15	Momentum and θ angle residual distributions between recoil system and anti-neutron candidate, in both ISR and NO ISR cases.	48
4.16	A representation displaying cluster split-off and pre-showering effects following $\bar{n}$ annihilation.	49
4.17	Distribution of the relative degree of anti-neutron candidates with an anti-neutron present in their ancestry chain.	50
4.18	Monte Carlo PDG codes of candidates with degrees of ancestry selection.	51

4.19	Distribution comparison of: recoil momentum (blue), generated momentum of $\bar{n}$ candidates associated to $\bar{n}$ by the Monte Carlo matching (red), generated momentum of $\bar{n}$ candidates associated to no $\bar{n}$ by the Monte Carlo matching (green) and generated momentum of $\bar{n}$ ancestors associated to $\bar{n}$ by the Monte Carlo matching (yellow). The black distribution is the sum of the second and fourth contributions. The recoil vector is the only reconstructed variable, while the others are obtained from Monte Carlo truth selection.	52
4.20	Residual distribution of primary $\bar{n}$ momentum (ancestor momentum) with respect to the recoil vector.	53
4.21	Monte Carlo Error code. 0 corresponds to a perfectly reconstructed particle, while 512 corresponds to a non valid MC match.	54
4.22	$\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: all $\bar{n}$ associated by Monte Carlo matching; red: $\bar{n}$ candidates correctly associated to true $\bar{n}$; green: $\bar{n}$ candidates with no $\bar{n}$ relatives; yellow: $\bar{n}$ candidates with an $\bar{n}$ mother particle.	54
4.23	$\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: $\bar{n}$ candidates detected in the forward region; red: $\bar{n}$ candidates detected in the barrel region; green: $\bar{n}$ candidates detected in the backward region.	55
4.24	Momentum and θ angle residuals distribution between recoil system and anti-neutron candidate, after the 1C Kinematic Fit procedure	56
4.25	Ideal representation of a cluster pattern, consisting of a 5×5 matrix in which each cell corresponds to a single crystal	57
4.26	Zernike Polynomials representation. Moments Z_4^0 and Z_5^1 are widely discussed in this thesis	59
4.27	Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed	60
4.28	Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (NO ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed	61
4.29	Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (ISR case)	63
4.30	Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (NO ISR case)	64
4.31	Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). Cut $clusterNHits > 10$ is performed	65

4.32	Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (NO ISR case). Cut $clusterNHits > 10$ is performed	66
4.33	Different ECL regions contribution in neutral cluster associated to $\bar{n}$ candidates, where 1 is the forward region, 2 is the barrel region and 3 is the backward region. As expected, the barrel region is the most involved	67
4.34	Comparison of reconstructed cluster variables for all $\bar{n}$ candidates in different ECL regions. The distributions are normalized to unit value	68
5.1	Tree level Feynman diagram of $e^+e^- \rightarrow q\bar{q}$	70
5.2	A scheme of the Belle II Data Model, starting from the nature processes information to the final results. As described, Monte Carlo truth information bypasses reconstruction, since the information is directly picked at generator level	71
5.3	A scheme of the Belle II Distributed Computing (The Grid) sites	72
5.4	Cross section of $e^+e^- \rightarrow Hadrons$, in Υ resonances energy region [5]. $\Upsilon(1S)$ resonance is very narrow, while $\Upsilon(4S)$ one is broader and lower	73
5.5	The selected channel with ROE made of a variable amount of particles.	74
5.6	Cluster energy distribution of $\bar{n}$ candidates, divided by the Monte Carlo ISR flag (<code>mcISR</code>). If <code>mcISR</code> = 0, the reconstructed particle is not an ISR photon. If <code>mcISR</code> = 1, the reconstructed particle is an ISR photon. Of course this is a level-generator information.	75
5.7	Reconstructed p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, NO ISR case.	76
5.8	First 10 channels of the TopoAna output in $u\bar{u}$ Monte Carlo cocktail, showing the different channel contributes, NO ISR case.	76
5.9	p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.	77
5.10	(a) p, π^- recoil mass and (b) p, π^-, γ_{ISR} recoil mas distribution in $u\bar{u}$ Monte Carlo cocktail	78
5.11	p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.	78
5.12	Figure of Merit (FOM) versus α angle for the $u\bar{u}$ cocktail.	80
5.13	Receiver Operating Characteristic (ROC) curve constructed from the signal efficiency (ϵ_{sig}) and the Background Efficiency (ϵ_{bkg}) for the $u\bar{u}$ Monte Carlo cocktail. where each point corresponds to a different cut value on the α angle. For clarity, the labels are shown only for $\alpha < 0.10$ rad.	80
5.14	ϵ_{sig} vs <i>Purity</i> curve for the $u\bar{u}$ Monte Carlo cocktail.	81
5.15	p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. $\alpha < 10$ rad cut is applied.	81

5.16	Top 5 channels from the TopoAna output for the $u\bar{u}$ Monte Carlo cocktails.	82
5.17	First 10 channels of the TopoAna output in $q\bar{q}$ Monte Carlo cocktail, showing the different channel contributes.	82
5.18	Top 5 channels from the TopoAna output respectively for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ Monte Carlo cocktails (from top to bottom). The <code>iDcyTr</code> index values may differ across samples since each TopoAna output is produced independently from its corresponding ROOT ntuples.	83
6.1	Decay file example powered by Evt_Gen	84
6.2	Usage of PHOKHARA&Evt_Gen, where simulation and reconstruction are performed	85

Chapter 0

Chapters subdivision

The anti-neutron in physics: the Chapter 1 describes the properties of the anti-neutron and its detection techniques in the high-energy physics (HEP) context. Its primary interactions -such as annihilation and hadronic shower production- are then exploited, discussing some hadronic models very quickly. The MANTRA (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) project is then described as a method to measure anti-neutron energies up to 10 GeV using an electromagnetic calorimeter.

The Belle II experiment: the Chapter 2 provides an overview of the SuperKEKB accelerator and Belle II detector. In the context of $\bar{n}$ studies, the Electromagnetic Calorimeter (ECL) plays a key role in hadron detection; a detailed discussion about the ECL is thus provided, covering the description of its inorganic crystals and the explanation of the clustering algorithm. The ECL performance are then discussed.

Then the Belle II Analysis Software Framework (basf2) is introduced, providing an overview of its operation method and its role in the Monte Carlo simulation and reconstruction processes at Belle II experiment. Afterwards, the Topology Analysis (TopoAna) independent software is described.

Study of the signal sample channel: the Chapter 3 is devoted to the study of the kinematic variables of anti-neutrons $\bar{n}$, including momentum, angular distributions and related observables. The analysis follows a semi-exclusive approach over the selected channel $e^+e^- \rightarrow p\pi^-\bar{n}$, studying both cases with and without Initial State Radiation (ISR) reconstruction, where the recoil system is made of $p\pi^-(\gamma_{ISR})$. Therefore, the recoil system variables are compared to those of the anti-neutrons candidates, studying the response of a Kinematic Fit as well.

Cluster split-off and pre-showering effects are investigated using studying anti-neutrons relatives chain. Several neutral cluster variables associated to anti-neutron candidates are then described and studied for both ISR and NO ISR cases, including the analysis of their relatives contributions.

Chapter 1

Introduction

The Standard Model is one of the greatest achievements of modern science, especially in the field of high-energy particle physics. Its reliability is the result of a long process that began in the 20th century and received its definitive confirmation in 2012 with the discovery of the Higgs boson.

However, there are some topics which remain unsolved, like the non-trivial neutrino mass or the presence of dark matter on cosmological scales. The Standard Model still represents an excellent phenomenological description of subatomic processes, at the $\lesssim 1$ TeV energy scale.

In this framework, the **Belle II** experiment, located at the SuperKEKB facility, plays a fundamental role in exploring the aforementioned topics. These are scenarios that go beyond the standard physics, thus entering the so-called *new physics*.

In parallel with experiments that maximize the energy, the aim of Belle II is to explore the intensity frontier. Its main goal is not to increase the center-of-mass energy, but rather to explore scenarios of high instantaneous luminosity up to $8 \times 10^{35} \text{ cm}^{-2} \text{ s}^{-1}$, around (and not limited to) the $\Upsilon(4S)$ resonance energy.

This approach enables flavour physics studies of heavy mesons such as B mesons, facilitating the identification of potential discrepancies with Standard Model predictions and opening the door to Beyond Standard Model physics.

1.1 High Luminosity B Factory

Since the end of the XX century, two asymmetric-energy e^+e^- B factories - KEKB for the Belle experiment at KEK and PEP-II for the BaBar experiment at SLAC- have achieved huge success, confirming the Standard Model in the quark flavour sector, establishing the presence of CP violation in the B meson system through the decay process $B^0(\bar{B}^0) \rightarrow J/\Psi K_S^0$ [6, 7, 8, 9]. This result fixed the main target of the present B factories. Experimental results confirm that the Kobayashi–Maskawa mechanism, a key component of the Standard Model of elementary particles, is the dominant source of CP violation observed in nature.

The Belle experiment proved its ability to measure a number of decay modes of the B meson and to extract Cabibbo-Kobayashi-Maskawa (CKM) matrix elements and other interesting observables. Precise measurements and new observations from Belle have overconstrained the quark flavor sector of the Standard Model. This sector remains self-consistent within the current experimental precision.

However, several fundamental questions persist in the quark and lepton flavor sectors; the Standard Model requires numerous parameters, including quark and lepton masses and mixing angles that must be determined experimentally.

Furthermore, from a cosmological perspective, the matter-antimatter asymmetry of the Universe poses a great challenge. While the CP violation constitutes

one of the necessary conditions for the evolution of a matter-dominated Universe [10], its magnitude cannot be explained solely by the CP violation within the Standard Model, which originates from quark mixing.

Exploring such scenarios requires substantial luminosity at present B factories. The SuperKEKB Higher Luminosity B Factory has been designed to investigate the physics of B mesons processes. Due to the very large production cross section of b -quarks in hadron collisions, certain measurements such as $B_s \rightarrow \mu\mu$ can be performed more effectively at hadron colliders, such as LHC. However, e^+e^- machines provide a much cleaner environment, essential for precision observables involving photons, π^0 's, K_L^0 's, or neutrinos in the final state.

At the $\Upsilon(4S)$ resonance, $B\bar{B}$ pairs are produced near threshold without associated particles. By fully reconstructing the four-momentum of one B ($\bar{B}$) meson from its decay products (full reconstruction technique), the missing momentum of the other $\bar{B}$ (B) meson decay can be inferred. This technique is essential for measuring channels with neutrinos in the final state. Therefore, SuperKEKB undoubtedly complements hadron collider experiments from both theoretical motivation and experimental perspectives.

1.2 Anti-neutron in HEP physics

The anti-neutron $\bar{n}$ ($\bar{u}\bar{d}\bar{d}$) is the anti-particle of the neutron n (udd), a neutral hadron of mass of around $0.939565 \text{ GeV}/c^2$ and spin $\frac{1}{2}$, with magnetic moment of $1.91\mu_N$, where μ_N is the nuclear magneton. Its existence was expected by the principle of invariance under charge conjugation and experimentally proved in 1956 [11], one year after the anti-proton $\bar{p}$ [12].

The anti-neutron $\bar{n}$ plays a key role in several physics measurements, both in collider physics and in astrophysical contexts. Since it is electrically neutral, it cannot be easily observed directly, allowing its detection via the products of its annihilation with ordinary matter.

For example, the measurement of the neutron electromagnetic form factor in the time-like region is strongly affected by the performance of the $\bar{n}$ detection. The BESIII experiment has shown [13] an oscillating behaviour in the electromagnetic structure of the neutron in the study of the $e^+e^- \rightarrow n\bar{n}$ annihilation process, where the two neutral hadrons in the final state were reconstructed using the well-known two-body kinematics. Direct detection of the $\bar{n}$ would further refine this analysis by enabling a more proper identification of the final state.

Another example, where an improved direct measurement of the $\bar{n}$ would significantly enhance the studies, is the analysis of hyperon decay channels at B-factories (and beyond). Some possible decay channels where the $\bar{n}$ is the only un-reconstructed particle in the final state are shown below:

$$\bar{\Lambda}^0 \rightarrow \pi^0 + \bar{n}, \quad \bar{\Sigma}^- \rightarrow \pi^- + \bar{n}, \quad \bar{\Lambda}_c \rightarrow K_s^0 + \pi^0 + \bar{n}$$

Furthermore, the anti-deuteron $\bar{D}$ hadronization is commonly described by a coalescence mechanism [14], where a $\bar{p}$ and a $\bar{n}$ sufficiently close in phase space

bind to form a $\bar{D}$. This latter is expected among the products of dark matter annihilation [15]; thus, the simultaneous measurement of the $\bar{n}$ - $\bar{p}$ correlation function in $\bar{D}$ production would significantly improve the understanding of this subject, allowing a more complete analysis of dark matter distribution.

A deeper knowledge on anti-neutrons kinematic and interaction would improve also Monte Carlo GEANT4 models of the hadronization process. GEANT4 is the main toolkit designed for simulating the passage of particles through matter [16]. It finds applications across high-energy, nuclear and accelerator physics, as well as in medical and space science research.

1.2.1 Anti-neutron interactions in physics

Due to isospin symmetry, it was believed that anti-neutrons strong interaction with nucleons (N) and matter would not provided much more informations than those obtained by anti-protons, for which obtaining proper beams was much easier.

The experimental landscape underwent a significant transformation in the mid-1980s with the development of dedicated anti-neutron beams, which enabled the observation of unique $\bar{n}$ characteristics, such as:

- The anti-nucleon nucleon $\bar{N}N$ interaction, in $\bar{p}p$ interactions, is a mixture of isospin $I = 0$ and $I = 1$ amplitudes, while the $\bar{n}p$ is a pure $I = 1$ eigenstate.
The same difference between $\bar{p}p$ and $\bar{p}n$ interactions could be obtained by means of the charge-symmetric system $\bar{p}n$ copiously produced at deuterium targets experiment. However, the use of deuterium introduces several drawbacks, such as re-scattering effects that may largely distort the final spectra [17]. This effect is absent if the $\bar{N}N$ system with pure $I = 1$ is formed by $\bar{n}$ interactions on protons of the material.
- The $\bar{n}p$ is a pure strong interaction, since it is free from Coulomb corrections, that are necessarily present in the $\bar{p}p$ interaction and may introduce further complications and ambiguities in the analysis of the data, especially at low-momenta.
- At low momentum, typically below 1 GeV/c, only S- and P-waves are present in the elementary $\bar{n}p$ interaction, described by the Dover-Richard parametrization [18]. Fig. 1.1 shows the trends of S-wave and P-wave annihilation cross sections according to the Dover-Richard parametrization, where in $\bar{n}p$ interactions the S-wave contribution diminishes with increasing energy, whereas the P-wave contribution grows.

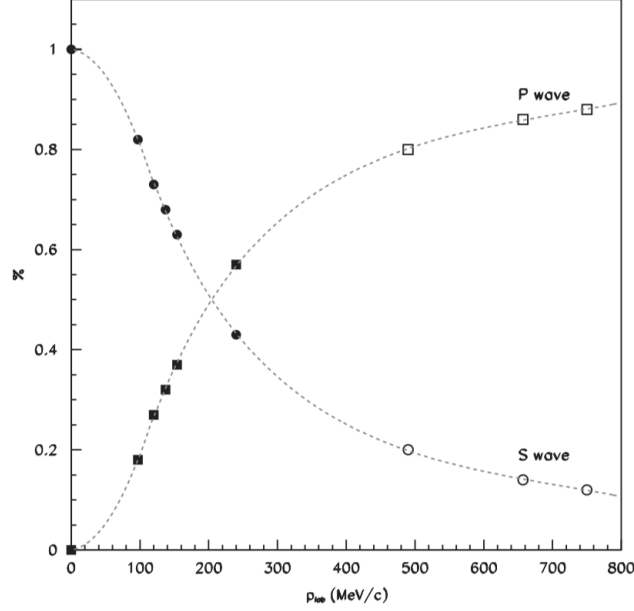


Figure 1.1: Trends of S- and P-wave annihilation cross sections according to annihilation models: the full markers (circles for S-wave, squares for P-wave) are from the Dover–Richard model [1], while the open ones have been obtained by [2] using a version of the optical model.

Therefore, the $\bar{n}p$ system can be represented as an isospin eigenstate $|I, I_3\rangle = |1, +1\rangle$. It is not a C-parity eigenstate, then only J (total angular momentum), P (spatial parity) and G-parity are good quantum numbers, where G-parity is defined as the combination of charge conjugation and a rotation of 180° in isospin space as $G = C e^{i\pi I_2}$.

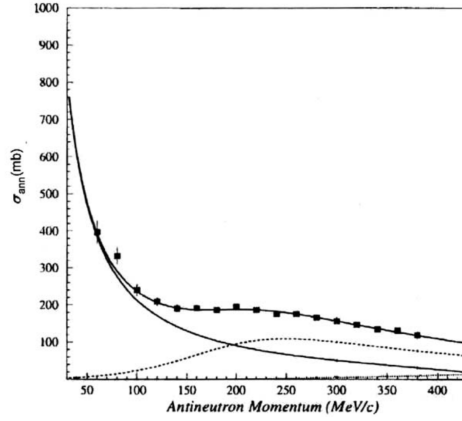
Since $\bar{n}p$ is an antifermion-fermion system, the following equations can be obtained:

$$P = (-1)^{L+1} \quad G = (-1)^{L+S+1}$$

where L is the relative angular momentum between the anti-neutron and the proton, while S is the spin of the system, namely 0 or 1. The $I^G(J^P)$ quantum numbers allowed for the lowest initial states are shown in table 1.

Even if the total cross section σ_T is the first piece of information on the interaction between two particles, for the case $\sigma_T(\bar{n}p)$ could be tough to measure it. The final value of σ_T published in [19] shows how the data could be fitted to a simple parametrization [20], obtaining the results showed in 1.2.

Wave	$2S+1L_J$	$I^G(J^P)$
S	1S_0	$1^-(0^-)$
	3S_1	$1^+(1^-)$
P	1P_1	$1^+(1^+)$
	3P_0	$1^-(0^+)$
	3P_1	$1^-(1^+)$
	3P_2	$1^-(2^+)$
D	1D_2	$1^-(2^-)$
	3D_1	$1^+(1^-)$
	3D_2	$1^+(2^-)$
	3D_3	$1^+(3^-)$

Table 1: Quantum number $I^G(J^P)$ on S , P e D waves.Figure 1.2: Fit of $\sigma_{ann}(\bar{n}p)$ measured by OBELIX up to 400 MeV, including S- (solid line), P- (broken line) and D- (point line) wave contribution.

1.2.2 The anti-nucleon–nucleon interaction

The first approach to describing the antinucleon–nucleon ($\bar{N}N$) interaction relies on **interaction models**. Unlike Quantum Electrodynamics (QED), where the coupling constant α grows with energy and diverges at finite energy, asymptotic freedom characterizes the Quantum Chromodynamics (QCD), the quantum field theory of the strong force between quarks and gluons. Quarks interact weakly at high energies, enabling perturbative calculations, but at low energies the interaction strengthens dramatically, confining quarks and gluons within composite hadrons. Here, the divergent coupling constant α_s renders perturbative approaches invalid, making interaction models essential for describing strong interactions in the $\lesssim 1$ GeV energy range.

The core of this approach is that, following the $\bar{N}N$ annihilation, a *fireball* (which represents the $\bar{N}N$ system) with definite energy and volume $\Omega = \frac{4}{3}\pi R_0^3$

is formed, where $R_0 \sim 1$ fm is the nucleon radius. All the possible $\bar{N}N$ annihilation final states -primarily light mesons- may be reached through the decay of the fireball, according to the statistical Fermi model, where the probability of a final state is determined solely by the available phase space [21].

The use of interaction models in $\bar{N}N$ interaction generally allows a correct description of annihilation, even if it doesn't reproduce the whole experimental scenario, such as differences between nucleon and anti-nucleon scattering and stopping power (Barkas effect) [22]. The reason why it happens is the that the potential formulation used to describe the short range interactions (where annihilation plays the most important role) is not sensitive to the details of the internal dynamics, namely the interactions between quarks and gluons.

The second approach is the **Effective Range** (ER) expansion one, which is independent on any hypothesis concerning the interaction potential and it is based on the obtained fit of experimental cross sections [23]. The annihilation cross section for two interacting particles may be expressed as a partial waves expansion as follows:

$$\sigma_{\text{ann}} = \sum_L \sigma_L^{\text{ann}} = \frac{4\pi}{k^2} \sum_L (2L+1) (\Im m f_L - |f_L|^2),$$

where k is the center-of-mass momentum of the incident particle, and the partial amplitude f_L may be written as a function of the δ_L wave phase shifts induced by the interaction in the system wave function:

$$f_L = \frac{1}{\cot \delta_L - i},$$

The partial waves may be expanded as a series in k , following the so-called *Effective Range (ER) expansion formalism*:

$$k^{2L+1} \cdot \cot \delta_L = \frac{1}{A} + \rho k^2 + O(k^4),$$

where A is the scattering length, and ρ is the interaction range. For S and P partial waves can be obtained:

$$k \cdot \cot \delta_0 = \frac{1}{a} + \frac{1}{2} r k^2$$

$$k^3 \cdot \cot \delta_1 = \frac{1}{b} - \frac{3}{2} \frac{1}{R} k^2$$

a and b are, respectively, the scattering length and nucleon volume, while r and R are the two effective ranges for the two partial waves. These parameters must be determined by a phenomenological fit to the annihilation cross section data. For example, the $I = 1$ source, relevant to the $\bar{n}p$ interaction case, it was found [20], respectively:

$$a_1 = (0.4 + 0.5i) \text{ fm and } r_1 = (-1.4 + 1.8i) \text{ fm for S-wave,}$$

$$b_1 = (0.8 + 0.1i) \text{ fm}^3 \text{ and } R_1 = (0.2 - 0.4i) \text{ fm for P-wave}$$

1.2.3 Electromagnetic and hadronic showers

In energy scenario beyond 10 MeV, Bremsstrahlung and pair production are the main processes which respectively involve photons and electrons/positrons, leading to the development of **electromagnetic showers** within the material. Therefore, to detect such phenomena, HEP experiments employ **electromagnetic calorimeters**. The electromagnetic calorimeter is a device whose size and density are designed to fully contain the electromagnetic shower, producing a signal proportional -or at least correlated- to the energy of the incident particle that initiated the shower. Since the radiation length X_0 determines the size of the electromagnetic shower, the typical longitudinal size for these detectors is of the order of $10\text{-}20 X_0$ ($\sim 30 - 40\text{cm}$).

Furthermore, **hadronic showers** can be originate from strong interactions of high energy (beyond $\sim \text{GeV}$) hadrons, such as protons, neutrons and pions, with the material, as well. To detect this kind of shower, an **hadronic calorimeter** may be employed; since the interaction length λ determines the hadronic shower size and $\lambda \gg X_0$, hadronic showers are larger and longer than electromagnetic ones, requiring hadronic calorimeters to be significantly bigger than electromagnetic calorimeters. Therefore, homogeneous calorimeters cannot be used for hadronic calorimetry as they are for electromagnetic calorimetry. Instead, the sampling calorimeter technique is mandatory, employing alternating layers of passive, dense absorber material and active detection material.

For example, approximately 95% of a hadronic shower is contained within a cylinder of radius equal to the hadronic interaction length ($\lambda_{had} \simeq 44.1\text{ cm}$, in CsI(Tl)). In contrast, about 90% of an electromagnetic shower is contained within a cylinder defined by the Molière radius ($R_M \simeq 3.6\text{ cm}$ in CsI(Tl)). Therefore, there is an order of magnitude difference between the two values. A shape comparison between these two type of shower can be observed in 1.3.

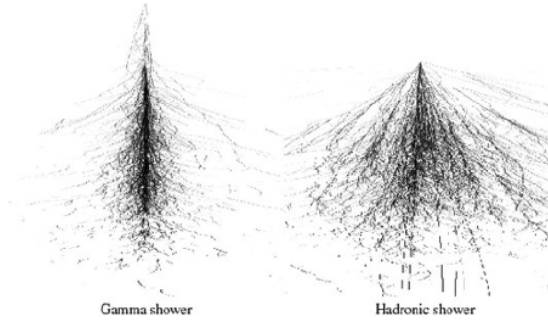


Figure 1.3: Electromagnetic shower vs hadronic shower comparison.

The $\bar{n}$ primarily interacts via the strong nuclear force, annihilating with nucleons in the material ($\sigma \sim 100\text{ mb}$), mainly producing light mesons such as pions which subsequently interact within the calorimeter:

$$\bar{n} + p/n \rightarrow \pi^\pm, \pi^0..$$

The π^0 mesons mainly ($\sim 98.8\%$) decays into a couple $\gamma\gamma$, producing electromagnetic showers that are fully contained in the calorimeter.

While charged pions π^+ and π^- undergo both electromagnetic and hadronic interactions, producing final products that are not fully contained within the calorimeter volume, this leads to a lack in the energy deposited in the detector. Both forward and backward directions are then involved, according to the reaction kinematics.

1.3 The MANTRA Project

It may be challenging for general-purpose experiments to directly identify and reconstruct anti-neutrons produced in e^+e^- collisions, such as those at Belle II and BESIII, which lack a hadronic calorimeter. Anti-neutrons generally annihilate with detector material nucleons such as that of an electromagnetic calorimeter, producing hadronic showers which only allows a loose measurement of the anti-neutron energy with extremely large uncertainty.

Since anti-neutron kinematics have so far been measured via missing mass methods or isospin symmetry with anti-protons, a direct anti-neutron reconstruction would be highly desirable, especially in studying processes with neutral hadrons, such as $B^+ \rightarrow K^+ n \bar{n}$ or $J/\Psi \rightarrow p \pi^- \bar{n}$.

This thesis fits in the larger context of the **MANTRA** (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) PRIN2022 project, an inter-university (University of Turin, University of Ferrara and INFN) effort aimed at developing a method to improve the reconstruction of anti-neutron in existing high-energy physics experiments. It provides a general approach to measure the energy of **anti-neutrons** $\bar{n}$ up to 10 GeV using an electromagnetic calorimeter.

The main idea is to combine the information from a high time-resolution time of flight detector (TOP and TOF in the Belle II and in the BESIII experiments, respectively) with that from an electromagnetic calorimeter (ECL and EMC in the Belle II and in the BESIII experiments, respectively), where the anti-neutrons annihilation occurs.

Due to its neutral charge and its immediate annihilation, tagging $\bar{n}$ can be very challenging. However, the annihilation can produce self-contained light mesons, mostly made of high-energy pions (π^-, π^0, π^+) and soft nuclear fragments. These can be detected in the Electromagnetic Calorimeter (detailed examination on next chapter), where charged pions can produce hadronic showers, while neutral pions immediately decay into two photons which produce electromagnetic showers in the calorimeter volume. The comparison of the cluster energy deposition patterns, the so called shower shapes variables, left in the ECL by anti-neutrons and other neutral particles, such as photons, can improve the characterization of neutral particles.

A crucial aspect of the analysis process is the role of simulation, where GEANT4 [16] (more details in the next chapter) plays a central role, modeling the hadronic interactions in the calorimeter; realistic simulation is essential to accurately reproduce real data. For $\bar{n}$ annihilation, parameterizations are typically derived from $\bar{p}$ measurements, necessitating validation against data.

This thesis aims to study the $\bar{n}$ hadronic interactions modeled by simulation software and compare them with those observed in Belle II data, focusing exclusively on signals from the Belle II electromagnetic calorimeter. To achieve this, $\bar{n}$ variables are first studied using a signal sample, where only the specifically selected processes with $\bar{n}$ in the final state are generated. Several background channels are then included to simulate the complete Belle II environment, concluding with a comparison between the results obtained from simulations and those from real data, with a Data/MC agreement.

Therefore, this work addresses the following question:

Are $\bar{n}$ hadronic showers correctly simulated in the Belle II software?

Chapter 2

The Belle II experiment

The following sections introduce the **SuperKEKB** [24] collider, describing its technical setup briefly. Then the **Belle II** [5] detector is presented, discussing it sub-detector by sub-detector. Afterwards, the Belle II Analysis Software Framework (**basf2**) and **TopoAna** program are briefly described, with attention to the role of Monte Carlo methods in collider experiment simulation and reconstruction.

2.1 The SuperKEKB collider

SuperKEKB [24] is an asymmetric e^+e^- double-ring collider, located at KEK (The High Energy Accelerator Research Organization), in Tsukuba, Japan. It is the upgrade of the previous collider KEKB. It has a circumference of 3016 m and collides bunches of electrons with bunches of positrons, respectively of energy 7 GeV (High Energy Ring -HER-) and 4 GeV (Low Energy Ring -LER-). Fig. 2.1 is a schematic image of the SuperKEKB design:

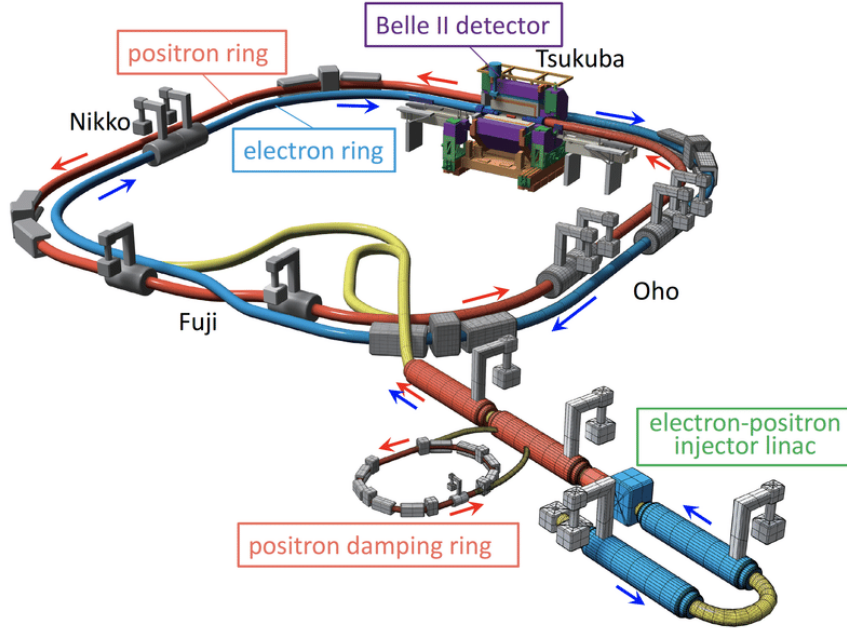


Figure 2.1: SuperKEKB design, where the Interaction Point (IP) is located in Tsukuba surrounded by the Belle II detector.

It is noteworthy that the positron damping ring reduces the e^+ beam emit-

tance, as positrons are obtained from a rotating tungsten target subjected. Electrons, by contrast, are injected directly from the linac since a controlled e^- beam can easily be obtained [25].

The resulting center-of-mass energy $\sqrt{s}$ in the IP can be modulated, but most of the data are acquired at the $\Upsilon(4S)$ resonance (10.58 GeV), an excited quarkonium state of $b\bar{b}$, which mainly decays into an entangled couple of $B\bar{B}$ ($\sim 100\%$). The $\Upsilon(4S)$ mass is just 20 MeV above the $B\bar{B}$ threshold, then the mesons are produced nearly in the rest frame. Since the lifetime of neutral B-mesons is short ($\tau = 1.510$ ps) and they are produced with low velocities, their path before decaying is short, making tough to separate the decays of the two produced mesons. The explicit asymmetry of bunches results in a relativistic forward Lorentz boost of the center-of-mass, where the decay length of the mesons is longer in the LAB frame, allowing the observation of the produced B mesons. Due to the huge amount of produced B mesons, SuperKEKB is regarded as an example of so-called (super) **B-factory**.

SuperKEKB has been running since 2018 and it achieved the world's highest instantaneous luminosity ($5.244 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$) in March 19, 2026. The design luminosity of SuperKEKB $8 \times 10^{35} \text{ cm}^{-2} \text{ s}^{-1}$, 40 times higher than Belle's one, with the goal to accumulate data corresponding to an integrated luminosity of 50 ab^{-1} . The March 19, 2026 SuperKEKB 24-Hour Operation Summary is shown in Fig. 2.2.

2.2 The Belle II detector

The only interaction point of SuperKEKB is surrounded by the **Belle II** detector [5], a general purpose high-energy particle layered detector with almost 4π of solid angle coverage. It presents a cylindrical shape to cover the e^+e^- collisions, composed of a barrel and two end-caps located along the z-axis.

The Belle II detector measures 8 m in both width and height, with a total weight of approximately 1400 tons. With the exception of the KLM, all its subsystems are immersed in a 1.5 T magnetic field.

The detector is composed of several sub-detectors arranged cylindrically, reflecting the asymmetric distribution of the final state particles. The cylindrical structure of the detector is the *barrel*, while the bases are made of the forward end-cap (along the direction of the e^- beam) and the backward end-cap (along the direction of the e^+ beam).

The Belle II detector consists of the following sub-detectors, arranged radially outward from the IP:

- (a) Beam Pipe made of Beryllium (Be),
- (b) Two layers of silicon Pixel Detector (PXD),
- (c) Four layers of double-sided silicon strip detector (SVD),
- (d) The Central Drift Chamber for tracking the charged particle (CDC),

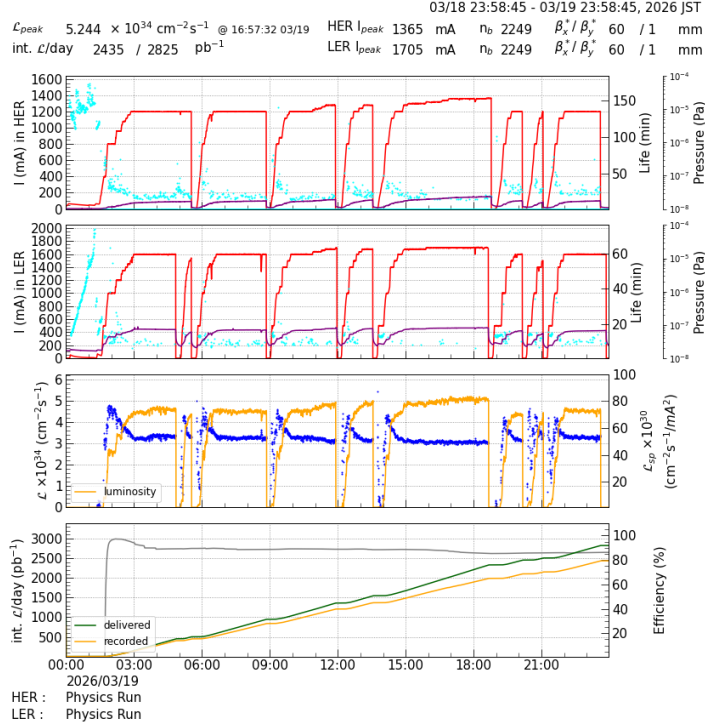


Figure 2.2: SuperKEKB operating parameters on March 18-19, 2026. From top to bottom: HER and LER ring currents (first two panels), instantaneous and integrated luminosity (bottom two panels).

- (e) Time of Propagation (TOP) Cherenkov counter in the barrel + Cherenkov ring imaging (ARICH) in the front end-cap region,
- (f) The Electromagnetic Calorimeter (ECL) to detect photons and to identify charged particles,
- (g) The superconductive magnet which provides 1.5 T axial magnetic field in order to bend charged particle,
- (h) The K_L and Muon detector (KLM) to track muons and K_L .

The electromagnetic calorimeter (ECL) has a central role in this elaborate since neutral ECL clusters associated to anti-neutron candidates are studied. A Belle II scheme is shown in Fig. 2.3.

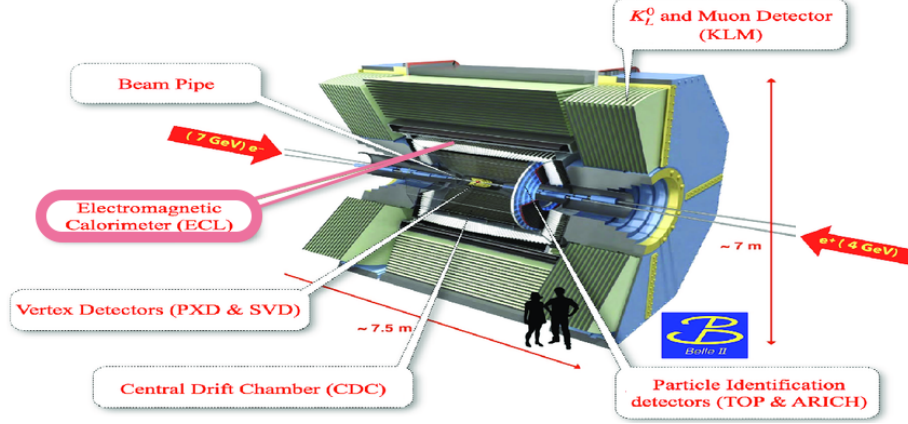


Figure 2.3: An artistic view of the Belle II detector.

2.2.1 The vertex detector VXD

Immediately following the beryllium beam pipe, which does not play a detection role, the innermost layer is the Vertex Detector (**VXD**).

It consists of two devices: the Silicon Pixel Detector (**PXD** - 2 layers) [26] and the Silicon Vertex Detector (**SVD** - 4 layers) [27]. Overall there are six layers, of whose the first two ($r_1 = 14\text{mm}$, $r_2 = 22\text{mm}$) are pixelated sensors of the DEPFET type and the remaining four ($r_3 = 38\text{mm}$, $r_4 = 88\text{mm}$, $r_5 = 115\text{mm}$, $r_6 = 140\text{mm}$) are double-sided silicon strip sensors.

The VXD provides coverage of $17^\circ < \theta < 150^\circ$ and full azimuthal coverage in ϕ , where θ is the polar angle relative to the z -axis -aligned with the e^- beam direction- and ϕ is the azimuthal angle in the xy plane using cylindrical coordinates. A scheme of the VXD is shown in Fig. 2.4.

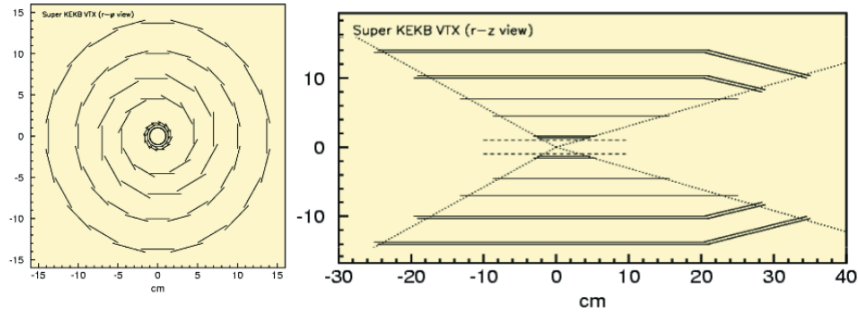


Figure 2.4: A scheme of the Belle II VXD.

The VXD purpose is to reconstruct the position of the primary vertex, the vertex of B-mesons decay which are directly produced by the decay of the $\Upsilon(4S)$, improving the resolution of the tracking system as well.

2.2.2 The Central Drift Chamber (CDC)

The Central Drift Chamber (**CDC**) [28] is a large volume composed of small drift cells and it is the main tracking system of the Belle II detector. The CDC reconstructs charged-particle tracks by measuring the hit positions, allowing the determination of their momenta. Furthermore, the specific energy loss dE/dx measurement provides PID (Particle IDentification) information, complemented by the 1.5 T magnetic field that enables charge separation.

The CDC has an internal radius of 16 cm and an external radius of 113 cm and it contains 14339 wires in total distributed in 56 layers, either parallel (axial orientation) or skewed (stereo orientation) to the magnetic field direction. Combining their information, it is possible to reconstruct a full tridimensional helix track. Inside the cell, the average drift velocity is $3.3\text{cm}/\mu\text{s}$.

The CDC has the same VXD angular (θ, ϕ) coverage and, in order to increase the acceptance at forward angles, it is implemented asymmetrically along the z axis. It is filled with a mixture of helium (He) and ethane (C_2H_6), which represents a proper trade-off between momentum resolution for tracking and dE/dx resolution for the PID. A schematic view of the CDC is shown in Fig. 2.5:

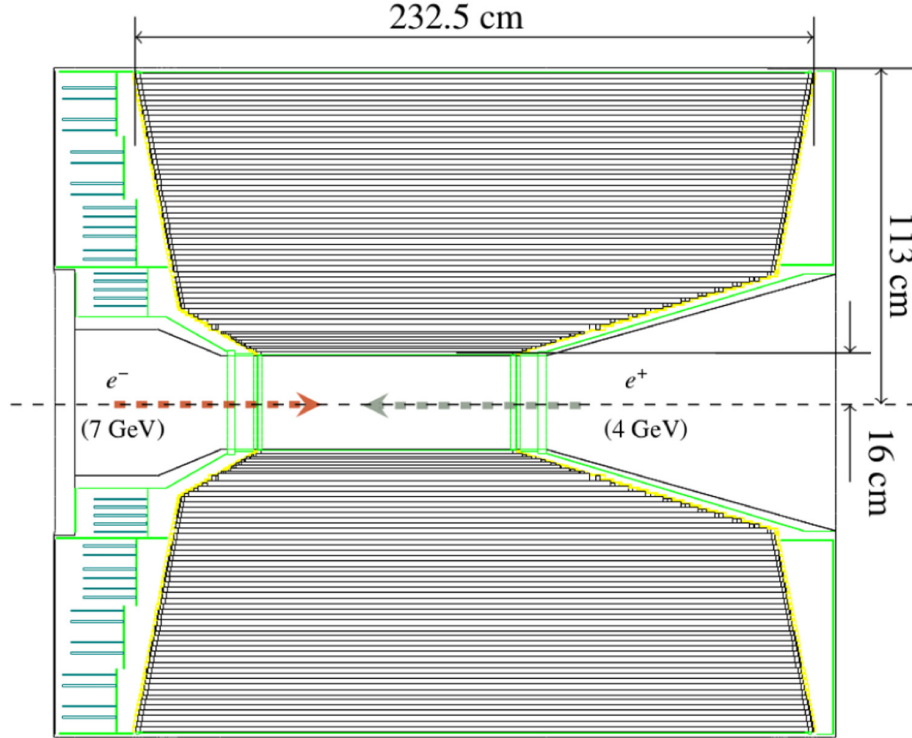


Figure 2.5: A scheme of the Belle II CDC.

2.2.3 PID systems: TOP and ARICH

Particles crossing a Cherenkov radiator material at speeds exceeding the phase velocity of light in that medium ($\beta > 1/n$, where n is the refractive index) emit photons at the Cherenkov angle $\theta_c = \arccos(1/\beta n)$. Particles with identical momentum but different masses produce Cherenkov radiation at distinct angles, enabling Particle IDentification (PID).

For particle identification in the barrel region, a Time of Propagation (**TOP**) [29] counter -a novel Cherenkov detector- is employed. In total, there are 16 modules, each consisting of a 45 cm wide and 2 cm thick quartz bar with a small expansion volume of about 10 cm long at the sensor end. There is an expansion wedge that behaves as a lens and improves the spatial resolution of the Cherenkov photon imaging. At the exit window of the wedge, two rows of sixteen fast multi-anode photon detectors are mounted.

Within the 2.6 m-long quartz, the particle produces Cherenkov radiation that travels via total internal reflection to the mirror. Then the radiation is reflected towards the photon detectors at the backward end, where a small prism is located, acting as a volume expansion and a light guide. The time of arrival and the impact position of the Cherenkov photons provide the PID information.

The hit distributions differ according to the particle species and their masses: different particles exhibit distinct Cherenkov angles, resulting in different photon paths. Consequently, they are detected at different positions within the PhotoMultiplier (PMT) array and at different times.

A scheme of TOP counter module is illustrated in Fig. 2.6 and a side view of the TOP counter in Fig. 2.7.

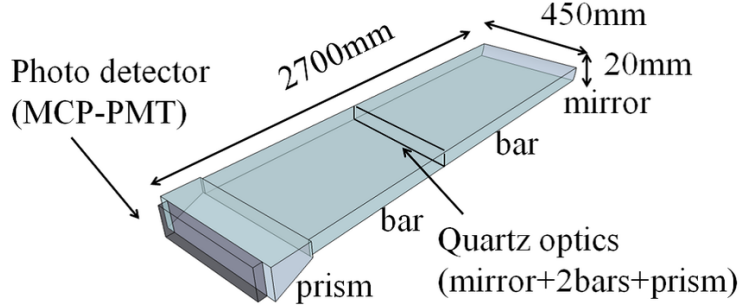


Figure 2.6: Scheme of the TOP counter module.

Thanks to a 16-channel Micro Channel Plate PhotoMultiplier (MCP-PMT), TOP counters are equipped with photosensors that provide a time resolution of about 100 ps.

The Aerogel Ring Imaging CHerenkov (**ARICH**) [30] is located in the forward end-cap region of the Belle II detector and is thus complementary to the aforementioned TOP. Its main goal is to separate kaons K from pions π across the full kinematic range of the experiment, from 0.5 to 4.0 GeV/c .

It consists of two 2 cm thick aerogel layers ($n_{up} = 1.045$, $n_{down} = 1.055$), which

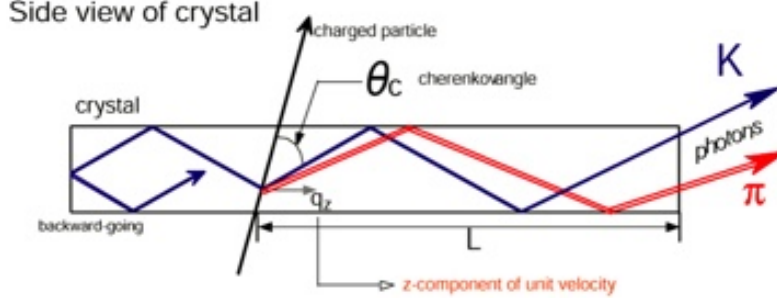


Figure 2.7: Representation of the side view of the TOP counter with internal reflection of the Cherenkov radiation.

increase the photon yield without degrading the Cherenkov angle resolution compared to using a single aerogel type [31]. A scheme of the ARICH working principle and an example of a ring produced by a cosmic-ray muon are shown in Fig. 2.8.

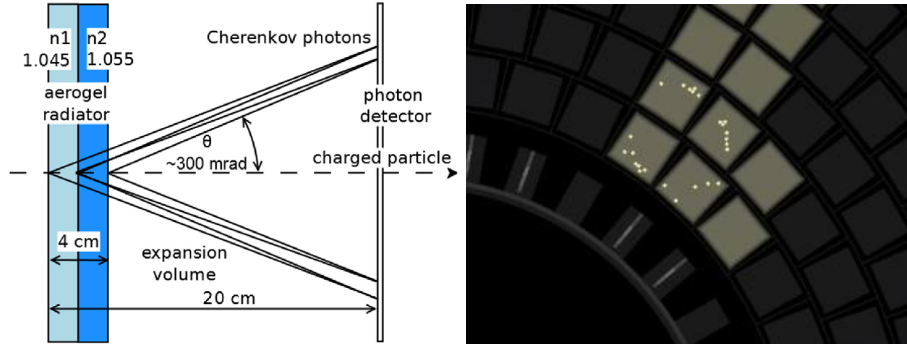


Figure 2.8: (a) Working principle of the ARICH. (b) Ring produced by a cosmic-ray muon.

2.2.4 The Electromagnetic Calorimeter (ECL)

The Electromagnetic CaLorimeter (**ECL**) [32] plays a central role in this thesis, since the neutral ECL clusters associated to anti-neutron candidates are studied.

The primary purpose of the ECL is to detect photons and electrons, providing precise measurements of their energy and position ($E, \vec{x}$), studying the produced electromagnetic shower. The ECL contributes to identifying neutral particles -such as photons and neutral hadrons- as well as separate the charged particles, primarily electrons and pions. Anti-neutrons can be detected through its annihilation products (mainly pions). A typical shape of an ECL shower caused by a 7 GeV photon is shown in Fig. 2.9.

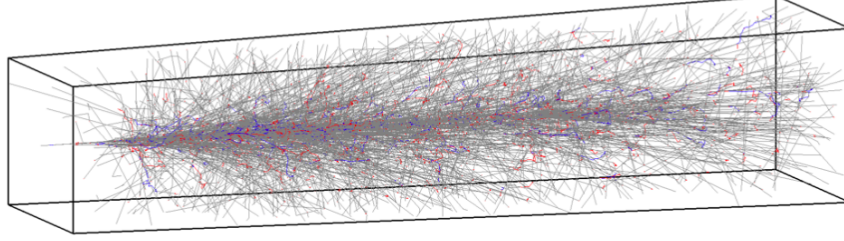


Figure 2.9: Typical ECL shower caused by a 7 GeV photon.

The Molière radius R_M is a characteristic constant of a material that defines the transverse scale of fully contained electromagnetic showers initiated by high-energy electrons or photons. Its definition, for an electromagnetic calorimeter, is that $\sim 90\%$ of electromagnetic shower is contained in a cylinder with radius R_m , while the $\sim 95\%$ is contained in a cylinder with radius $2R_m$.

The radiation length X_0 is defined as the mean length (in cm) over which an electron's energy is reduced by a factor of $1/e$. The critical energy E_c is defined as the energy at which radiation losses equal ionization losses for electrons/photons. X_0 and E_c determine the Molière radius R_M as follows:

$$R_m = \frac{21MeV}{E_c} \times X_0$$

CsI(Tl) crystals

The thallium-doped caesium iodide **CsI(Tl)** crystals have $X_0 = 1.860$ cm and $R_m = 3.53$ cm. This material is one of the brightest known scintillator, providing a light yield Y of ~ 52000 photons/MeV, with a photon emission spectrum at ~ 550 nm, ideal for photodiode readout. Properties of typical inorganic scintillators used in calorimetry are shown in 2.10.

Material	Density (g/cc)	Wavelength of Maximum Emission	Refractive Index	Principal Decay Constant (μ-Sec)	Total Light Yield	Pulse Rise Time (μ-Sec)	Absolute Scintillation Efficiency	Relative Pulse Height
Nal(Tl)	3.67	415	1.85	0.23	38000	0.5	11.3	1.00
CsI(Tl)	4.51	540	1.80	1.0	52000	4.0	11.9	0.49
LiI(Eu)	4.08	470	1.96	1.4	11000		2.8	0.23
BGO	7.13	505	2.15	0.30	8200	0.8	2.1	0.13
BC-400	1.032	423	1.58	0.002	10000		3.0	0.25
BC-454	1.026	425	1.58	0.0022	10000		3.0	0.25

Figure 2.10: Properties of inorganic scintillators used in calorimetry [3].

The ECL is composed of 8736 array of **CsI(Tl)** crystals. Each is a truncated

pyramid with dimension $6 \times 6 \times 30 \text{ cm}^3$, where the radial length is around $16.2X_0$. Each crystal is wrapped with a layer of $200 \text{ }\mu\text{m}$ thick Gore-Tex porous teflon and then covered with a $50 \text{ }\mu\text{m}$ thick aluminized polyethylene. Assuming the z -axis is aligned with the e^- beam momentum direction, with $z = 0$ coinciding with the nominal IP position, the ECL is made of a 3 m long barrel section and the end-caps at $z = 2.0 \text{ m}$ (forward) and $z = -1.0 \text{ m}$ (backward). An ECL scheme is shown in Fig. 2.11:

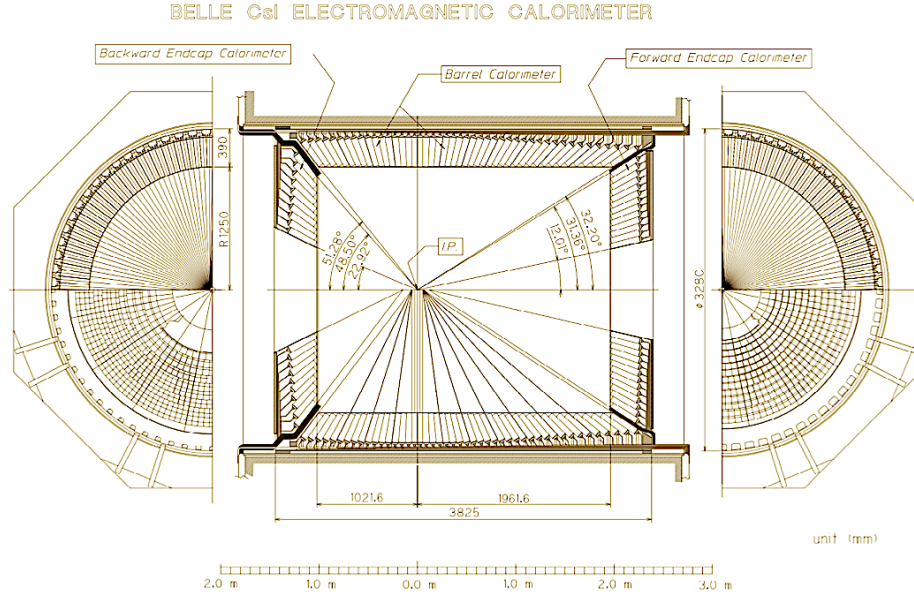


Figure 2.11: A scheme of the Belle II ECL detector.

It covers about 90% of the solid angle in the center-of-mass system (polar coverage: $12^\circ \leq \theta \leq 155^\circ$); there is a $\sim 1^\circ$ gap between barrel and each end-cap.

An image showing the ECL θ coverage and segmentation for each region is shown in Fig. 2.12.

	θ coverage	θ segments	ϕ segments	# of counters
Forward Endcap	$12.01^\circ - 31.36^\circ$	13	48 - 144	1152
Barrel	$32.2^\circ - 128.7^\circ$	46	144	6624
Backward Endcap	$131.5^\circ - 155.03^\circ$	10	64 - 144	960

Figure 2.12: ECL coverage and segmentation for each ECL region.

Light from each crystal is detected by two $10 \times 20 \text{ mm}^2$ *Hamamatsu S2744*-

08 photodiodes [33], glued to the rear surface of the crystal and coupled to charge-sensitive preamplifiers.

2.2.5 The solenoid and iron structure

After the ECL, an axial magnetic field of 1.5T is generated by a 23 tons superconducting solenoid. The curvature of charged-particle trajectories (Larmor radius) allows the determination of their momentum and charge, which, when combined with additional detector information, enables particle identification. The magnet is made of niobium-titanium-copper (NbTi/Cu) alloy and is cooled by a liquid helium cryostat to achieve superconducting conditions.

2.2.6 The K_L and Muon Detector (KLM)

The K_L and Muon Detector (**KLM**) [34] represents the outermost layer of Belle II and it is dedicated to the detection of K_L mesons and muons, which are very penetrating particles. It is composed of alternating layers of 4.7 cm thick iron plates and active detector elements; the former acts as a return yoke for the magnetic flux. The alternating layers provide at least 3.9 interaction lengths (compared to 0.8 in the ECL), allowing the K_L to interact and produce hadronic showers.

KLM covers the barrel region (15 layers) and the end-caps regions (forward 14 layers and backward 12 layers). The two inner barrel layers and the entire end-caps are instrumented with scintillator strips, while the 13 outer barrel layers are instrumented with Resistive Plate Chambers (RPC).

As compared to charged hadrons, muons travel longer distances and with smaller deflections inside the KLM, which allows discrimination between the two.

Chapter 3

The Belle II Analysis Software Framework



To perform the simulations, reconstruction and the analyses of this thesis, the Belle II Analysis Software Framework (**basf2**) [35] has been employed. It is the primary software used in all aspects of the data-processing chain at Belle II, such as generating simulated data, unpacking raw data, performing reconstruction (tracking, clustering) and carrying out high-level analysis. The final analysis steps (as histogramming, fitting distributions, etc.) are usually performed in other frameworks as well, such as **ROOT** or the Matplotlib Python library.

3.1 Monte Carlo simulation

The Monte Carlo method is a technique used to compute probabilities and related quantities by exploiting sequences of pseudo-random numbers. These sequences are "pseudo-random" because sequences of random numbers possess meaning, not individual ones -though the latter are frequently referenced. Randomness characterizes the sequence, not single numbers.

Monte Carlo methods are widely used in HEP physics for tasks such as **event generation** and **detector simulation**. The first one regards the production and the decays of particles with a certain set of properties, while the second one concerns the simulation of the particle interactions during the path in the detector materials.

In Belle II, electron-positron collisions produce a certain amount of particles, according to the adopted physics model, which may follow the Standard Model or alternatives such as supersymmetry or dark matter production. Example of a physics generator is *PYTHIA* for the generation of high-energy physics collision events, *EvtGen* which simulates the decays of particles, primarily B and D mesons, and *Phokhara* which simulates the leptonic and hadronic final states at the next-to-leading order (NLO) accuracy, namely simulating Initial State Radiation (**ISR**), which has a primary role in the analysis, and so forth.

After the generation, the particles are propagated through the detector, accounting for their interactions with material and the effects of the magnetic

field, using GEANT4. This is an open source software which performs the interaction between particles and the detector systems.

3.2 Reconstruction

The reconstruction represents the process in which particles are measured and identified. The closer the reconstructed quantities (such as initial positions and four-momenta) are to the generated-level information, the better the reconstruction performance in simulated MC and real data. Since several products of e^+e^- collisions are short living particles, it is necessary to reconstruct them indirectly via their daughter particles. Furthermore, intrinsic noise must be taken into account, as it may generate spurious random signals unrelated to actual particles. Interactions between beam electrons or positrons, either with each other or with other accelerator components -known as beam background- must be considered. The same reconstruction algorithms are employed in data and MC, in order to obtain the same reconstruction. To test the correct response of the reconstruction, generator-level Monte Carlo truth of the simulated data pool is use (the so called **MC truth**). The Monte Carlo matching (MC-matching) procedure identifies the generator-level particle responsible for a reconstructed candidate, crucial for understanding reconstruction effects and backgrounds. This corresponds to the association of reconstructed particles with MC particles in Belle II nomenclature.

3.2.1 The ECL reconstruction algorithm

For each ECL crystal, a 32-bit word is recorded as the digitized output from the electronics. This 32-bit word, referred to as **ECLDigit**, encodes:

- energy information as Flash Analog-to-Digital Converter (FADC) counts (18 bits),
- time information in clock ticks (12 bits),
- 2 status bits (signal quality flags).

To reconstruct the shower, an algorithm processes the ECLDigit objects into two new data objects:

- **ECLShowers**, an ECL-internal object containing additional debug information,
- **ECLClusters**, the primary analysis object.

Only ECLShowers with $E > 20$ MeV are converted to ECLClusters.

The first step identifies **connected regions**, which rapidly reduces the combinatorial complexity. These are specific cell collections determined by a dedicated clustering procedure.

An example of this procedure is shown in Fig. 3.1.

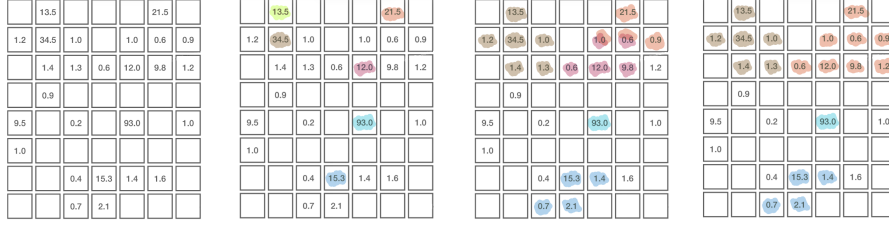


Figure 3.1: Description of ECL clustering process: (a) Step 0; (b) Step 1; (c) Step 2; (d) Step 3.

Each cell represents one ECLDigit, with its value corresponding to the deposited energy in MeV. Empty cells indicate ECLDigit values below the readout threshold (0.1 MeV in this example).

In step 1, every ECLDigit above 10 MeV is selected. In step 2, any neighbors above 0.5 MeV are attached to the highest-energy seed found in step 1. Sometimes two seeds compete for the same neighbor, and sometimes clusters have no neighbors (i.e. 93.0). In step 3, overlapping neighbor regions trigger a merge. Additionally, any cells above 1.5 MeV recruit their next neighbors iteratively. After this process, the connected regions are obtained, as shown in Fig. 3.2.

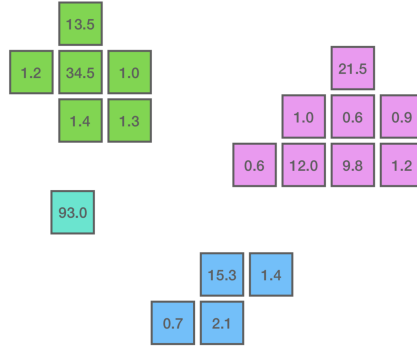


Figure 3.2: Connected regions obtained in the example (threshold: 0.1 MeV).

The final step is to pick the crystals with energy above 10 MeV, namely the Local Maxima (LM). Every connected region has one LM at least. The connected regions are furtherly processed in order to obtain the position and the energy of the cluster, processing the LM with a splitting-weighting process among the connected regions crystals [36].

The main idea is that an ECLShower object is produced from every LM. If a connected region contains only one LM, the total energy of its crystals is assigned to the corresponding ECLShower. Otherwise, if multiple local maxima are present, an iterative procedure is performed, where the weight for each crystal depends on the distance of the crystal from the center-of-gravity of each

ECLShower (center-of-gravity is calculated using weighted energies). Connected regions are used to minimize the number of times the computationally intensive splitting procedure must be performed.

3.2.2 ECL performance

The observed ECL **energy resolution** can be written as follow:

$$\frac{\sigma_E}{E} = \sqrt{\left(\frac{0.81\%}{\sqrt[3]{E}}\right)^2 + \left(\frac{0.066\%}{E}\right)^2 + (1.34\%)^2},$$

where the first term represents stochastic fluctuations (Poissonian statistics) in the shower generation, the second term accounts for electronic noise, and the third is a constant term dependent on calibration.

For example, $\frac{\sigma_E}{E} = 4\%$ at 100 MeV and $\frac{\sigma_E}{E} = 1.4\%$ at 8 GeV [37]). The energy resolution behavior is shown in Fig. 3.3.

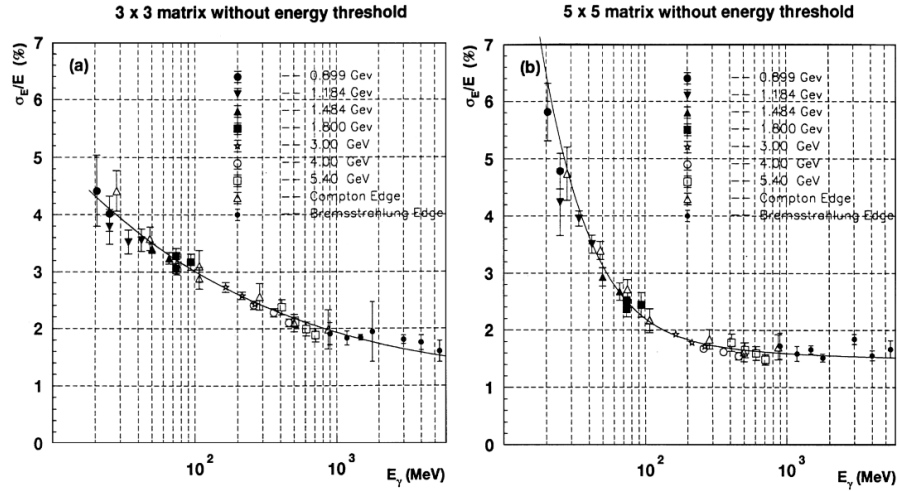


Figure 3.3: The energy resolution as a function of incident photon energy for the (a) 3Y3 and (b) 5Y5 matrix with total energy sum. The error bars are not statistical but are the rms of four measurements taken with the crystals (3,3), (3,4), (4,3) and (4,4) into the photon beam [4].

3.3 basf2 modules and path

A **basf2 module** is the smallest building block of the framework and it is a script, usually written in C++, designed to process a specific unit of data. A typical data chain consists of a linear arrangement of modules. All the work is done in modules, including reading and writing data to and from disk or more complex operations such as full detector simulation.

Modules are linearly arranged in a **path**, which is a container for them. The

linear order depends on the specific task defined by the user. Therefore, the modules are executed one at a time, in the same order they were added into the path, from the first to the last. A schematic view of processing flow is shown in Fig. 3.4.

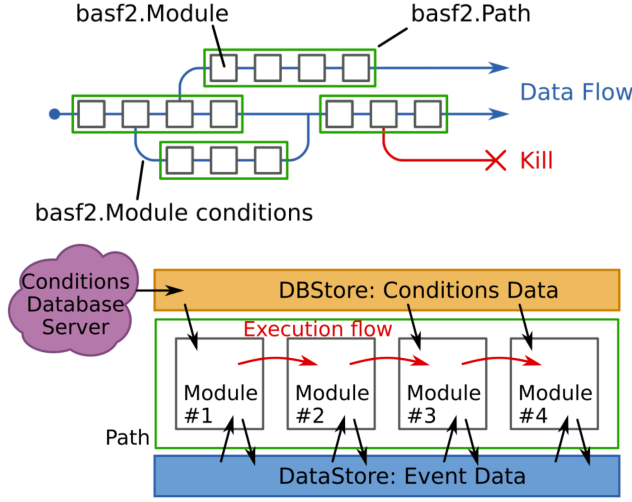


Figure 3.4: Scheme of the processing chain in basf2.

The modules process data stored in the DataStore, with each module having access to read from and write to it. There are also non-event data, the so called conditions, which are loaded from a central conditions database, the DBStore. These databases store the information about the specific detector conditions at each run of the experiment.

A **package** is a logical collection of code in basf2, consisting of several modules and python scripts that configure paths to do common things. The packages have meaningful names related to their specific operation (tracking, reconstruction, ecl, klm, etc...). All the packages are stored in the Belle II GitLab repository. The most used package in this thesis is the **analysis package**, which contains python functions and C++ basf2 modules, such as modularAnalysis. The **steering file** is the core script in which analysis and data-processing tasks are performed, with a path being declared and modules added to it. Some highlights of the steering files used are provided in the appendix.

3.4 TopoAna: Topology Analysis

In high-energy physics data analysis, a comprehensive understanding of Monte Carlo samples is crucial to select signals with higher efficiency while suppressing backgrounds. This is exactly the case of the present elaborate. Knowledge of the physics processes enables identification of the main back-

grounds. Selection criteria can then be further optimized by analyzing differences between these backgrounds and the signal. An example of topology diagram is provided in 3.4.

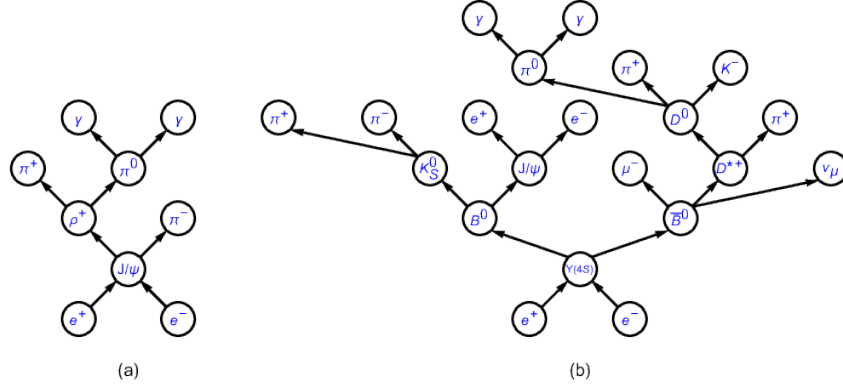


Figure 3.5: Topology diagrams of (a) $e^+e^- \rightarrow J/\psi$ and (b) $e^+e^- \rightarrow \Upsilon(4S) \rightarrow B^0 \bar{B}^0$, two typical processes at e^+e^- colliders, such as Belle II and BESIII.

The **TopoAna** program, developed in C++ and ROOT, provides a convenient interface for verifying Monte Carlo topology information beyond raw data inspection.

The program input consists of one or more ROOT files containing a **TTree** with raw topology truth information from the MC samples. This TTree stores the processes that occurred in each event, including particle multiplicity (number of final state occurrences), PDG codes and mother indices (parent-daughter relationships).

TopoAna is an offline tool independent of basf2. It extracts human-readable topology information from raw MC truth data. Examples of raw MC information (Fig. 3.6) and human-readable MC information (Fig. 3.7) are shown, both depicting the topology diagram in Fig. 3.5 (b).

```

Number of particles      : 63
PDG codes of particles  : 300553,
                        -511, 511, -433, 421, 211, 22, -413, 111, 111, 113,
                        211, -431, 22, -323, 213, -421, -211, 22, 22, 22,
                        22, 211, -211, 333, 11, -12, 22, -311, -211, 211,
                        111, 221, 331, 321, -321, 310, 22, 22, 111, 111,
                        111, 111, 111, 221, 111, 111, 22, 22, 22, 22,
                        22, 22, 22, 22, 22, 22, 22, 22, 22,
                        22, 22
Mother indices of particles : -1,
                        0, 0, 1, 1, 1, 1, 2, 2, 2, 2,
                        2, 3, 3, 4, 4, 7, 7, 8, 8, 9,
                        9, 10, 10, 12, 12, 12, 12, 14, 14, 15,
                        15, 16, 16, 24, 24, 28, 31, 31, 32, 32,
                        32, 33, 33, 33, 36, 36, 39, 39, 40, 40,
                        41, 41, 42, 42, 43, 43, 44, 44, 45, 45,
                        46, 46
    
```

Figure 3.6: Raw MC information example from 3.5 b.

0	$e^+e^- \rightarrow \Upsilon(4S)$	-1	9	$\rho^+ \rightarrow \pi^0\pi^+$	6
1	$\Upsilon(4S) \rightarrow B^0\bar{B}^0$	0	10	$K^{*-} \rightarrow \pi^-\bar{K}^0$	6
2	$B^0 \rightarrow \pi^0\pi^0\rho^0\pi^+D^{*-}$	1	11	$D_s^- \rightarrow e^-\bar{\nu}_e\phi\gamma$	7
3	$\bar{B}^0 \rightarrow \pi^+D^0D_s^{*-}\gamma$	1	12	$\eta \rightarrow \pi^0\pi^0\pi^0$	8
4	$\rho^0 \rightarrow \pi^+\pi^-$	2	13	$\eta' \rightarrow \pi^0\pi^0\eta$	8
5	$D^{*-} \rightarrow \pi^-\bar{D}^0$	2	14	$\bar{K}^0 \rightarrow K_S^0$	10
6	$D^0 \rightarrow \rho^+K^{*-}$	3	15	$\phi \rightarrow K^+K^-$	11
7	$D_s^{*-} \rightarrow D_s^-\gamma$	3	16	$\eta \rightarrow \gamma\gamma$	13
8	$\bar{D}^0 \rightarrow \eta\eta'$	5	17	$K_S^0 \rightarrow \pi^0\pi^0$	14

Figure 3.7: Highly readable MC information example from 3.5 b, as presented by the TopoAna program.

Chapter 4

Study of the Monte Carlo signal channel

The first part of the present work aims to characterize the kinematics of the $\bar{n}$, using a Monte Carlo signal sample generated specifically for the selected reference channel:

$$e^+ + e^- \rightarrow p + \pi^- + \bar{n} (+ \gamma_{ISR})$$

This is selected because it features simple kinematics, with the $\bar{n}$ as the only missing particle, enabling a semi-exclusive analysis.

The Feynman diagram for this process, without and with ISR emission, is shown in Fig. 4.1.

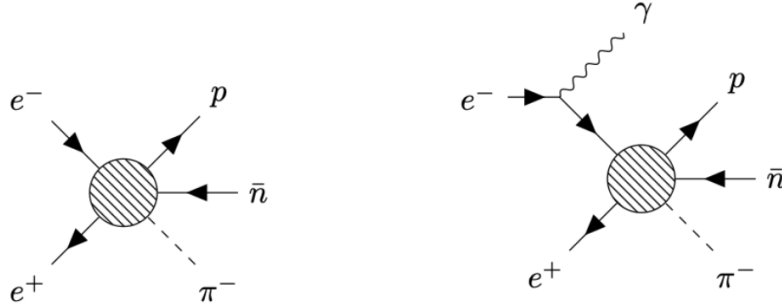


Figure 4.1: Feynman Diagram of the channel of interest. The ISR gamma is a pre-interaction radiation emitted by the e^+ and/or e^- . The recoil system is made of $p\pi^-(\gamma_{ISR})$.

In particular, the kinematic relation between the $\bar{n}$ and the recoil system $p\pi^-(\gamma_{ISR})$ will be exploited, considering both configurations: in the presence of Initial State Radiation (ISR) photon emission and in its absence. ISR corresponds to the emission of photons from incoming particles before they collide in high-energy physics experiments. This reduces the effective center-of-mass energy available, which will no longer be ~ 10.58 GeV, in the Belle II case. The ISR energy and angle distributions will be discussed in detail, providing a comparison with a simple phase space model emission.

Then, the effect of a one constraint (1C) kinematic fit over the recoil mass will be discussed, in order to obtain a higher recoil momentum resolution.

In total, 100000 events are generated and to perform the event generation and the analysis, **basf2** is employed. Script code highlights for this section are shown

in Appendix 6, where the decay file, MC generation file, and steering file are explained in detail.

The basf2 **Phokhara_evtgen_combination** generator has been adopted. It is made of two generators: **Phokhara** and **EvtGen**.

Phokhara is a Monte Carlo event generator that simulates e^+e^- annihilation into hadrons at next-to-leading order (NLO) accuracy. This includes virtual and soft photon corrections to single-photon events, plus double hard photon emission.

EvtGen is a Monte Carlo event generator that simulates decays of heavy-flavor particles, primarily B and D mesons. It includes decay models for intermediate and final states with scalar, vector, and tensor mesons/resonances, as well as leptons, photons, and baryons. Decay amplitudes generate each branch of the full decay tree, accounting for angular and time-dependent correlations to simulate CP-violating processes.

Therefore, these two generators have been adopted as depicted in Fig. 4.2.

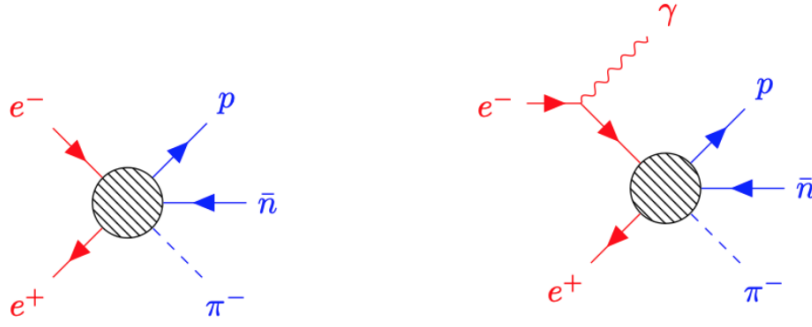


Figure 4.2: Feynman diagram of the channel of interest. *Phokhara* handles the initial hard photon emission (red), while *EvtGen* simulates the decay to final-state particles (blue).

Importantly, the *Phokhara_evtgen_combination* generator produces both events with and without ISR emission, as explicitly illustrated by the TopoAna overview of generated events in Fig. 4.3.

A *vpho* (virtual photon) is now introduced as a fictitious particle used by the generator *Phokhara* to produce ISR photons before the e^+e^- collisions and then enforce (*evt.gen*) the selected decay channel, approaching the following scheme:

$$e^+ + e^- \rightarrow vpho (+\gamma_{ISR}) \text{ (Phokhara generator)}$$

$$vpho \rightarrow p + \pi^- + \bar{n} \text{ (EvtGen generator)}$$

4.1 Kinematic variables comparison

Two scenarios can be studied. If the reconstructed final state particles are a proton (p), a negative pion (π^-) and an anti-neutron ($\bar{n}$), it is referred to as the **NO ISR case**. If the ISR photon (γ_{ISR}) is also reconstructed, it is referred to as the **ISR case**. Therefore, in the NO ISR case, the recoil system is defined as the $p\pi^-$ system, while in the ISR case it is defined as the $p\pi^-\gamma_{\text{ISR}}$ system.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p$	0	35291	35291
2	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p \gamma^I \gamma^I$	2	22971	58262
3	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p \gamma^I$	1	18735	76997
4	$vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^F$	3	10005	87002
5	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F$	6	5274	92276
6	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F$	4	4621	96897
7	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F$	7	1503	98400
8	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F$	8	700	99100
9	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F$	5	597	99697
10	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	9	167	99864
11	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F \gamma^F$	12	63	99927
12	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F \gamma^F$	10	61	99988
13	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F \gamma^F \gamma^F$	11	4	99992
14	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F \gamma^F \gamma^F$	15	4	99996
15	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	14	2	99998
16	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F \gamma^F$	13	1	99999
17	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F \gamma^F \gamma^F \gamma^F$	16	1	100000

Figure 4.3: Topology Analysis (TopoAna) output of the generated 100k events. The fourth column is the unambiguous decay tree index associated to the correspondent channel, the fifth the number of entries per channel, and the last the cumulative number of entries.

As a first step, the kinematic variables are discussed through the recoil system consisting of $p\pi^-(\gamma_{\text{ISR}})$. In parallel, the neutral $\bar{n}$ cluster candidate is reconstructed and its properties are compared to those from the recoil, comparing different kinematic variables. Through the recoil analysis, the antineutron can be identified as the missing particle in the final state.

The first three contributes in Fig. 4.3 shows that events without ISR emission are the most frequently generated ($\sim 35\%$), followed by those with two ($\gamma^I \gamma^I$) ($\sim 23\%$) and one (γ^I) ISR photon ($\sim 19\%$).

Additionally, numerous Final State Radiation (FSR) photons are emitted by the charged final state particles due to next order perturbative QED processes.

4.1.1 ISR distributions

The ISR energy and angular distributions are discussed below. The primary effect of ISR emission is the reduction of the center-of-mass energy $\sqrt{s}$ below the $\Upsilon(4S)$ resonance.

Therefore, the energy E and θ distributions of ISR radiation are shown in Fig. 4.4.

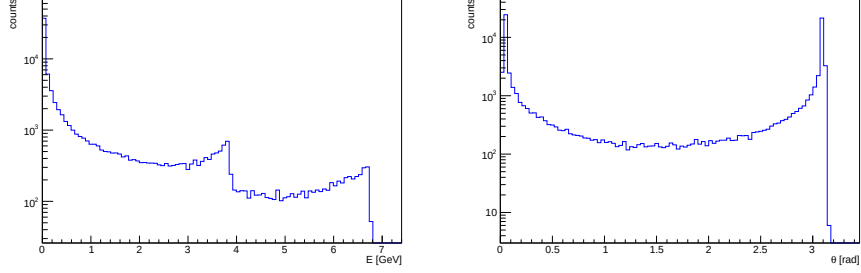


Figure 4.4: (a) ISR energy distribution. (b) ISR θ distribution. Both distributions are shown on a logarithmic y -axis scale.

The 2D energy E and θ distribution of ISR radiation are shown in Fig. 4.5.

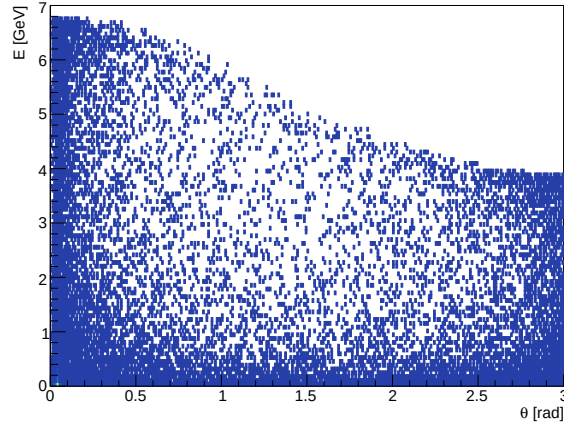


Figure 4.5: 2D energy E and θ distribution of ISR radiation.

In the ISR energy distribution, two peaks are observed slightly below 4 GeV and 7 GeV, corresponding to the energies of the e^+ and e^- beams, respectively. The angular distribution presents a U-shape, indicating that forward and backward directions are the most probable.

Furthermore, as can be observed in Fig. 4.5, the higher the energy of the ISR radiation, the closer $|\cos(\theta)|$ is to 1, indicating that high-energy γ_{ISR} photons are emitted preferentially in the forward and backward directions. In fact, peaks are observed at ~ 7 GeV for $\theta \sim 0$ rad (e^- beam direction) and at ~ 4 GeV for $\theta \sim \pi$ rad (e^+ beam direction).

4.1.2 Event selections

In the ISR case, the final state $p\pi^-\bar{n}\gamma_{\text{ISR}}$ is reconstructed; the recoil of $p\pi^-\gamma_{\text{ISR}}$ is then built. In the NO ISR case, the final state $p\pi^-\bar{n}$ is reconstructed; the recoil of $p\pi^-$ is then built. Therefore, the $\bar{n}$ can be identified through the recoil mass peak at the anti-neutron mass.

To properly select the signal in the recoil mass distribution, several cuts are applied in the steering file as follows:

1. Recoil system: $0 \text{ GeV}/c^2 < \text{recoil mass} < 2 \text{ GeV}/c^2$ and in ECL acceptance
2. $\bar{n}$: neutral clusters from ECL
3. p : $\text{protonID} > 0.9$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
4. π^- : $\text{pionID} > 0.1$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$

The first one is a cut that selects the region of interest in the recoil mass distribution, corresponding to the interval $[0, 2] \text{ GeV}/c^2$ around the antineutron mass ($\sim 939, 56 \text{ MeV}/c^2$), requiring that the recoil vector lies within the ECL acceptance.

To analyze only $\bar{n}$ signals from the calorimeter, the second selection requires that the neutral $\bar{n}$ cluster candidates are reconstructed within the electromagnetic calorimeter (ECL), avoiding the $\bar{n}$ candidates from other detectors (mainly from KLM).

The third and fourth criteria are standard selections recommended by the basf2 framework. The ID requirement imposes a constraint on the protonID or pionID based on a maximum likelihood criterion as follows:

$$\text{protonID} = \frac{L_p}{L_e + L_\mu + L_\pi + L_K + L_d} \quad \text{pionID} = \frac{L_\pi}{L_e + L_\mu + L_p + L_K + L_d},$$

where L_p is the likelihood for the proton hypothesis, L_π is the likelihood for the pion hypothesis, and so on.

The cuts on dr and $|dz|$ select only tracks originating from the interaction point (IP) area, where dr is defined as the transverse distance with respect to IP and dz is defined as the axial distance, along the z-axis, with respect to IP.

Since multiple $\bar{n}$ neutral cluster candidates may exist per event, only the candidate with the smallest angle -defined as α - relative to the recoil system is selected. Therefore, a selection is imposed using the basf2 **rankByLowest** method. It allows to select only the highest rank candidates, based on a specific variable; in this case, the angle between the neutral cluster and the recoil vector.

Using spherical coordinate system, after some algebraic steps, the 3D angle α between the two vectors can be obtained:

$$\alpha = \arccos[\sin(\theta_1)\sin(\theta_2)\cos(\phi_1 - \phi_2) + \cos(\theta_1)\cos(\theta_2)]$$

Here vector **1** denotes the recoil vector and the vector **2** denotes the $\bar{n}$ direction vector. The respective α distributions, before and after the aforementioned selection, are shown in Fig. 4.6, in both ISR and NO ISR cases:

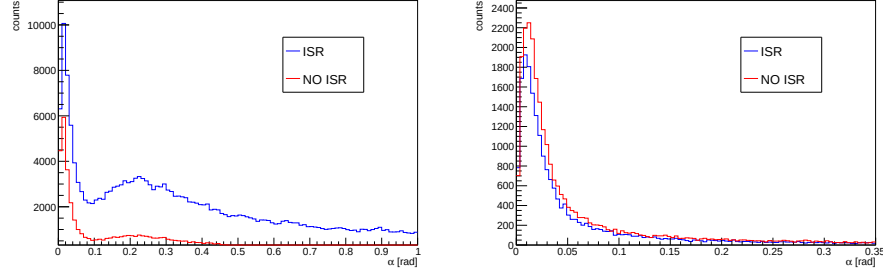


Figure 4.6: (a) α distribution. (b) α distribution after the rank selection.

To select the most probable $\bar{n}$ candidates, only those with $\alpha < 0.35$ ($\sim 20^\circ$) are stored.

The efficiency ϵ is defined as follows:

$$\epsilon = \frac{n^\circ \text{ of reconstructed candidates}}{n^\circ \text{ of generated events}}$$

Referring to Fig. 4.3, 46969 events are generated without ISR emission ($iDcyTr \in 0, 3, 7, 9, 14, 13$) and 53031 with ISR emission ($iDcyTr \in 2, 1, 6, 4, 8, 5, 12, 10, 11, 15, 16$).

Since 20146 candidates are reconstructed for the ISR case and 25013 for the NO ISR case, an efficiency respectively of $\sim 53\%$ and 38% can be observed. Since most of the ISR radiation is emitted in the backward ($\theta \sim 180^\circ$) and in the forward ($\theta \sim 0^\circ$) directions -where detection coverage is absent- the efficiency in the ISR case is slightly lower than in the NO ISR case.

4.1.3 Recoil mass distribution

To select events with anti-neutron, the recoil system $p\pi^-(\gamma_{ISR})$ is studied through the recoil mass. Specifically, the recoil mass should peak at the nominal anti-neutron mass, since no background channels are present in this sample. The only background consists of secondary particles produced during the propagation of primary particles through the detectors.

The recoil mass is defined as follows:

$$\text{recoil mass} = \sqrt{(p_{e^+e^-} - p_{\text{sum}})^2}$$

where $p_{e^+e^-}$ is the total four-momentum of the initial e^+e^- system at the $\Upsilon(4S)$ resonance, and p_{sum} is the sum of 4-momenta of $p\pi^-(\gamma_{ISR})$.

The recoil mass distribution, for both ISR and NO ISR cases, is shown in Fig. 4.7.

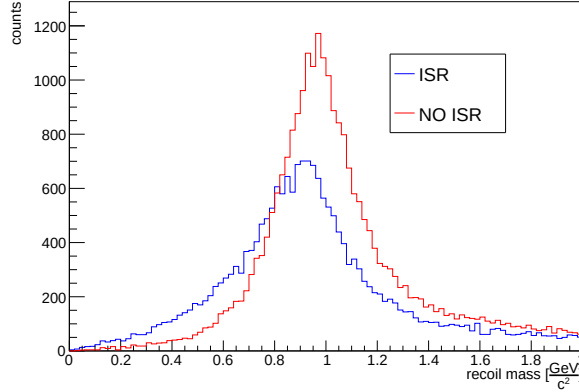


Figure 4.7: Recoil mass distribution, in both ISR and NO ISR cases.

The recoil mass distribution peaks on the $\bar{n}$ mass (~ 939.56 MeV), indicating that the the recoil mass is well reconstructed in both ISR ($p \pi^- \gamma_{ISR}$) and NO ISR ($p \pi^-$) cases. Therefore, the variables associated with the $\bar{n}$ clusters can be compared with the reconstructed recoil variables (p , θ).

In the event simulation the generator produces more than one γ_{ISR} per event, reflecting into the obtained recoil mass distribution; the one obtained in ISR case shows more entries in the left tail, since only the most energetic gamma is considered in the recoil reconstruction, while the others are not considered.

The distribution obtained in the NO ISR case exhibits a higher peak as the recoil system reconstruction consists only of $p\pi^-$, allowing better resolution than the ISR case, since fewer tracks are considered.

By selecting events around the peak, $\bar{n}$ candidates events can be identified. The recoil direction is expected to provide a good estimate of the $\bar{n}$ momentum.

4.1.4 The recoil kinematic

Before analyzing the recoil kinematic distribution, the reconstructed momentum and $\cos(\theta)$ distributions of each particle composing the recoil system is examined. For particles with MC association, the MC variables can be studied for a given list of candidates as well.

For example, considering the proton candidate list, both the **reconstructed** momentum and the Monte Carlo (**generated**) momentum of the proton list -which is not affected by the reconstruction process- are available. The latter will be referred to as the "generated" momentum.

Since a candidate list is the result of the reconstruction process, it also contains particles that have been mis-identified as the main particle (for example, π^+ mis-identified as p or γ mis-identified as $\bar{n}$).

Therefore, the considered "generated" variables represent the Monte Carlo values for a given reconstructed particle list, which differs from the Monte Carlo

values for a generator-level particle list that contains only the selected particle type.

The relative reconstructed momentum and $\cos(\theta)$ distributions of proton and pion are shown in Fig. 4.8, while the generated one are shown in in Fig. 4.9.

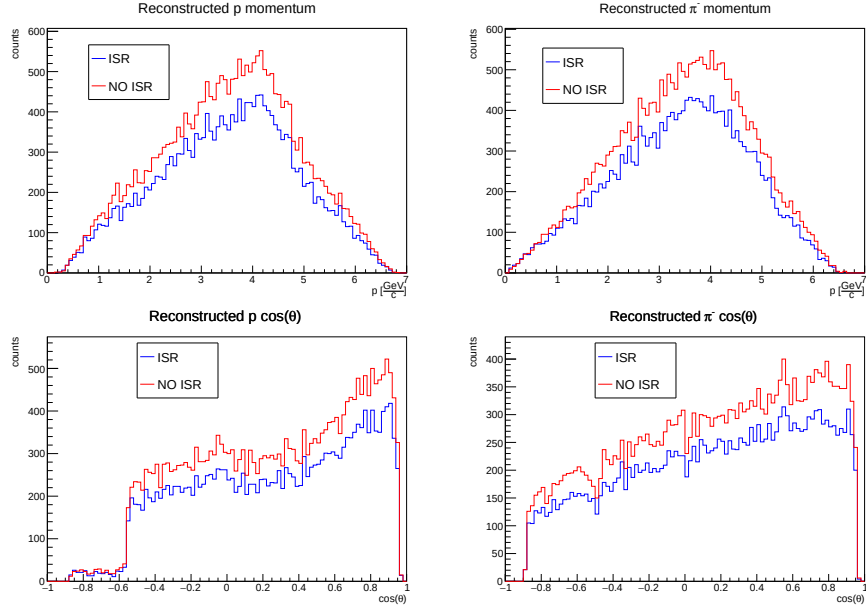


Figure 4.8: Reconstructed momentum and $\cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.

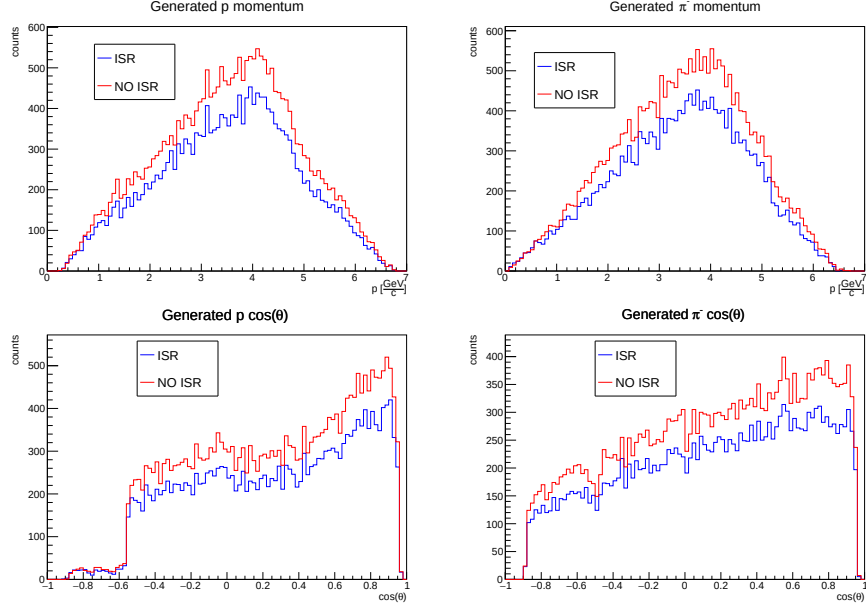


Figure 4.9: Generated momentum and $\cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.

The distributions clearly exhibit the typical phase-space shapes, as expected from the adopted generator model. The $\cos(\theta)$ distribution of protons presents low counts before θ value such as at $\cos(\theta) \lesssim -0.6$. In this region, proton identification performance is insufficient for effective selection, failing the protonID selection. This occurs especially for protons with momentum higher than 2 GeV/c, as shown in Fig. 4.10.

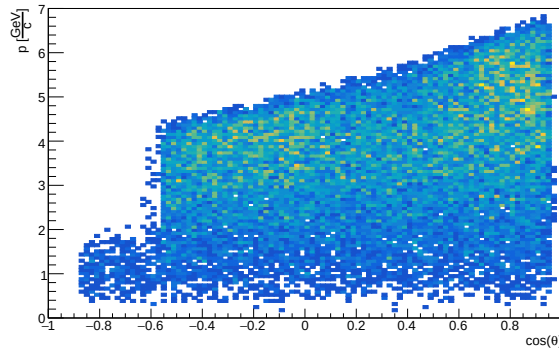


Figure 4.10: 2D distribution of proton momentum versus proton $\cos \theta$, in the NO ISR case.

The recoil momentum can be defined as:

$$\text{recoil momentum} = |\sum_i \vec{p}_i|$$

The recoil momentum distributions can be obtained in both ISR and NO ISR cases, as illustrated in Fig. 4.11.

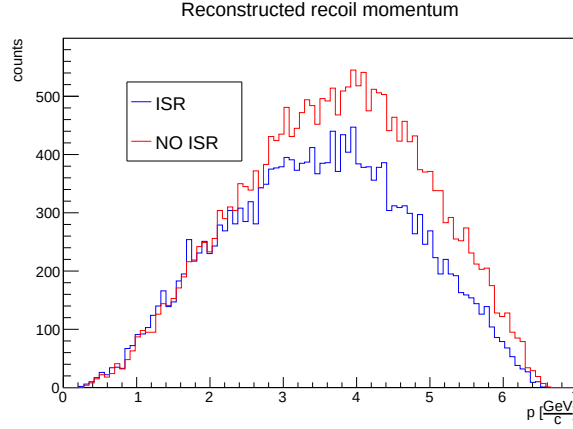


Figure 4.11: Momentum distribution of the recoil system, in both ISR and NO ISR cases.

Since the recoil system is well reconstructed through the recoil mass, this distribution can be compared with that of the $\bar{n}$, both from reconstruction and from generator level.

The $\bar{n}$ candidate list originates from neutral clusters detected in the ECL. Therefore, since the calorimeter measures the energy of the clusters, the $\bar{n}$ momentum $p_{\bar{n}}$ can be defined as follows:

$$p_{\bar{n}}^2 = E_{ECL}^2 - m_{\bar{n}}^2,$$

where E_{ECL} is the energy of the $\bar{n}$ candidate cluster measured by the calorimeter and $m_{\bar{n}}$ is the mass of the anti-neutron.

The $\bar{n}$ momentum distribution are shown in Fig. 4.12.

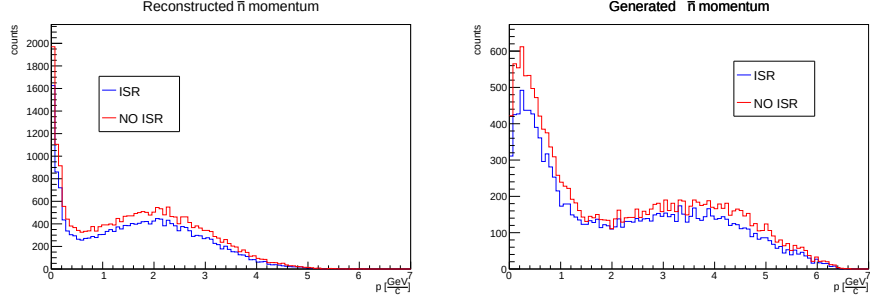


Figure 4.12: Reconstructed (left) and generated (right) momentum distribution of anti-neutron candidates, in both ISR and NO ISR cases.

Notably, the reconstructed $\bar{n}$ momentum is systematically underestimated compared to the generator-level one, indicating missing energy within neutral clusters. This topic will be addressed in detail below through the momentum residual distributions.

Both the reconstructed and the generator-level distributions exhibit two structures, indicating two contributions within $\bar{n}$ candidate neutral clusters. To investigate this, the reconstructed recoil momentum is compared with the generator-level momentum of all $\bar{n}$ candidates, using MC truth matching. Since the $\bar{n}$ PDG code is **-2112**, the neutral clusters associated to $\bar{n}$ list can be matched to **true** (truth = 1) $\bar{n}$ particles, using Monte Carlo PDG = -2112, or to **false** (truth = 0) candidates with Monte Carlo PDG $\neq$ -2112. The obtained result is shown in 4.13.

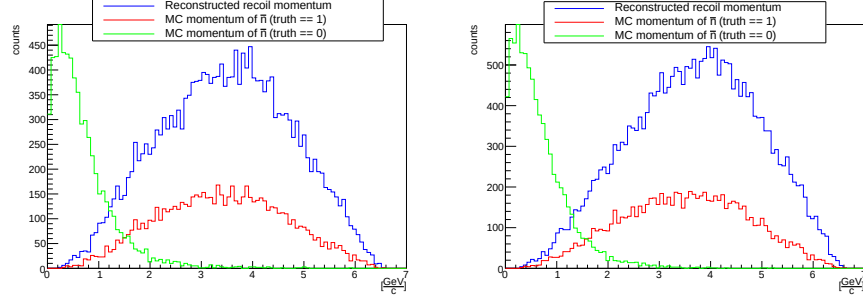


Figure 4.13: Recoil momentum and generated $\bar{n}$ list momentum. The blue distribution shows the recoil momentum, the red distribution corresponds to the generated $\bar{n}$ with MC PDG = -2112 (associated to $\bar{n}$ by MC matching), while the green distribution corresponds to the generated $\bar{n}$ with MC PDG $\neq$ -2112 (associated to no $\bar{n}$ by MC matching).

The low-momentum structure corresponds to candidates not associated to $\bar{n}$ by the Monte Carlo truth, exhibiting significant deviation from the recoil momentum. This is likely due to cluster split-off and/or pre-showering effects

as will be discussed later. Furthermore, the distribution corresponding to the generated $\bar{n}$ with MC PDG = -2112 has less entries than the recoil momentum distribution, suggesting that several anti-neutrons are not correctly reconstructed. All these aspects will be discussed in the next section through analysis of $\bar{n}$ relatives (ancestors).

To obtain a measure of the recoil momentum and recoil θ resolution, these variables can be compared with those of the Monte Carlo matched $\bar{n}$ (truth = 1), in both reconstructed and generated cases. To address this comparison, the linear correlation coefficient ρ is studied, defined as:

$$\rho_{x,y} = \frac{cov[x,y]}{\sigma_x \sigma_y} \quad \text{where } cov[x,y] = E[xy] - E[x]E[y]$$

Therefore, the correlations in θ and in momentum are shown in Fig. 4.14, remembering that the momentum is obtained starting from the energy of the cluster as follow:

$$p_{rec}^2 = E_{ECL}^2 - m_{\bar{n}}^2$$

where E_{ECL} is the energy of the $\bar{n}$ candidate cluster measured by the calorimeter and $m_{\bar{n}}$ is the mass of the anti-neutron.

A good correlation can be highlighted between the two θ distributions in both reconstructed and generator-level cases, indicating that the anti-neutron θ direction is properly described.

A good correlation can be observed between the reconstructed recoil momentum and the generated $\bar{n}$ one, confirming how the first is properly describing the $\bar{n}$ kinematic.

However, a poor resolution is observed in reconstructed $\bar{n}$ momentum, in both ISR and NO ISR cases, presenting an underestimation of the reconstructed $\bar{n}$ momentum as well.

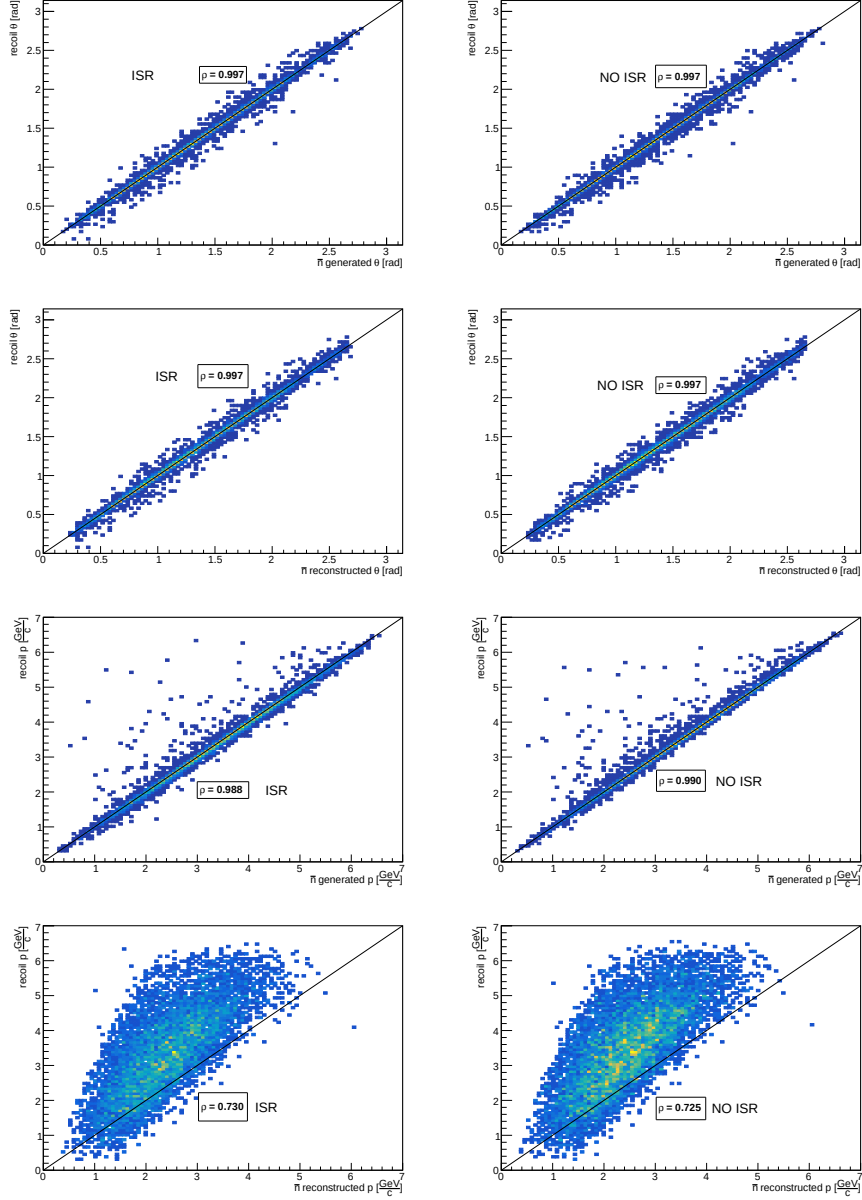


Figure 4.14: Obtained momentum and θ correlation distributions between recoil system and anti-neutron candidate, in both reconstructed and generated case, in both ISR and NO ISR cases.

In order to directly study the observed poor resolution in reconstructed $\bar{n}$ momentum, the momentum residual distributions can be analyzed in Fig. 4.15.

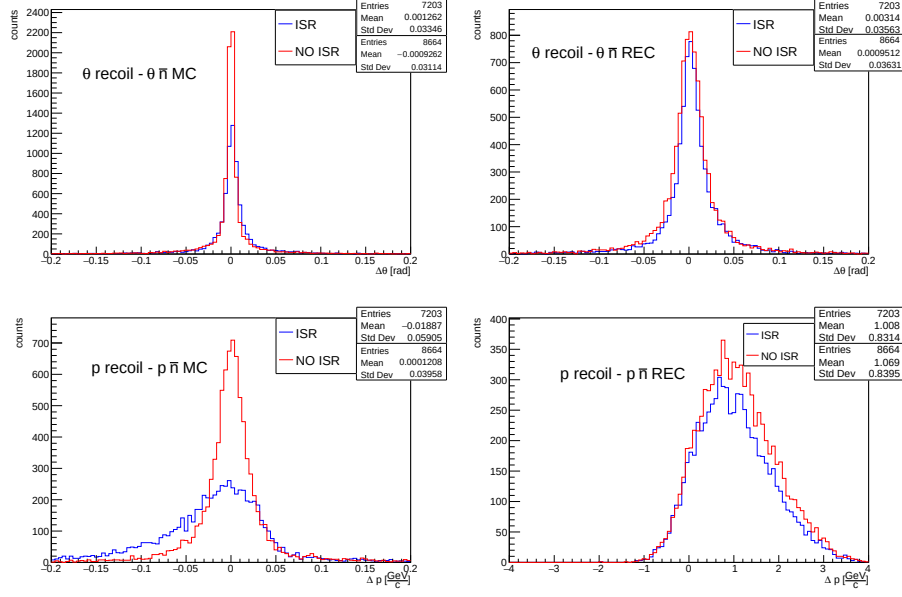


Figure 4.15: Momentum and θ angle residual distributions between recoil system and anti-neutron candidate, in both ISR and NO ISR cases.

Besides exhibiting poor resolution, the reconstructed momentum of the $\bar{n}$ is underestimated as the residuals distributions show, suggesting a missing energy in the shower. Therefore, the $\bar{n}$ momentum is poorly reconstructed using only the ECL information. Most of anti-neutrons undergo in annihilation with hadrons such as $\bar{n}n$ and $\bar{n}p$, producing pions which escape the calorimeter. Therefore, the reconstructed momentum (or energy) in the calorimeter is not directly comparable with that of anti-neutrons, making the reconstructed $\bar{n}$ momentum an unreliable variable.

Furthermore, considering only the θ residual distributions and those obtained with the Monte Carlo momentum of the $\bar{n}$, the distributions in the NO ISR case exhibit higher resolution, explicitly confirming what was noted for the recoil mass distribution in Fig. 4.7.

4.2 The $\bar{n}$ relatives analysis

The origin of ECL clusters that are reconstructed as $\bar{n}$ candidates but are associated with other particles according to MC truth information is investigated in this part.

Cluster split-off and/or pre-showering effects can occur distorting the $\bar{n}$ reconstructed distributions. Cluster split-off occurs within the ECL when a portion of the primary shower produces signals in non-contiguous crystals. Pre-showering occurs when the primary interaction takes place in the TOP and its products travel towards the ECL, where they are detected as the primary signal.

Since the ISR and NO ISR results are analogous, only the ISR case is analyzed on the following section. A representation of cluster split-off and pre-showering is shown in Fig. 4.16.

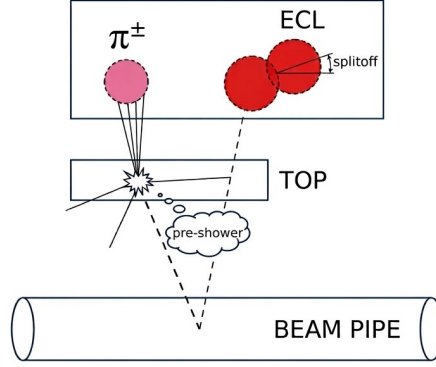


Figure 4.16: A representation displaying cluster split-off and pre-showering effects following $\bar{n}$ annihilation.

One of the secondary particles from $\bar{n}$ annihilation can be detected as the primary ECL cluster, resulting in information loss and requiring the TOP time information to improve the anti-neutron detection. Anti-neutrons, as neutral particles, do not emit Cherenkov radiation but can annihilate within the TOP volume. The resulting charged products then radiate in the bars. The Cherenkov signal is correlated with the anti-neutron time of flight and can be used to improve the energy estimation performed by the calorimeter [38].

4.2.1 The basf2 ancestor variables

Since $\bar{n}$ particles are selected along the recoil direction, there are ECL clusters in that direction that are not associated with $\bar{n}$ by MC matching. Their origin is studied, considering the possibility that $\bar{n}$ interacted with material.

In order to study the decay chain and kinship degree of the neutral cluster associated to $\bar{n}$ candidates detected in the ECL, the basf2 **ancestor** variables are used. These identify whether a given particle originates from the decay of another particle or a subsequent decay chain. The following ancestor variables are employed:

- **varForFirstMCAncestorOfType(type, variable)**: this metavariable returns the requested **variable** of the first ancestor of the given **type** -a specific particle- found in a certain point of the chain
- **hasAncestor(PDG, abs)**: this metavariable returns a positive integer if an ancestor -a certain relatives in the chain- with the given **PDG** code is found, 0 otherwise. It means that 1 = first mother, 2 = grandmother,

etc. Second argument **abs** is optional: 1 means that the sign of the PDG code is taken into account, default is 0

A typical scenarios of $\bar{n}$ as ancestor could be the following:

- 0) $\bar{n}$ from IP $\rightarrow$ 1) $\bar{n}$ annihilation in TOP (pions produced) $\rightarrow$ 2) pions detected
- 0) $\bar{n}$ from IP $\rightarrow$ 1) $\bar{n}$ annihilation in ECL (pions produced) $\rightarrow$ 2) pions detected

The variables **varForFirstMCAncestorOfType(anti-n0, p)** and **hasAncestor(-2112, 1)** of $\bar{n}$ cluster candidates are discussed. Given the close similarity between the results, only the ISR case is discussed. The hasAncestor distribution is shown in Fig. 4.17.

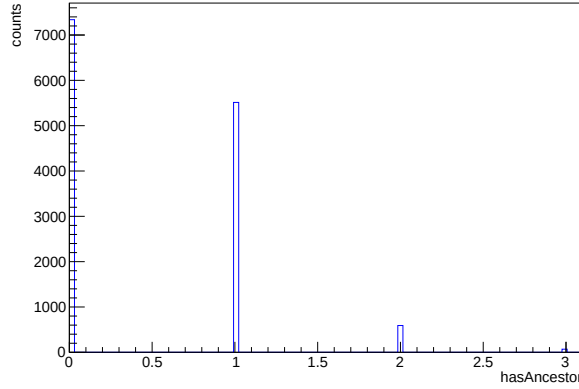


Figure 4.17: Distribution of the relative degree of anti-neutron candidates with an anti-neutron present in their ancestry chain.

From the distribution in Fig. 4.17, it is evident that the majority of candidates do not have a neutron ancestor, lacking an $\bar{n}$ in their relative chain (hasAncestor=0), suggesting they are the primary $\bar{n}$ themselves. The second contribution is made of $\bar{n}$ candidates having $\bar{n}$ as mother etc.

In order to directly study the contribution of each term, the Monte Carlo truth is accessed via the corresponding **mc.PDG**. This is the PDG code of the selected candidates as associated by the Monte Carlo.

Therefore, the $\bar{n}$ candidates' mc.PDG code is studied for each HasAncestor contribution. The obtained distributions are shown in Fig. 4.18.

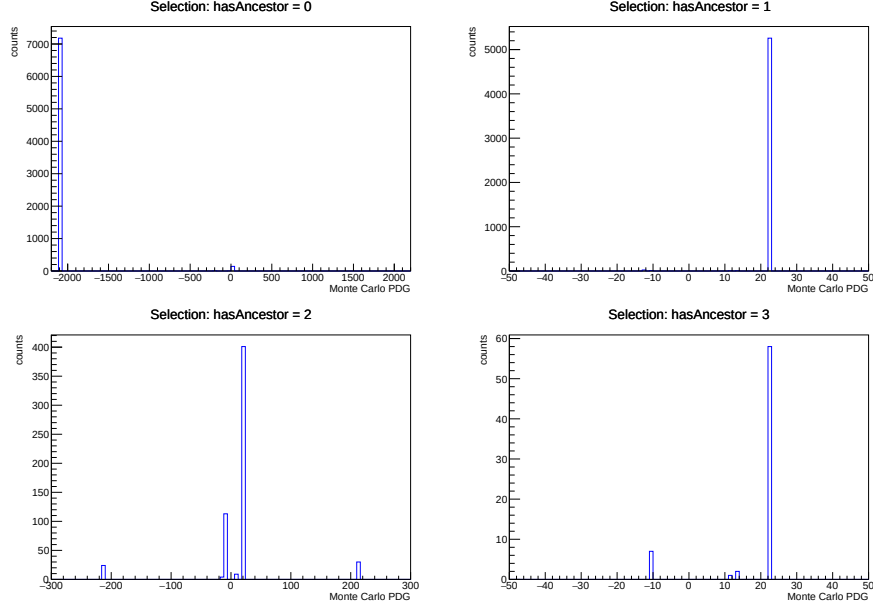


Figure 4.18: Monte Carlo PDG codes of candidates with degrees of ancestry selection.

- **hasAncestor = 0** clearly shows that $\bar{n}$ candidates without an $\bar{n}$ as relatives are mainly true anti-neutrons, as their Monte Carlo PDG code is primarily -2112. This is expected since the anti-neutron comes from the IP and it is directly detected in the ECL through its own annihilation.
- **hasAncestor = 1** shows that the $\bar{n}$ candidates with an $\bar{n}$ as mother are predominately gammas (Monte Carlo PDG = 22), suggesting they result from $\bar{n}$ interactions prior to $\bar{n}$ candidate ECL cluster formation.
- **hasAncestor = 2** suggests that the $\bar{n}$ candidates with an $\bar{n}$ as grandmother are mainly gammas, with some spurious pions (-211 & +211) and electrons or positrons (+11 & -11). The case with hasAncestor = 3 yields analogous results.

4.2.2 Ancestor momentum

The figure Fig. 4.13 can be updated, by including the momentum of the ancestors obtained with `varForFirstMCAncestorOfType(anti-n0, p)`. The idea is to check whether the information from the momentum of the ancestors is consistent with that obtained from the primary $\bar{n}$. The obtained distributions are shown in Fig. 4.19.

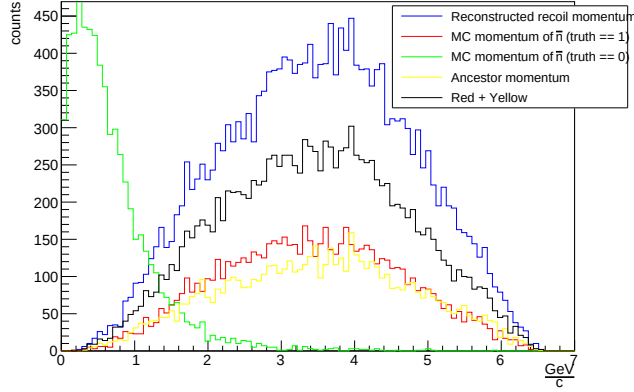


Figure 4.19: Distribution comparison of: recoil momentum (blue), generated momentum of $\bar{n}$ candidates associated to $\bar{n}$ by the Monte Carlo matching (red), generated momentum of $\bar{n}$ candidates associated to no $\bar{n}$ by the Monte Carlo matching (green) and generated momentum of $\bar{n}$ ancestors associated to $\bar{n}$ by the Monte Carlo matching (yellow). The black distribution is the sum of the second and fourth contributions. The recoil vector is the only reconstructed variable, while the others are obtained from Monte Carlo truth selection.

- The recoil momentum is made of 20146 entries (blue),
- the momentum distribution of $\bar{n}$ associated is made of 7203 entries (red),
- the momentum distribution of no $\bar{n}$ associated is made of 6308 entries (green),
- the momentum distribution of ancestors is made of 6173 entries (yellow),
- the distribution of the sum of the second and fourth contributions if is made of 13376 entries (black).

This indicates that 98% of the first structure shown in Fig. 4.13 relates to ancestor momentum, suggesting that phenomena prior to the neutral cluster associated with $\bar{n}$ candidates can occur, such as cluster split-off and pre-showering. The quality of the ancestor momentum can be discussed by studying the residual distribution with respect to the reconstructed recoil momentum. The obtained distribution is shown in Fig. 4.20.

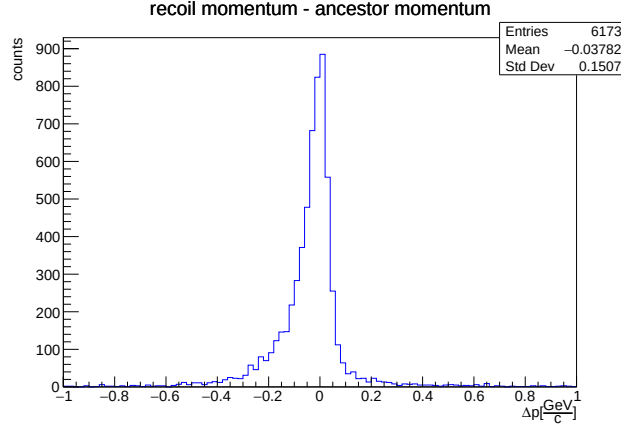


Figure 4.20: Residual distribution of primary $\bar{n}$ momentum (ancestor momentum) with respect to the recoil vector.

Therefore, the reconstructed recoil momentum is a good estimator of the primary $\bar{n}$ (ancestor) momentum as well, confirming that cluster split-off or pre-showering may occur prior the ECL cluster formation.

Furthermore, clusters not associated with either primary $\bar{n}$ or their decay products lack PDG codes, indicating that the MC association has failed for these candidates. They are associated with a Not a Number (NaN) particle by the Monte Carlo matching. Referring to the recoil momentum (20146 entries) and the sum of the second and fourth contributions (13376 entries), their difference equals 6779 entries, which exactly matches the number of $\bar{n}$ candidates associated with a NaN value. These are particles that did not receive a correct Monte Carlo association during MC truth matching, then they cannot be recovered.

This can be summarized using the meta-variable (mcErrors), which encodes as a bit code the reason why the Monte Carlo matching fails. This variable explicitly illustrates the behaviour shown in Fig. 4.21, where the first peak is on the code **0**, which means that "This particle and all its daughters are perfectly reconstructed", while the second peak is on the code **512**, which confirms that "There was an error in MC matching. Not a valid match. Might indicate fake/background track or cluster", as expected.

4.2.3 $\bar{n}$ production vertex

To verify the $\bar{n}$ production position, the Monte Carlo production vertex of the $\bar{n}$ is studied. To address it, the radial production distance ρ is introduced, defined as follows:

$$\rho = \sqrt{x^2 + y^2},$$

which is the Monte Carlo radial distance of $\bar{n}$ candidates, from the z-axis.

The distribution is splitted in two parts: the left part -close to the IP- ($\rho \in$

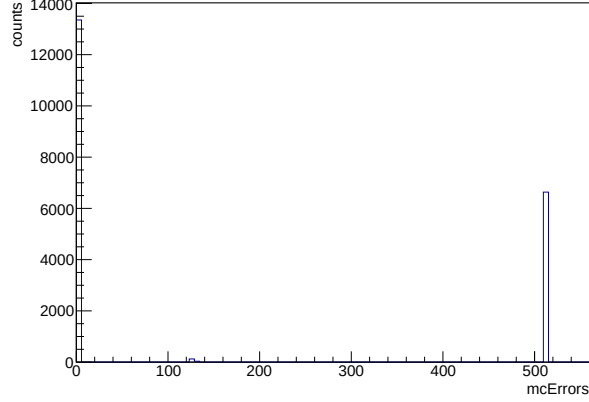


Figure 4.21: Monte Carlo Error code. 0 corresponds to a perfectly reconstructed particle, while 512 corresponds to a non valid MC match.

$[0, 0.1]$ cm) and the right part -far from the IP- ($\rho \in (0.1, 220]$ cm). The obtained result is shown Fig. 4.22.

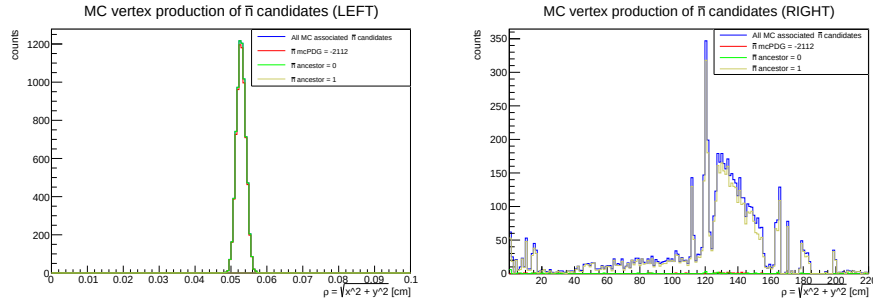


Figure 4.22: $\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: all $\bar{n}$ associated by Monte Carlo matching; red: $\bar{n}$ candidates correctly associated to true $\bar{n}$; green: $\bar{n}$ candidates with no $\bar{n}$ relatives; yellow: $\bar{n}$ candidates with an $\bar{n}$ mother particle.

The correctly associated anti-neutrons overlay the $\bar{n}$ candidates without an anti-neutron ancestor ($\bar{n}$ ancestor = 0), originating from the interaction point (left panel). This distribution is not centered around zero because the generator uses a default non-zero starting radial distance.

The $\bar{n}$ candidates with an $\bar{n}$ mother particle -that are mainly photons (Fig. 4.18)- are produced throughout the Belle II detector.

The first structures are in the vertex detector region ($\rho < 16$ cm) and the two peaks located at the TOP edges, mainly around 118 cm (where the TOP begins) and 125 cm (where the TOP ends and the ECL starts), explicitly showing

how cluster split-off and pre-showering events occur in the events. In fact, $\bar{n}$ candidates with $\bar{n}$ as mother particle ($\bar{n}$ ancestor = 1), may be produced before the ECL (pre-showering) or within the ECL (split-off). The production vertex variable is now examined, dividing each contribute from the different ECL regions; backward region, barrel region and forward region. The obtained result is shown Fig. 4.23.

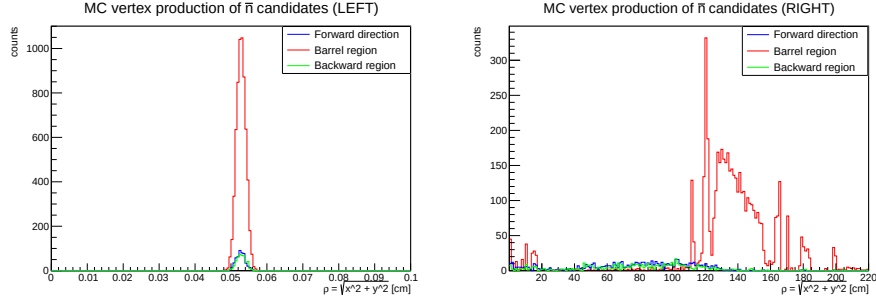


Figure 4.23: $\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: $\bar{n}$ candidates detected in the forward region; red: $\bar{n}$ candidates detected in the barrel region; green: $\bar{n}$ candidates detected in the backward region.

In the forward and backward regions, the two peaks located at the TOP edges disappear, as expected, since the TOP provides no coverage in those regions. The same conclusions hold for the first structures in the vertex detector region, where the PXD and SVD are located.

4.3 Kinematic Fit effects

A **Kinematic Fit** is a constrained optimization technique used in particle physics to refine the measured four-momenta (or other kinematic quantities) of event particles by imposing known physical constraints, such as energy-momentum conservation or recoil mass. Measured quantities like track momenta, energies from calorimeters, or decay vertices come with resolution-limited errors from detectors. The fit adjusts these within their error bars to best satisfy the imposed constraint.

There are several type of Kinematic Fit that could yield improved resolution. The one constraint **1C Kinematic Fit** has been tested over the $p\pi^-(\gamma_{ISR})$ system recoil mass, forcing it to the anti-neutron mass (~ 0.9385 GeV) and then studying the response on the recoil kinematic variables resolution versus the anti-neutron candidate ones. The obtained residuals are shown below:

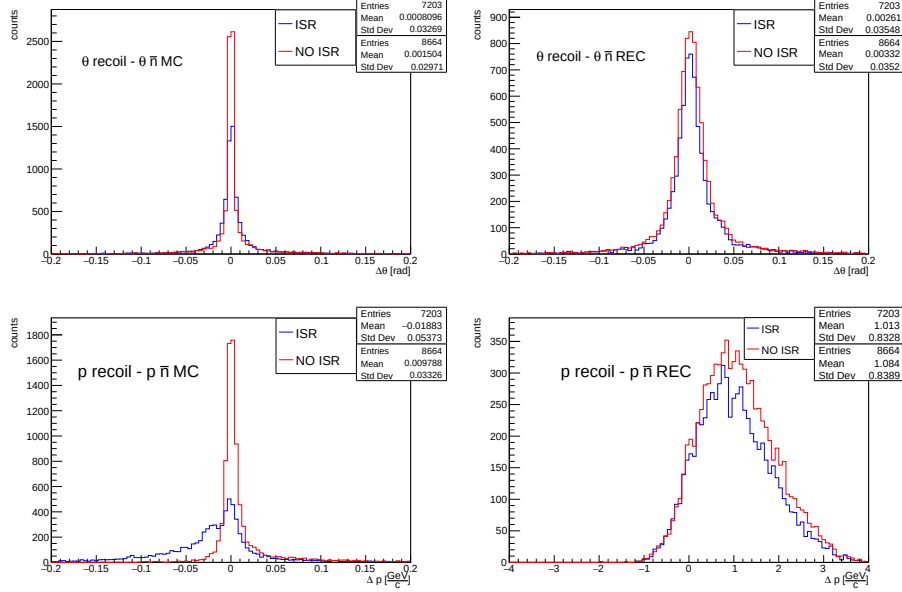


Figure 4.24: Momentum and θ angle residuals distribution between recoil system and anti-neutron candidate, after the 1C Kinematic Fit procedure

As can be easily seen from the distributions, no significant improvement in recoil momentum resolution was observed post 1C Kinematic Fit compared to the unconstrained baseline 4.14. This confirms that the recoil system was already optimally reconstructed in the signal sample Monte Carlo. Further kinematic fit studies using a Monte Carlo cocktail are discussed in the next section.

4.4 Shower shape variables in the signal sample

The following chapter introduces the core of this thesis: the analysis of the **shower shape variables** of neutral ECL clusters associated with anti-neutron candidates. Since the final goal is to assess Data/MC agreement for these variables, the shower shape variables are first studied in an MC signal sample with no background channels present. The main idea is that particle identification can be achieved by studying the shower shapes and their characteristics.

4.4.1 The analyzed shower shape variables

The selected variables are the **cluster energy**, the **E1E9** variable, the **E9E21** variable, **Lateral Momentum**, **Second Moment**, **Zernike Moments** (50 and 41), and **Number of Hits**, which are described first.

Cluster Energy

The Cluster Energy (*clusterE* in the basf2 software), as the name suggests, represents the total energy deposited by the cluster associated with a given candidate particle, in this case the anti-neutron. It has units of GeV and only clusters with Cluster Energy above 20 MeV are stored by the reconstruction software.

Cluster E1E9 and E9E21 In order to provide a proper definition of these variables (respectively *clusterE1E9* and *clusterE9E21* in the basf2 software), the following figure is shown below:

10	11	12	13	14
25	2	3	4	15
24	9	1	5	16
23	8	7	6	17
22	21	20	19	18

Figure 4.25: Ideal representation of a cluster pattern, consisting of a 5×5 matrix in which each cell corresponds to a single crystal

Cluster E1E9 and cluster E9E21 are defined as follow:

$$E1E9 = \frac{E_1}{\sum_{i=1}^9 E_i} \quad E9E21 = \frac{\sum_{j=1}^9 E_j}{\sum_{j=10}^{25} E_j}, \text{ with } j \neq 10, 14, 18, 22$$

The first one returns the ratio of the energy in the central crystal (E_1) to the total energy in the surrounding 3×3 crystal grid (E_9). The second one returns the ratio of the energy in the 3×3 crystal grid around the central crystal (E_1) to the total energy in the 5×5 crystal grid around the central crystal, excluding the corners.

These ratios are always less than 1 and tend toward larger values for photons and smaller values for hadrons, as the latter produce more spread-out showers. These variables are dimensionless and are included within the range $[0, 1]$.

Lateral Momentum

The cluster Lateral Momentum or Lateral Energy distribution (*clusterLAT* in the basf2 software) is defined as follow:

$$C_{LM} = \frac{\sum_{i=2}^n E_i r_i^2}{E_0 r_0^2 + E_1 r_0^2 + \sum_{i=2}^n E_i r_i^2},$$

with $r_0 \simeq 5$ cm (average distance between two crystals centres). n is the number of crystals associated to the shower, $E_i = (E_0, E_1, \dots)$ are the crystal energies sorted by descending energy and r_i is the distance of the i -th digit to the shower center projected onto a plane perpendicular to the shower axis. For radially-symmetric electromagnetic showers, this variable peaks at ~ 0.3 , where it is larger for hadronic particles. This variable is dimensionless and it is included within the range $[0, 1]$.

Second Moment

The cluster Second Moment (*clusterSecondMoment* in the basf2 software) is defined as follow:

$$C_{SM} = \frac{1}{S_0} \frac{\sum_{i=0}^n E_i r_i^2}{\sum_{i=0}^n E_i},$$

where E_i and r_i are the same variables as used for the lateral momentum, while S_0 is a normalization factor that normalizes to 1 for isolated photons, making C_{SM} dimensionless. The Second Moment measures the dispersion of cluster size.

Zernike Moments

The Zernike Moments (Zernike Polynomials) are a set of orthogonal functions which are defined on the unit disk. Since each moment represents a specific pattern shape, they are widely used to represent smooth, two-dimensional functions (especially wavefronts) over circular apertures.

On the unit disk parametrized by polar coordinates (ρ, ϕ) with $0 \leq \rho \leq 1$, the Zernike modes are usually written as:

$$Z_n^m(\rho, \phi) = R_n^m(\rho) \cos(m \phi)$$

$$Z_n^{-m}(\rho, \phi) = R_n^m(\rho) \sin(m \phi)$$

Where:

$$R_n^m(\rho) = \sum_{k=0}^{\frac{n-m}{2}} \frac{(-1)^k (n-k)!}{k! \left(\frac{n+m}{2} - k\right)! \left(\frac{n-m}{2} - k\right)!} \rho^{n-2k}$$

n represents the radial degree and m represents the azimuthal frequency ($m = 0$ for spherical Zernike polynomials). A figure of the Zernike polynomials is given below:

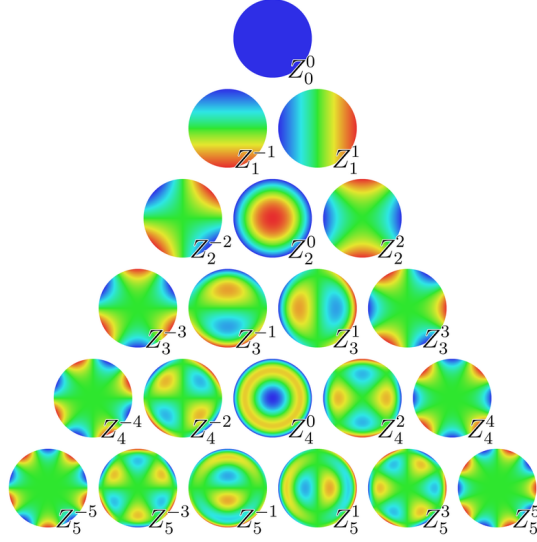


Figure 4.26: Zernike Polynomials representation. Moments Z_4^0 and Z_5^1 are widely discussed in this thesis

Precisely, Zernike Moments Z_4^0 and Z_5^1 (*clusterAbsZernikeMomen40* and *clusterAbsZernikeMoment51* in the basf2 software) are studied in this thesis, where the first one represents the radial spread distribution structure, while the second one represents the level of dipolar asymmetry.

Number of Crystals hit

The variable Number of Crystals hit (*clusterNhits* in the basf2 software) returns sum of weights w_i , with $w_i \leq 1$ of all crystals in the cluster. For non-overlapping clusters this is equal to the number of crystals in the cluster. However, in the case of overlapping clusters where individual crystal energies are split among them, this can be a non-integer value.

4.4.2 Anti-neutrons ECL cluster variables

In order to isolate the contribution of the neutral clusters which are correctly associated with $\bar{n}$ particles (Monte Carlo PDG = -2112) (**red distribution**) from all $\bar{n}$ candidates (**blue distribution**), the two distributions are overlaid for each cluster variable. The former is naturally included within the latter, naturally.

The obtained cluster variables in ISR case are shown below:

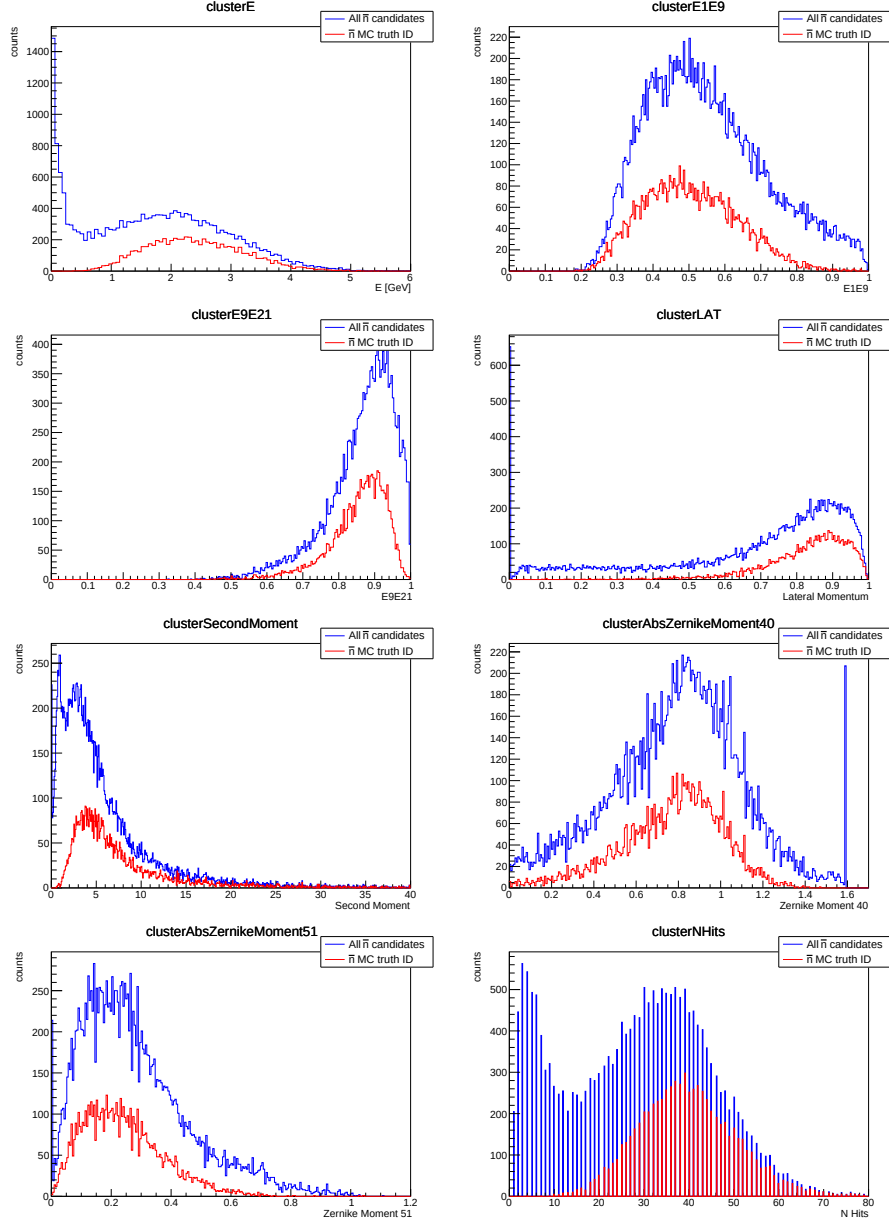


Figure 4.27: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed

The obtained cluster variables in NO ISR case are shown below:

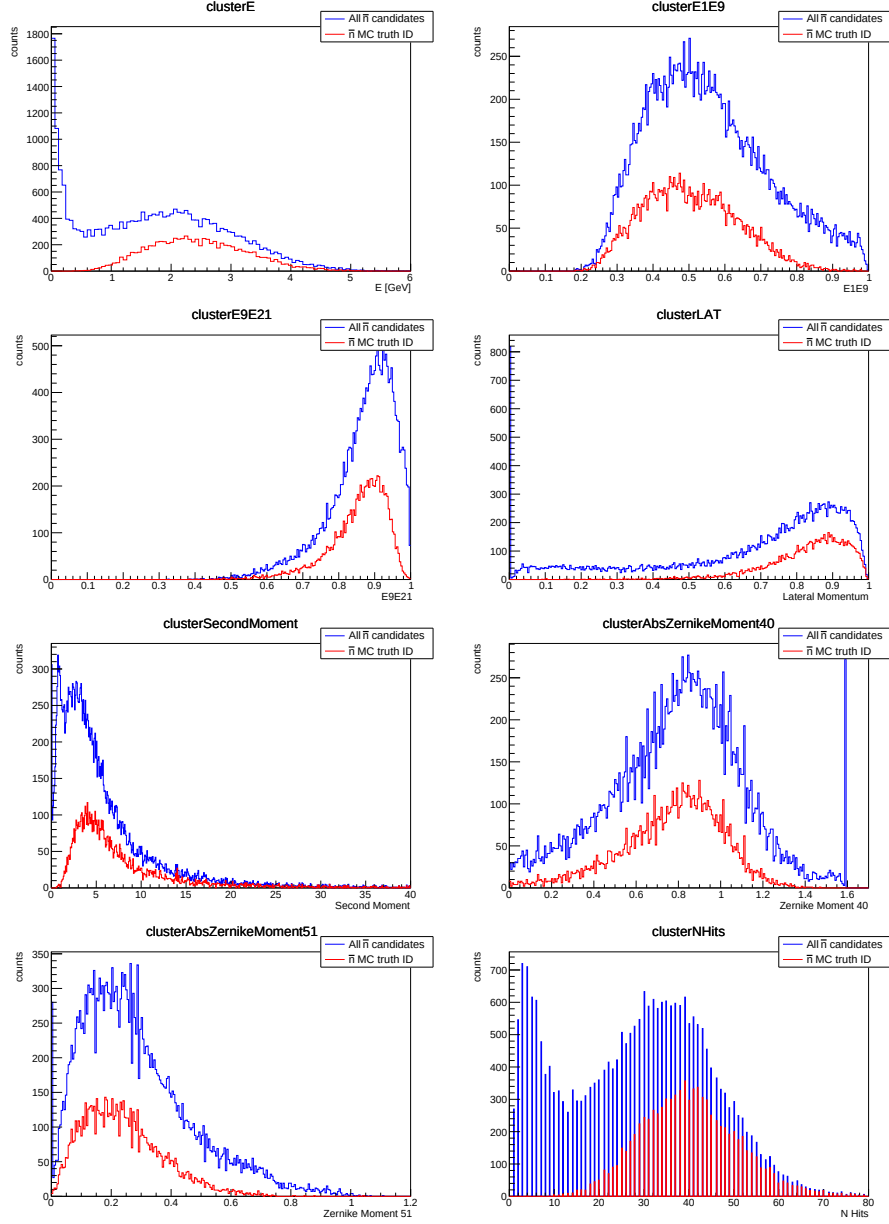


Figure 4.28: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (NO ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed

First, it is important to highlight that the ISR and non-ISR distributions carry the same information, as expected. Indeed, the measurement of cluster variables is independent of the recoil system, which could be made of either $p\pi^-\gamma_{ISR}$ or $p\pi^-$ respectively.

It is also noteworthy that the [blue distribution](#) shows some peculiar structures respect the [red distribution](#). The reconstructed cluster energy for all $\bar{n}$ candidates exhibits a first structure at low energy values that resembles those discussed for the anti-neutron momentum distributions. This is also evident in the second moment distribution, where a low-energy structure is clearly visible. The most prominent difference appears in the number of hits ($N Hit$) distribution, where a distinct structure at $N Hit < 10$ is evident.

The nature of these characteristic structures is analyzed in the following subparagraph.

4.4.3 Ancestor cluster variables

It would be interesting to investigate the nature of this structure using the same approach as before, namely studying the ancestor contributions. The shower shapes with the following selections are compared:

- $\bar{n}$ Monte Carlo PDG = -2112 (neutral clusters correctly associated with $\bar{n}$ particles)
- $\bar{n}$ Monte Carlo PDG = NaN (neutral clusters not associated with any particles)
- $\bar{n}$ hasAncestor = 0 (neutral clusters with no $\bar{n}$ as relative)
- $\bar{n}$ hasAncestor != 0 (neutral clusters with $\bar{n}$ as relative)

Referring to figures 4.29 and 4.30, as expected the neutral clusters correctly associated with $\bar{n}$ particles carry the same information of those without $\bar{n}$ as relative, in both ISR and NO ISR cases.

Furthermore, it is evident how reconstructed neutral clusters not associated with any particles by the Monte Carlo and those with $\bar{n}$ as relative are primarily responsible for the structures observed at low values of energy, second moment, and number of crystals.

A further selection on number of crystals ($clusterNHits > 10$) can be performed to study the corresponding cluster variables. The result is shown in figure 4.31 and 4.32.

As observed, these structures arise from small-sized clusters produced by $\bar{n}$ particles that undergo phenomena -such as cluster split-off and pre-showering- prior to neutral cluster formation for $\bar{n}$ candidates, as confirmed by ancestor studies. The requirement of at least 10 crystals eliminates these structures.

The obtained cluster variables in ISR case are shown below:

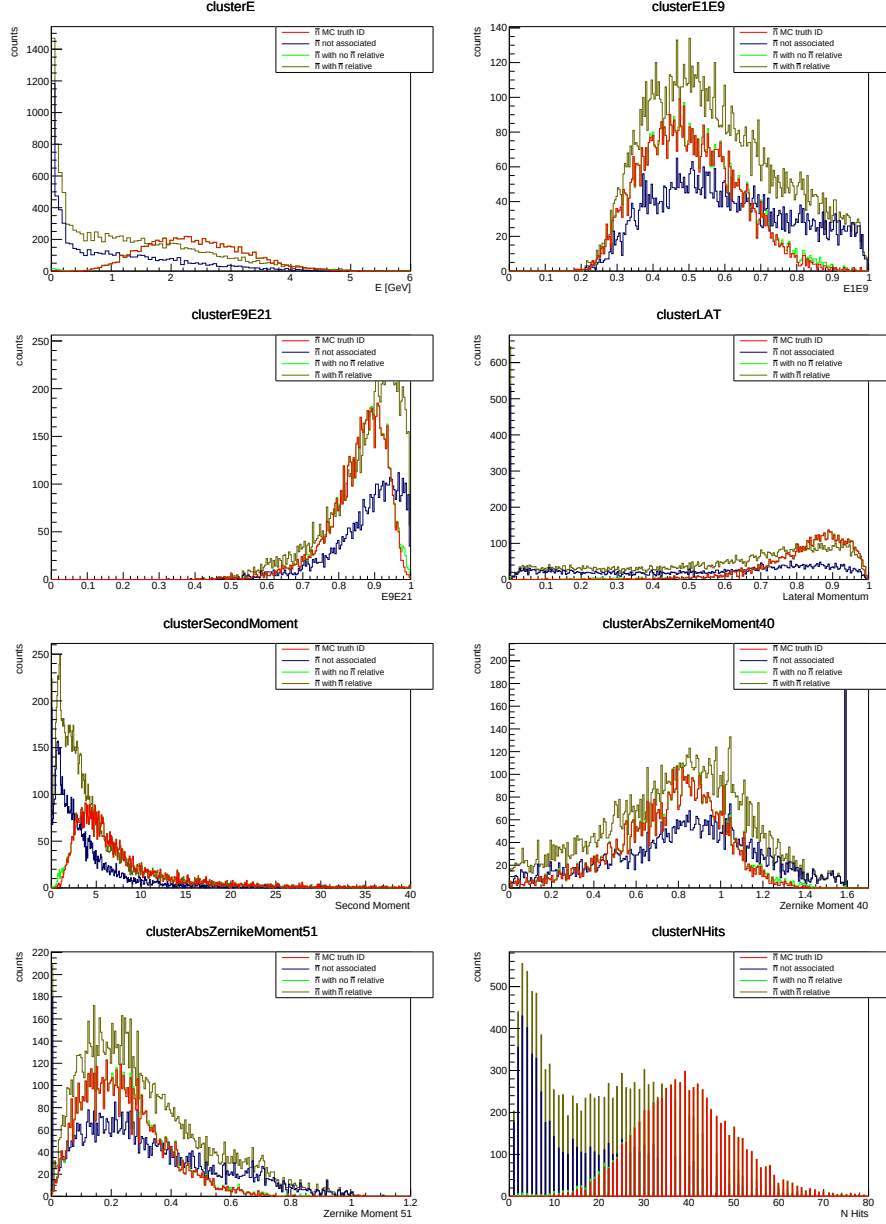


Figure 4.29: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (ISR case)

The obtained cluster variables in NO ISR case are shown below:

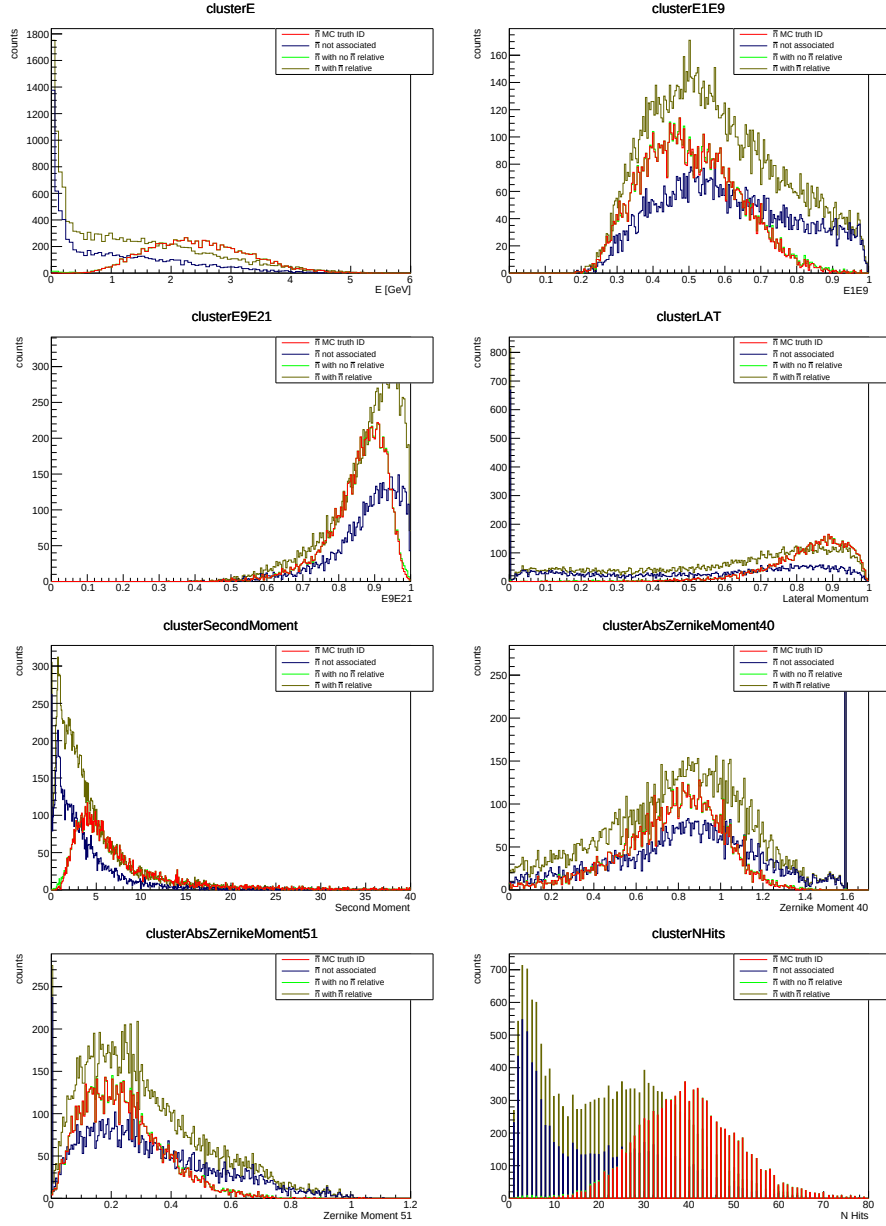


Figure 4.30: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (NO ISR case)

The obtained cluster variables in ISR case are shown below:

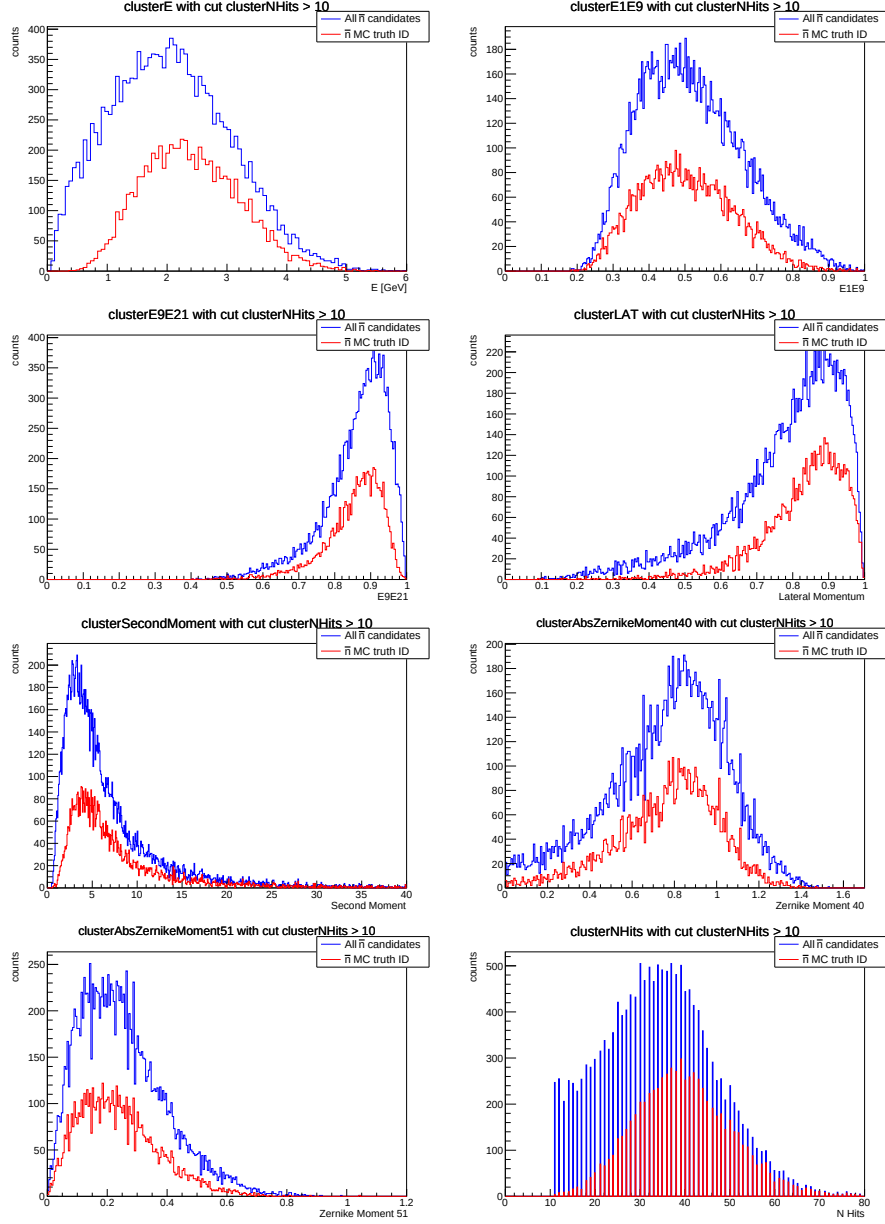


Figure 4.31: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). Cut $clusterNHits > 10$ is performed

The obtained cluster variables in NO ISR case are shown below:

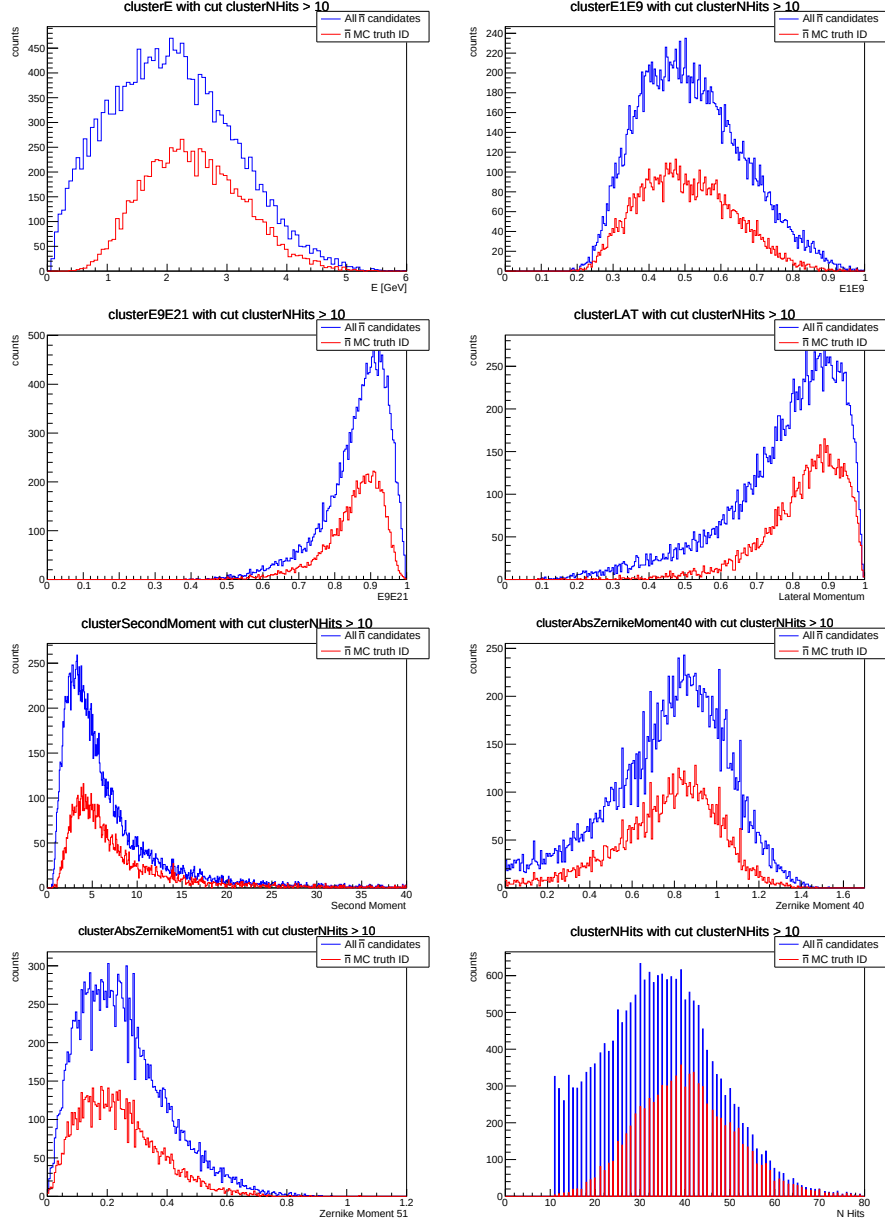


Figure 4.32: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -1112 (NO ISR case). Cut $clusterNHits > 10$ is performed

4.4.4 Cluster variables in different ECL regions

As described in Chapter 3, the Belle II detector is divided into barrel and forward/backward end-cap regions. It would be interesting to study the reconstructed cluster variables in each ECL region separately. Since the ISR and NO ISR cases exhibit analogous shower shapes, only the latter is discussed here. The region hit distribution is shown below:

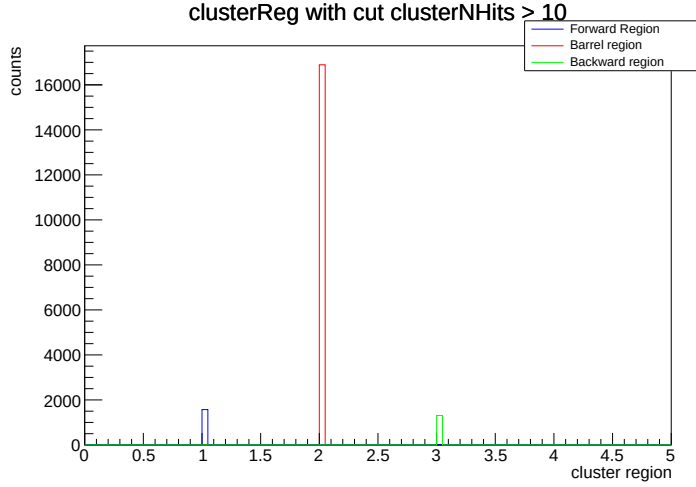


Figure 4.33: Different ECL regions contribution in neutral cluster associated to $\bar{n}$ candidates, where 1 is the forward region, 2 is the barrel region and 3 is the backward region. As expected, the barrel region is the most involved

As observed in Fig. 4.34, the cluster variable associated to all the reconstructed $\bar{n}$ shapes are consistent across all ECL regions.

However a slight difference can be noted for cluster energy and number of hits variables. In fact, these two exhibit a shift toward lower values in the forward region, reflecting the Lorentz boost of the center-of-mass. In any case, the barrel region contains the majority of ECL clusters, as expected.

The obtained cluster variables in NO ISR case are shown below:

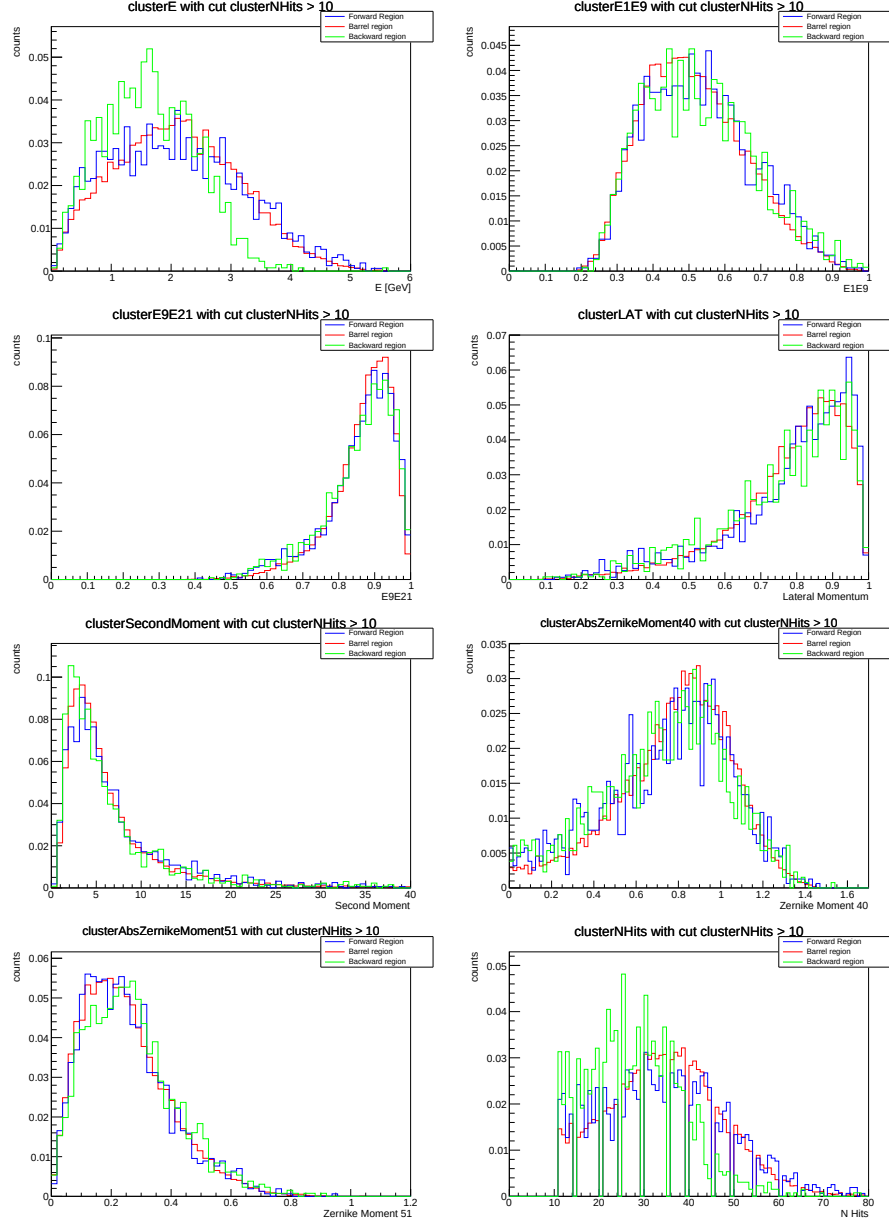


Figure 4.34: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates in different ECL regions. The distributions are normalized to unit value

4.5 Summary of Signal Channel Analysis

The channel $e^+ + e^- \rightarrow p + \pi^- + \bar{n}$ has been studied, with both ISR and NO ISR emission.

The recoil system $p\pi^-(\gamma_{ISR})$ is correctly reconstructed by the analysis, as evidenced by the peak above the anti-neutron mass in the recoil mass distribution. Furthermore, the kinematic distributions (p and θ) of the recoil system are consistent with those of generator-level $\bar{n}$ candidates, in both ISR and NO ISR cases. At reconstructed level, good agreement is observed in the θ distribution, while the reconstructed $\bar{n}$ candidate momentum shows underestimation and poor resolution compared to the recoil system momentum, as observed in correlations and residuals distributions. Furthermore, a low-momentum value structure is observed in reconstructed and generated momentum of $\bar{n}$ candidates compared to the recoil system momentum.

Therefore, energy loss occurs within ECL clusters, suggesting leakage processes such as cluster split-off and/or pre-showering effects. Ancestor studies reveal that photons related to $\bar{n}$ particles are often misidentified as main $\bar{n}$ candidates, accounting for the low-momentum structure observed. In fact, the ancestors momentum can be partially recovered when good Monte Carlo associations occur, showing good agreement with the recoil system momentum as well. A study of $\bar{n}$ production reveals that candidates with $\bar{n}$ ancestors are generated far from the IP, before the ECL (pre-showering) and within the ECL (cluster split-off).

Furthermore, a one-constraint (1C) Kinematic Fit was tested, revealing no significant improvement in the reconstructed recoil momentum and confirming that the recoil system was already optimally reconstructed by the analysis.

Eventually, the reconstructed cluster variables associated with $\bar{n}$ candidates - such as cluster energy, lateral momentum, Zernike moments, etc.- were studied in the Monte Carlo signal sample. Similar structures to those observed in the momentum distributions were identified, particularly in the cluster energy and number of hits distributions. Therefore, ancestor studies confirm the momentum distribution observations: sub-clusters from secondary particles with $\bar{n}$ ancestry produce small-sized, low-energy structures in the cluster distributions.

In particular, a cut on the cluster number of hits yields reconstructed cluster variables closer to those expected from Monte Carlo $\bar{n}$ candidate selection, in both ISR and NO ISR cases.

Chapter 5

Study of the Monte Carlo cocktail production

The second part of the analysis of this thesis is devoted to the study of **Belle II Monte Carlo cocktail simulations**. A Belle II "cocktail" MC production (typically generated with Pythia8) simulates all relevant physics channels simultaneously. Therefore, the analyzed channel $e^+e^- \rightarrow p\pi^-\bar{n}(\gamma_{\text{ISR}})$ is now embedded within all other background channels in the cocktail; the first task is to correctly reconstruct it amidst the surrounding backgrounds, studying the recoil mass distribution.

To achieve this, a first attempt uses the complete RUN1 $u\bar{u}$ production. Building on the signal sample discussed in the last chapter, further optimal cuts and selection criteria are developed to properly isolate the signal in the recoil mass distribution. Therefore the corresponding shower shape variables of neutral cluster associated to anti-neutron candidates are studied.

Eventually, the full Belle II $q\bar{q}$ ($u\bar{u}$, $d\bar{d}$, $c\bar{c}$, $s\bar{s}$) production is analyzed. The resulting cluster variables are compared with those from real data to assess Data/MC agreement.

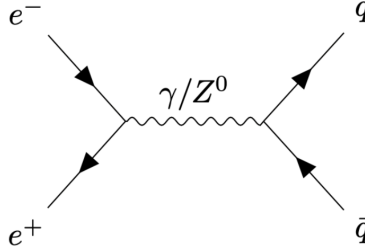


Figure 5.1: Tree level Feynman diagram of $e^+e^- \rightarrow q\bar{q}$

First of all, it could be useful start describing the Belle II Distributed Computing, namely the Belle II Grid.

5.1 Introduction to the Belle II Distributed Computing

Belle II datasets are categorized into two main types: **Data** and **Monte Carlo**. The Data are obtained from calibrated collision events processed by the experiment. Monte Carlo samples are generated using the Monte Carlo method according to various event generators, as discussed in Chapter 2. These are further classified into **Signal MC**- a specific process generated with a dedicated decay file upon analyst request (as in the previous chapter)- and **Generic MC**, or "cocktail", containing all processes expected in data with their correct relative

fractions (as in the present chapter). These are usually in mdst (Mini Data Summary Table) format. There is cdst (Calibration Data Summary Table) as well, but it is used mainly to address calibration tasks.

Data and Monte Carlo samples can be “skimmed” by applying analysis-dependent offline pre-selections called “skims”. Skims reduce data volume by selecting events of interest for specific analyses and retaining only the necessary information, producing smaller, more manageable datasets.

Using these Data and MC collections, analyses are performed via custom steering files with basf2, producing final ROOT ntuples. Subsequent offline analysis is then conducted outside the basf2 framework using ROOT, matplotlib, etc.

A scheme of the Belle II Data Model is shown below:

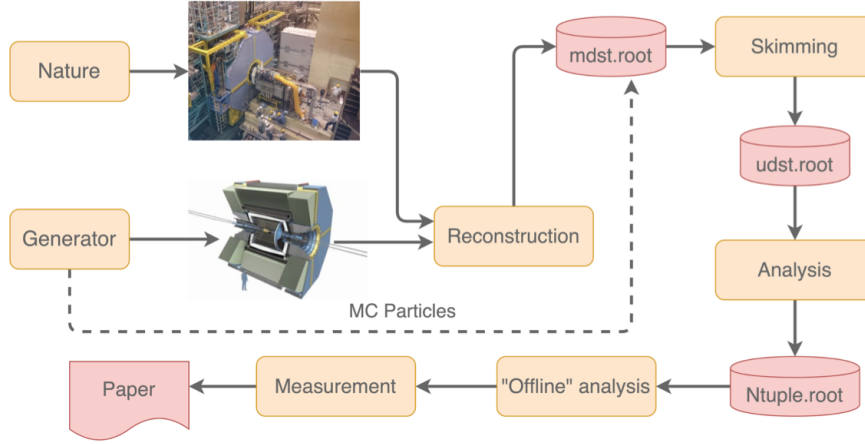


Figure 5.2: A scheme of the Belle II Data Model, starting from the nature processes information to the final results. As described, Monte Carlo truth information bypasses reconstruction, since the information is directly picked at generator level

Belle II Data and Monte Carlo samples are distributed across numerous storage sites worldwide. This Distributed Computing System, known as **the Grid**, consists of a network of loosely coupled computers with varying local conditions (operating systems, resources, etc.).

Two full copies of the raw data are maintained at dedicated data centers. Raw data is processed, skimmed, and distributed over other storage sites:

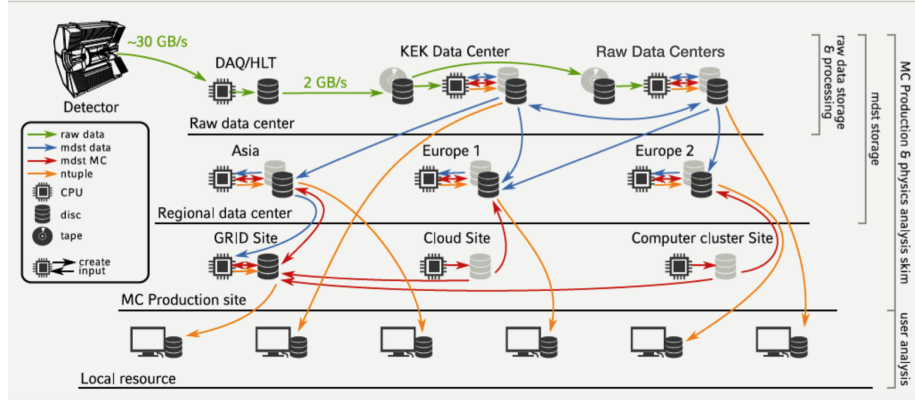


Figure 5.3: A scheme of the Belle II Distributed Computing (The Grid) sites

Each site interact with **DIRAC**, an interware that manages interactions between Belle II and its computing resources. **BelleDIRAC** is an extension that enables users to interact with DIRAC for job submission, data placement, monitoring, and other tasks.

5.1.1 Types of processes at the Belle experiment enviroment

Many processes can occur during an e^+e^- collision. These may mimic the selected signal and be accidentally reconstructed as signal events, though they represent background processes.

The types of processes present depend on if beams collide on-resonance or off-resonance in reference to the $\Upsilon(4S)$ resonance. $\Upsilon(4S)$ production occurs at $\sqrt{s} = 10.58$ GeV, yielding $\Upsilon(4S) \rightarrow BB$ processes. The B^+B^- and $B^0\bar{B}^0$ final states are referred to as charged and neutral $B\bar{B}$ pairs, respectively.

$e^+e^- \rightarrow q\bar{q}$ ($q = u, d, s, c$) processes, known as continuum production, also occur. The production cross section of the Belle II processes is shown below:

Process	Cross section (nb)
$e^+e^- \rightarrow \Upsilon(4S)$	1.10
$e^+e^- \rightarrow u\bar{u}(\gamma)$	1.61
$e^+e^- \rightarrow d\bar{d}(\gamma)$	0.40
$e^+e^- \rightarrow s\bar{s}(\gamma)$	0.38
$e^+e^- \rightarrow c\bar{c}(\gamma)$	1.30
$e^+e^- \rightarrow \tau^+\tau^-(\gamma)$	0.92
$e^+e^- \rightarrow e^+e^-(\gamma)$	300
$e^+e^- \rightarrow \mu^+\mu^-(\gamma)$	1.15
$e^+e^- \rightarrow \gamma\gamma(\gamma)$	4.99
$e^+e^- \rightarrow e^+e^-e^+e^-$	39.7
$e^+e^- \rightarrow e^+e^-\mu^+\mu^-$	18.9

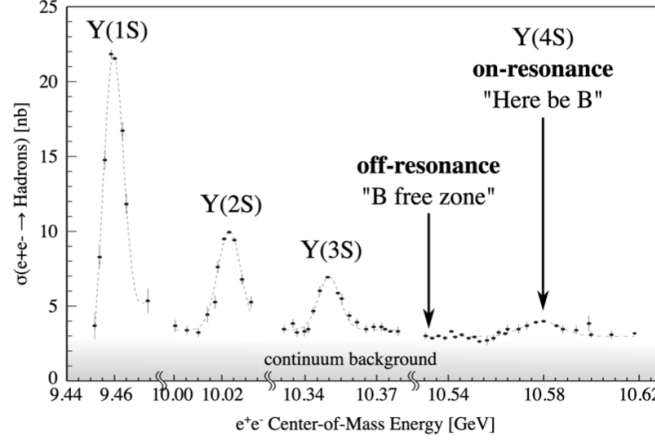


Figure 5.4: Cross section of $e^+e^- \rightarrow \text{Hadrons}$, in Υ resonances energy region [5]. $\Upsilon(1S)$ resonance is very narrow, while $\Upsilon(4S)$ one is broader and lower

5.2 Applied selections and cuts

To follow the same approach as used in the signal sample analysis, the recoil mass is first studied using the complete $u\bar{u}$ cocktail production, which corresponds to a $\sim 1429.23 \text{ fb}^{-1}$ integrated luminosity of data.

The following real cuts have been applied:

1. Recoil system: $0 \text{ GeV}/c^2 < \text{recoil mass} < 2 \text{ GeV}/c^2$
2. $\bar{n}$: neutral clusters from ECL and number of hits > 10 and energy $< 3 \text{ GeV}$
3. p : $\text{protonID} > 0.9$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
4. π^- : $\text{pionID} > 0.1$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
5. $\alpha < 0.20 \text{ rad} \sim 11.5^\circ$
6. No further charged particles in the Rest of Event (ROE)

Four additional cuts are applied beyond those used for the signal sample, such as cluster number of hits > 10 GeV, $\alpha < 0.20 \text{ rad}$, the request of no further charged particles in the Rest of Event (ROE) and the energy of the neutral cluster associated to $\bar{n}$ candidates must be less than 3 GeV.

The first cut arises from observations in the previous chapter; to achieve cleaner selection of neutral clusters associated with anti-neutron candidates -reducing cluster split-off and pre-showering contributes- a requirement on the number of lighted crystals is applied.

The second cut allows a stricter selection of anti-neutron candidates through a smaller α angle. α is defined as the 3D angle between the recoil vector and the

neutral cluster associated with the $\bar{n}$. As will be explained later, this threshold will be lowered further.

To explain the third cut, the concept of **Rest of Event** (ROE) must first be introduced. The ROE is built around the reconstructed target particles and consists of all other particles in the rest of event. In the present case, the reconstructed particles are $p, \pi^-, \bar{n}$ (and γ_{ISR}); therefore, the ROE includes all particles outside this reconstructed system. Consequently, the third selection criterion requires that no charged particles (i.e. tracks associated with charged particles) be present in the Rest Of Event, providing a cleaner event reconstruction.

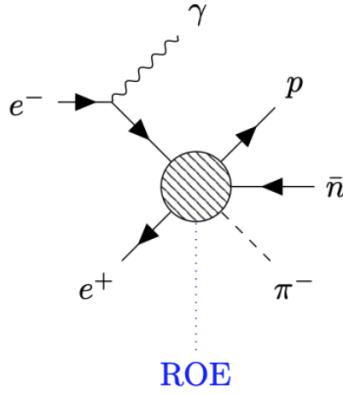


Figure 5.5: The selected channel with ROE made of a variable amount of particles.

The fourth cut arises from cluster energy studies performed on the first chunk of $u\bar{u}$ cocktail production, corresponding to $\sim 215 fb^{-1}$. The $\bar{n}$ list can be affected by several mis-identified γ_{ISR} . In order to study this phenomenon, the Monte Carlo ISR flag (mcISR) can be used to separate the two contributions in reconstructed $\bar{n}$ list. mcISR variable is a Monte Carlo flag; If mcISR = 0, the reconstructed particle is not an ISR photon. If mcISR = 1, the reconstructed particle is an ISR photon mis-identified as an anti-neutron. The obtained cluster energy distribution for $\bar{n}$ candidates is shown in figure 5.6.

It is easy to note that two structures at ~ 4 GeV and ~ 6 GeV, due to γ_{ISR} mis-identified as anti-neutrons, arise. In order to remove them, a cut on cluster energy such as those applied is mandatory.

However, γ radiation represents one of the most abundant produced particles in HEP experiments such as Belle II, making a major contribution to the background. In addition to the ISR ($E \sim 10$ MeV - 5 GeV) and FSR ($E < 10$ MeV) radiation described above, bremsstrahlung ($E < 50$ MeV) and secondary photons -such as those from $\pi^0 \rightarrow \gamma\gamma$ decays ($E < 400$ MeV)- are also present. Therefore, NO ISR channel where the recoil system is made of p, π^- should

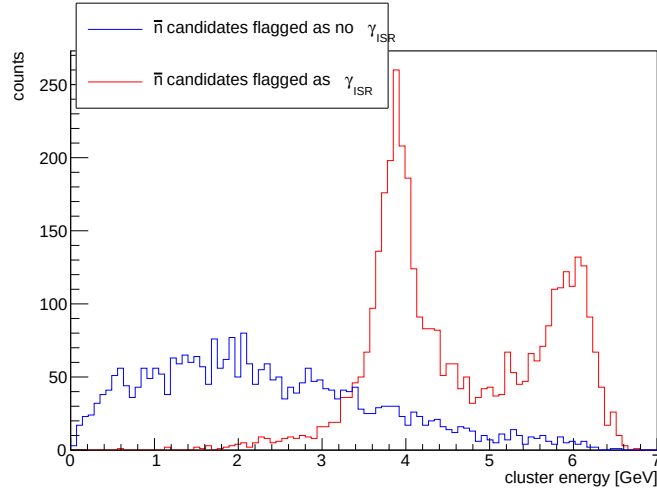


Figure 5.6: Cluster energy distribution of $\bar{n}$ candidates, divided by the Monte Carlo ISR flag (mcISR). If $\text{mcISR} = 0$, the reconstructed particle is not an ISR photon. If $\text{mcISR} = 1$, the reconstructed particle is an ISR photon. Of course this is a level-generator information.

represents a cleaner system to reconstruct than the ISR one where the recoil system is made of p, π^-, γ , since several background γ 's can be mis-identified as the ISR photon, which must be accounted for in the recoil system.

In order to compare the feasibility of selecting the channel in both ISR and NO ISR cases separately, the recoil mass distribution must be first studied using the same selection criteria. The better the selection of the signal from the background, the more consistent the shower shape variables will be for neutral clusters associated with $\bar{n}$ candidates, eventually allowing proper Data/MC agreement for these variables.

5.3 Recoil mass in $u\bar{u}$ cocktail production (NO ISR case)

The recoil mass distribution obtained from the $u\bar{u}$ cocktail is first studied in NO ISR case. If the recoil system p, π^- is properly reconstructed among all possible background channels, then a peak should be observed at the anti-neutron rest mass ($939.5 \text{ MeV}/c^2$). The obtained result is shown in 5.7.

A distinct structure is observed above the $\bar{n}$ mass value, suggesting that the selected channel is properly reconstructed. For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted 5.8. TopoAna does not account for particle-matter interactions; thus, all displayed particles originate directly from the e^+e^- collision at the IP.

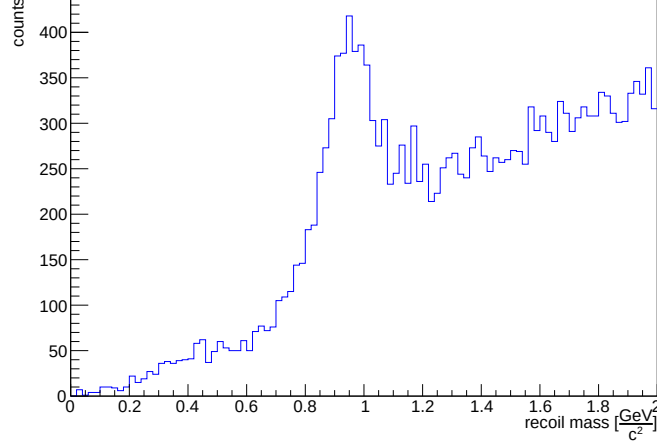


Figure 5.7: Reconstructed p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, NO ISR case.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p$	24	2405	2405
2	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p$	$\pi^0 \pi^- \bar{n} p$	2	2317	4722
3	$e^+e^- \rightarrow \pi^- \bar{n} p \gamma^I$	$\pi^- \bar{n} p \gamma^I$	16	1980	6702
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p \gamma^I$	$\pi^0 \pi^- \bar{n} p \gamma^I$	6	1676	8378
5	$e^+e^- \rightarrow \pi^- \bar{n} p \gamma^I \gamma^I$	$\pi^- \bar{n} p \gamma^I \gamma^I$	12	751	9129
6	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p \gamma^I \gamma^I$	$\pi^0 \pi^- \bar{n} p \gamma^I \gamma^I$	3	612	9741
7	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I \gamma^I$	$\pi^+ \pi^- \gamma^I \gamma^I$	33	467	10208
8	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I$	$\pi^+ \pi^- \gamma^I$	0	381	10589
9	$e^+e^- \rightarrow \pi^- \bar{\Delta}^0 p, \bar{\Delta}^0 \rightarrow \pi^0 \bar{n}$	$\pi^0 \pi^- \bar{n} p$	28	313	10902
10	$e^+e^- \rightarrow \pi^0 p \bar{p}$	$\pi^0 p \bar{p}$	67	287	11189

Figure 5.8: First 10 channels of the TopoAna output in $u\bar{u}$ Monte Carlo cocktail, showing the different channel contributes, NO ISR case.

Accessing the Monte Carlo information from TopoAna, it is clear that the main contribution comes from the signal channel with $p\pi^-\bar{n}$ in the final state, immediately followed by the one with $p\pi^-\bar{n}\pi^0$ in the final state. Further contribution arise from $p\pi^-\bar{n}\gamma_{ISR}$, which could be interpreted as signal, even if, in NO ISR case, the γ_{ISR} is not explicitly reconstructed within the recoil system. The fourth contribute is due to the one with $p\pi^-\bar{n}\pi^0\gamma_{ISR}$ and so on. It is important to note that the contributions are sorted from most to least abundant.

5.3.1 Recoil mass components analysis

The recoil mass distribution components can be studied by correlating the ntuple information obtained from the steering file with that produced by the TopoAna program.

Each channel is identified by its own `iDcyTr` (index of the decay tree), which allows one to represent the contribution of a given channel (or a combination of channels) to a specific variable, such as the recoil mass of the system $p\pi^-$. The components are therefore denoted as follows:

1. $e^+e^- \rightarrow \pi^-\bar{n}p$ (`iDcyTr` == 24)
2. $e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$ (`iDcyTr` == 2)
3. $e^+e^- \rightarrow \pi^-\bar{n}p\gamma_{ISR}$ (`iDcyTr` == 16)
4. $e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma_{ISR}$ (`iDcyTr` == 6)
5. $e^+e^- \rightarrow \text{other}$

The obtained stacked component to the recoil mass are shown in figure 5.9.

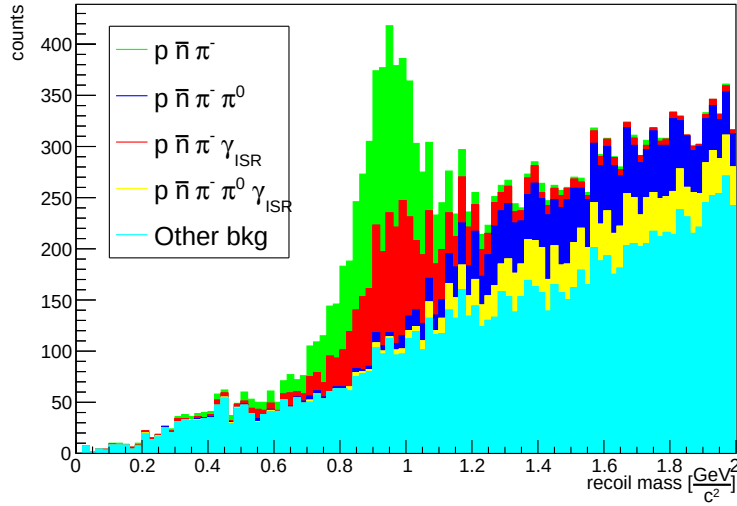


Figure 5.9: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.

Defining the *Signal Region* as the interval $[0.8, 1.2]$ GeV, it can be observed that contributions from the channel with `iDcyTr` = 2 predominantly lie to the right of the signal region. This is expected, as the recoil system computation excludes the π^0 . Same conclusion can be made for the with `iDcyTr` = 6. As expected, the signal channel (`iDcyTr` = 24) predominantly populates the signal region. The $\pi^-\bar{n}p\gamma_{ISR}$ channel (`iDcyTr` = 16) also mainly populates the signal region, indicating that the γ_{ISR} is negligible and has minimal impact on the recoil mass, as shown by the ISR energy distribution in Fig. 4.4. This can be directly observed from the single contribute from these two channels displayed in figure 5.10.

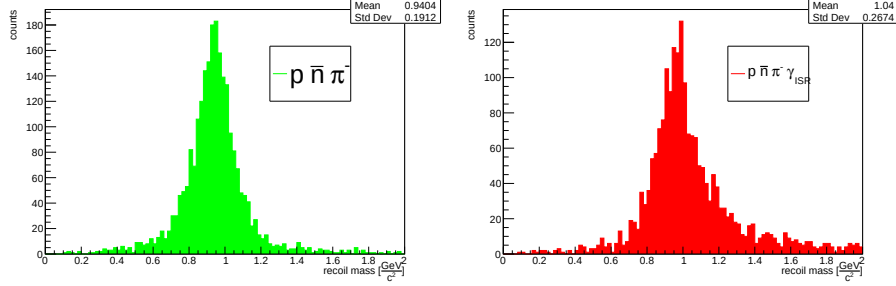


Figure 5.10: (a) p, π^- recoil mass and (b) p, π^-, γ_{ISR} recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail

Thus, despite the γ_{ISR} not being directly reconstructed, the $\pi^- \bar{n} p \gamma_{ISR}$ channel (iDcyTr = 16) can be considered a signal channel. This leads to the recoil mass composition shown in figure . 5.11.

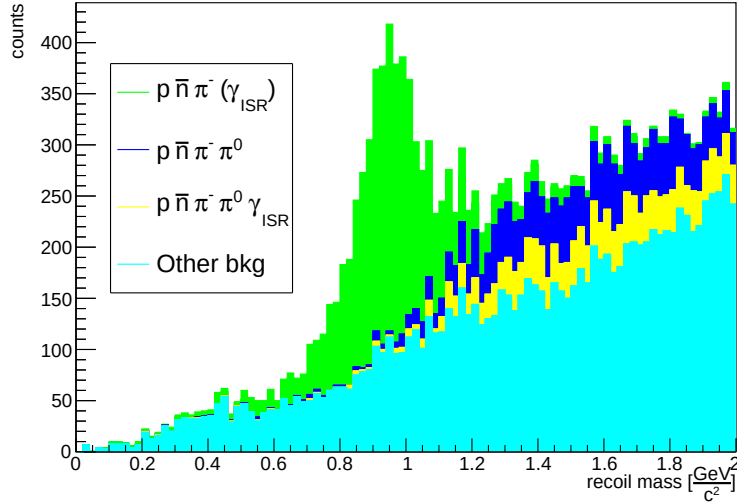


Figure 5.11: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.

The contributions from the remaining channels represent a flat background without discernible structures.

To further suppress the background, a tighter α cut can be studied. In the next subsection, the performance of the metrics and their usage are investigated in order to find the most appropriate cut in α .

5.3.2 Performance metrics study on α cut

In general, a **performance metric** quantifies how a classification process is optimized based on a specific parameter, which could, for example, be represented by a threshold.

In this specific HEP context, a performance metric quantifies how well the selection cuts distinguish the signal from the background. In this analysis, the signal can be assumed to be the channel with $\pi^- \bar{n} p(\gamma_{ISR})$ in the final state, while the background is represented by all the other channels.

It could be useful to investigate whether a stricter cut on the α angle -defined as the three-dimensional angle between the recoil system and the neutral cluster associated with the anti-neutron candidate- could improve the signal selection over the background.

To address this kind of analysis, performance metrics such as the **Figure of Merit** (FOM), the Signal Efficiency (ϵ_{sig}), the Background Efficiency (ϵ_{bkg}) and the **purity** can be employed. These are defined as follow:

- $FOM = \frac{SIGNAL(\alpha)}{\sqrt{SIGNAL(\alpha) + BACKGROUND(\alpha)}}$,
- $\epsilon_{sig} = \frac{SIGNAL(\alpha)}{SIGNAL(\alpha=0.20 \text{ rad})}$,
- $\epsilon_{bkg} = \frac{BACKGROUND(\alpha)}{BACKGROUND(\alpha=0.20 \text{ rad})}$,
- $Purity = \frac{SIGNAL(\alpha)}{SIGNAL(\alpha) + BACKGROUND(\alpha)}$,

where

- $SIGNAL(\alpha)$ is defined by the number of entries from the signal channel, depending on α .
- $BACKGROUND(\alpha)$ is defined by the number of entries from the other channels within the signal region, depending on α .

$SIGNAL(\alpha) + BACKGROUND(\alpha) = TOTAL(\alpha)$, the total amount of entries.

Assuming the signal region corresponds to the interval $[0.8, 1.2]$ GeV, the FOM characterizes the statistical significance of the applied threshold, namely the selected cut on α . The higher the FOM, the more signal is selected over the background (the figure is maximized). For each possible cut value of α , the Figure of Merit (FOM) is computed and the cut corresponding to the maximum FOM (or to a stable plateau) is selected. A higher FOM indicates a better trade-off between signal efficiency and background rejection, so the selection is optimized when the FOM is maximized. The obtained FOM is shown in Fig. 5.12.

One α value corresponding to the plateau can be chosen in order to maximize the FOM. The value $\alpha < 0.1$ rad ($\sim 5.73^\circ$) is therefore chosen, corresponding to $FOM = 43.4185$

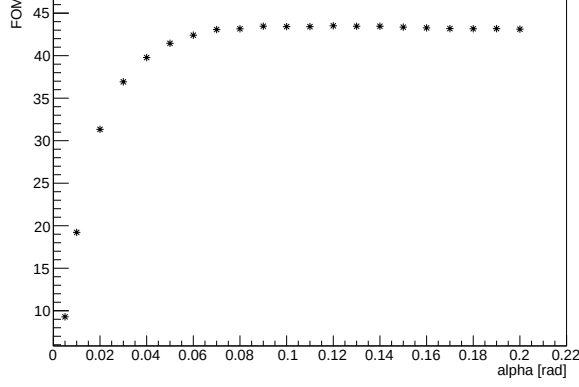


Figure 5.12: Figure of Merit (FOM) versus α angle for the $u\bar{u}$ cocktail.

Furthermore, the signal efficiency ϵ_{sig} and the background efficiency ϵ_{bkg} performance metrics can be adopted to study the Receiver Operating Characteristic (ROC), a curve that shows how well a binary classifier distinguishes signal from background as the decision threshold (α cut) is varied. Each point on the curve corresponds to a different threshold (a cut on the α angle). The Area Under the ROC Curve (AUC/ROC-AUC) is a single number summarizing the overall performance: 1.0 indicates perfect discrimination, while 0.5 corresponds to no better performance than random. The obtained ROC is shown in Fig. 5.13.

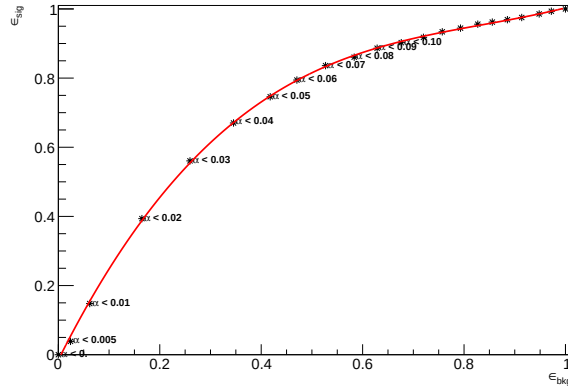


Figure 5.13: Receiver Operating Characteristic (ROC) curve constructed from the signal efficiency (ϵ_{sig}) and the Background Efficiency (ϵ_{bkg}) for the $u\bar{u}$ Monte Carlo cocktail. where each point corresponds to a different cut value on the α angle. For clarity, the labels are shown only for $\alpha < 0.10$ rad.

The Receiver Operating Characteristic (ROC) curve for the α angle cut yields an area under the curve (AUC) of 0.71, indicating good discrimination power between signal and background.

Therefore, the obtained performance metrics for $\alpha < 0.1$ rad are:

$$FOM = 43.4185, \epsilon_{sig} = 90\%, \epsilon_{bkg} = 68\%, Purity = 63\%$$

The obtained ϵ_{sig} vs $Purity$ is shown in Fig. 5.14.

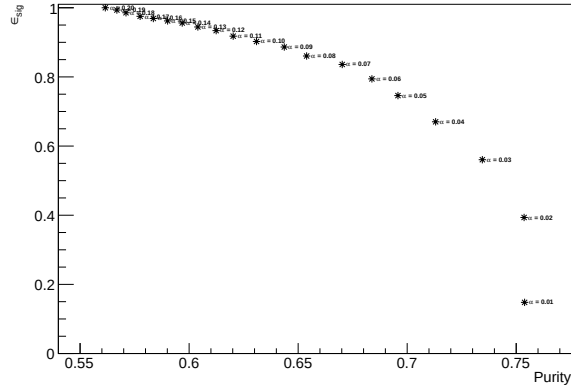


Figure 5.14: ϵ_{sig} vs $Purity$ curve for the $u\bar{u}$ Monte Carlo cocktail.

Following the optimization described, the recoil mass channel components distribution can now be analyzed using the cut $\alpha < 0.10$ rad.

The obtained stacked component to the recoil mass are shown in figure 5.15.

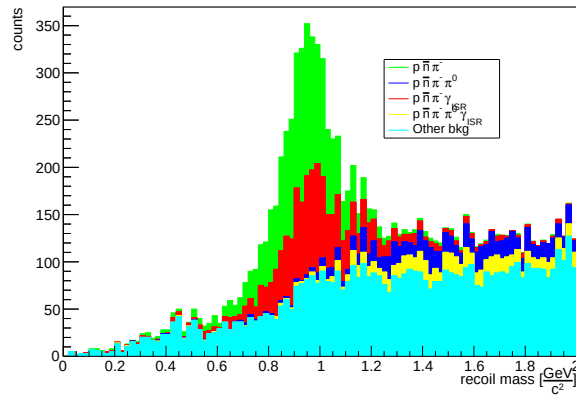


Figure 5.15: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. $\alpha < 10$ rad cut is applied.

As expected, the signal is more prominent relative to the background than in the previous case. Background components containing π^0 peak outside the signal region $[0.8, 1.2]$ GeV on the right side, while other sources exhibit a lower, flat distribution across the recoil mass spectrum. The first 5 channels of $u\bar{u}$ TopoAna output with the $\alpha < 10$ rad cut is shown in 5.16.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n}p$	$\pi^- \bar{n}p$	14	2195	2195
2	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I$	$\pi^- \bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p$	$\pi^0 \pi^- \bar{n}p$	2	907	4777
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^- \bar{n}p\gamma^I \gamma^I$	9	598	6073

Figure 5.16: Top 5 channels from the TopoAna output for the $u\bar{u}$ Monte Carlo cocktails.

After applying the $\alpha < 0.10$ rad cut, the dominant contribution remains from the signal channel $p\pi^- \bar{n}$, now immediately followed by $p\pi^- \bar{n}\gamma_{ISR}$. As discussed, both contributions can be identified as signal since they fall within the signal region $[0.8, 1.2]$ GeV.

5.4 $q\bar{q}$ cocktail production (NO ISR case)

In order to fully characterize the Monte Carlo simulation, all $q\bar{q}$ continuum components must be studied. The Run 1 Monte Carlo productions for $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ are therefore analyzed. Each of them corresponds to an integrated luminosity of $\sim 1429.23 \text{ fb}^{-1}$ of data.

For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted in Fig. 5.17.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n}p$	$\pi^- \bar{n}p$	2	2794	2794
2	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I$	$\pi^- \bar{n}p\gamma^I$	4	2163	4957
3	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p$	$\pi^0 \pi^- \bar{n}p$	1	1267	6224
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I$	9	944	7168
5	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^- \bar{n}p\gamma^I \gamma^I$	5	767	7935
6	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I \gamma^I$	$\pi^+ \pi^- \gamma^I \gamma^I$	10	443	8378
7	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I$	$\pi^+ \pi^- \gamma^I$	7	424	8802
8	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I \gamma^I$	3	315	9117
9	$e^+e^- \rightarrow \pi^0 p\bar{p}$	$\pi^0 p\bar{p}$	98	244	9361
10	$e^+e^- \rightarrow K^- p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0 \bar{n}$	$\pi^0 K^- \bar{n}p$	226	216	9577

Figure 5.17: First 10 channels of the TopoAna output in $q\bar{q}$ Monte Carlo cocktail, showing the different channel contributes.

The TopoAna output reveals that the dominant contribution is from the signal channel $p\pi^- \bar{n}$ in the $q\bar{q}$ continuum. A secondary contribution comes from $p\pi^- \bar{n}\gamma_{ISR}$, which could be treated as signal despite the γ_{ISR} not being

explicitly reconstructed. The next contribution comes from $p\pi^-\bar{n}\pi^0$ in the final state. The fourth contribution is due to $p\pi^-\bar{n}\pi^0\gamma_{\text{ISR}}$, and so on. The situation is very similar to that observed in the $u\bar{u}$ case.

It would be interesting to discuss the individual TopoAna outputs separately for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ samples. The first 5 contributes for each sample are shown in 5.18.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^-\bar{n}p$	$\pi^-\bar{n}p$	14	2195	2195
2	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	907	4777
4	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	9	598	6073
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^-\bar{n}p$	$\pi^-\bar{n}p$	1	599	599
2	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	3	488	1087
3	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	0	360	1447
4	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	8	246	1693
5	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	4	169	1862
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow K^-p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p$	3	120	120
2	$e^+e^- \rightarrow K^-p\bar{\Lambda}\gamma^I, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I$	1	102	222
3	$e^+e^- \rightarrow K^0K^-\bar{n}p, K^0 \rightarrow K_L^0$	$K_L^0K^-\bar{n}p$	9	66	288
4	$e^+e^- \rightarrow \pi^0K^-p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0\pi^0K^-\bar{n}p$	15	52	340
5	$e^+e^- \rightarrow K^-p\bar{\Lambda}\gamma^I\gamma^I, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I\gamma^I$	50	48	388
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow D^{*0}p\bar{\Lambda}_c^-\gamma^I, D^{*0} \rightarrow D^0\gamma, \bar{\Lambda}_c^- \rightarrow \pi^+\pi^-\bar{\Sigma}^-, D^0 \rightarrow \pi^0K_S^0, \bar{\Sigma}^- \rightarrow \pi^0\bar{p}, K_S^0 \rightarrow \pi^0\pi^0$	$\pi^0\pi^0\pi^0\pi^0\pi^+\pi^-\bar{p}\bar{p}\gamma^I\gamma$	0	1	1

Figure 5.18: Top 5 channels from the TopoAna output respectively for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ Monte Carlo cocktails (from top to bottom). The iDcyTr index values may differ across samples since each TopoAna output is produced independently from its corresponding ROOT ntuples.

Chapter 6

Signal sample analysis scripts

The scripts related to decay files, generation files and steering files are presented in the appendix A. Only the key highlights of the complete macros are reported, in order to ensure a smoother reading.

6.1 The decay file and the generation steering file

Monte Carlo (MC) truth is widely used in High Energy Physics (HEP) experiments. It allows the generation of datasets that follow a specific physical process, propagating particles through the detector and mimicking its behaviour layer by layer. These datasets are processed like real data during the reconstruction phase, with the advantage that the true information from the MC generator is known. The more realistic the MC simulation, the better the understanding of the real dataset.

The production of a MC sample is divided in three main steps:

- The generation of the MC particles with a generator
- The simulation of the detector structure and response to a particle (propagation)
- The reconstruction of tracks, cluster etc.

The **decay file** represents the first step to generate events specifying the decay channels of interest. It is, as the name suggests, the script in which the decay channel of a specific particle is explicitly set to a desired mode. It allows to impose several constraints on particle decay modes, in order to study a specific detector response or the kinematics of a given particle. This is called in jargon "generating signal MC events". The related file has *.dec* extension.

An example from section 3 is explained below:

```
1 Decay vpho
2 1.0 p+ anti-n0 pi- PHSP;
3 Enddecay
4
5 End
```

Figure 6.1: Decay file example powered by Evt_Gen

The declaration of a decay begins with the keyword *Decay* and ends with *Enddecay*. In this case, the dummy particle *vpho* is constrained to decay into a proton (p+), an antineutron (anti-n0) and a negative pion (pi-) according to the Phase Space Model (PHSP), with branching ratio set to 1.0. Since the channel is the following one

$$e^+ + e^- \rightarrow p + \pi^- + \bar{n} (+ \gamma_{ISR})$$

in order to produce this specific channel with ISR emission, both *PHOKHARA* and *Evt_Gen* generators are employed. The first one is acting as ISR generator by generating:

$$e^+ + e^- \rightarrow vpho (+ \gamma_{ISR})$$

Finally, the virtual photon is decayed by *Evt_Gen*, according to the shown above decay file:

$$vpho \rightarrow p + \pi^- + \bar{n}$$

To assign a particle the same final states as its charge-conjugated partner (assuming charge conjugation), the *CDecay* identifier can be used, followed by the charge-conjugated state. However it is not employed in the present case. The **generation steering file** is the operative script that generates a MC sample. An example is given below:

```

9  main = b2.Path()
10
11 # Define number of events (-n number -o output in terminal) and experiment number
12 main.add_module('EventInfoSetter', explist=[0])
13
14 print("***** CHANNEL e+ e- (isr) --> vpho --> p+ pi- nbar ***** ")
15 final_state = ['p+', 'pi-', 'anti-n0']
16 decfile = "../dec_file/dec_isr.dec"
17 #Generate isr events with phokhara_evt_gen
18 ge.add_phokhara_evtgen_combination(path=main, final_state_particles = final_state, user_decay_file = decfile,)
19
20
21 main.add_module('PrintMCParticles', onlyPrimaries = False)
22
23 # Simulate the detector response
24 si.add_simulation(path=main)
25
26 # Reconstruct the objects
27 re.add_reconstruction(path=main)
28
29 mdst.add_mdst_output(path=main)
30 # Process the steering path
31 b2.process(path=main)
32
33 # Finally, print out some statistics about the modules execution
34 print(b2.statistics)

```

Figure 6.2: Usage of PHOKHARA&Evt_Gen, where simulation and reconstruction are performed

As previously discussed, basf2 scripts are structured as modules, which are sequentially arranged within a path, which is called *main* in this example (row9). The first module (row 12) sets the desired amount of events and the specific experiment number, which is provided via keyboard input prior to script interpretation. Then, the signal from the decay file is added (row 18), with **add_phokhara_evtgen_combination** from generators basf2 modules ¹. From row 24 to 27, the simulation and the reconstruction, as described in section 3.3.1, are added to the path. Then the output file is created to store the MC truth, namely the generated event with a specific signal sample. Finally the steering path is processed in row 31.

¹<https://software.belle2.org/development/sphinx/generators/doc/index.html>

References

- [1] C. B. Dover et al. Antinucleon-nucleon interactions and annihilation. *Progress in Particle and Nuclear Physics*, 29:87–175, 1992.
- [2] T. Ueda. Theoretical study of $\bar{N}n$ interactions. *Progress of Theoretical Physics*, 72(6):1670–1682, 1979.
- [3] S G Nedilko et al. Properties of scintillators. *Nuclear Instruments and Methods in Physics Research Section A*, 605(1-2):20–23, 2009.
- [4] H. Ikeda, A. Satpathy, B. S. Ahn, V. M. Aulchenko, et al. The belle csi(tl) calorimeter. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 441(1-2):401–424, 2000.
- [5] E. Kou, P. Urquijo, W. Altmannshofer, et al. The Belle II physics book. *Progress of Theoretical and Experimental Physics*, 2019, 2019.
- [6] A. Abashian and others (Belle Collaboration). Measurement of the CP violation parameter $\sin 2\phi_1$. *Phys. Rev. Lett.*, 87:091802, 2001.
- [7] K. Abe and others (Belle Collaboration). Observation of large CP violation in the neutral b meson system. *Phys. Rev. D*, 66:032007, 2002.
- [8] B. Aubert and others (BaBar Collaboration). Measurement of CP-violating asymmetries in b^0 decays. *Phys. Rev. Lett.*, 86:2515, 2001.
- [9] B. Aubert and others (BaBar Collaboration). A study of time-dependent CP-violating asymmetries and flavour oscillations in neutral b decays at the $v(4s)$. *Phys. Rev. D*, 66:032003, 2002.
- [10] A. D. Sakharov. Violation of CP invariance, c asymmetry, and baryon asymmetry of the universe. *JETP Lett.*, 5:24, 1967.
- [11] B. Cork et al. Antineutron detection via annihilation. *Phys. Rev.*, 104:1193–1199, 1956.
- [12] O. Chamberlain et al. Discovery of the antiproton. *Phys. Rev.*, 100:947–950, 1955.
- [13] M. Ablikim et al. [BESIII]. *Nature Physics*, 17, 2021.
- [14] L. Šerkšnytė et al. *arXiv 2201.00925*, 2022.
- [15] F. Donato, N. Fornengo, and P. Salati. *Phys. Rev. D*, 62:043003, 2000.
- [16] Geant4 Collaboration. Geant4: A simulation toolkit. Technical Report CDS-EXT-2003-005, CERN, 2003.
- [17] P. S. Eastman et al. Antineutron beam properties. *Nucl. Phys. B*, 51:57–60, 1973.

- [18] Tullio Bressani and A. Filippi. Antineutron physics. *Physics Reports*, 383(3-6):213–297, 2003.
- [19] T. Armstrong et al. Antineutron annihilation cross sections. *Phys. Rev. D*, 36(3):659–672, 1987.
- [20] A. Bertin et al. Antineutron physics at lear. *Nuclear Physics B (Proc. Suppl.)*, 56A:227–231, 1997.
- [21] E. Fermi. Quantum theory of interference in atomic beams. *Progress of Theoretical Physics*, 5(6):570–579, 1950.
- [22] W. H. Barkas et al. Barkas effect in antiproton-nucleus scattering. *Phys. Rev.*, 105(4):1037–1044, 1957.
- [23] J. Mahalanabis et al. Antineutron-proton interaction studies. *Nucl. Phys. A*, 485(4):546–560, 1988.
- [24] Kazunori Akai, Kazuro Furukawa, and Haruyo Koiso. Superkekb collider. *Nuclear Instruments and Methods in Physics Research A*, 907:188–199, 2018.
- [25] M. Kikuchi. Design of damping ring for superkekb. In *Proceedings of 2005 Particle Accelerator Conference (PAC’05)*, pages 773–775, 2005.
- [26] H. Ye et al.. Commissioning and performance of the belle ii pixel detector. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 987:164875, 2021.
- [27] L. Zani et al.. The silicon vertex detector of the belle ii experiment. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 1038:166952, 2022.
- [28] N. Taniguchi. Central drift chamber for belle-ii. *Journal of Instrumentation*, 12(06):C06014, jun 2017.
- [29] Yasuyuki Horii. TOP Detector for Particle Identification at the Belle II Experiment. *PoS*, EPS-HEP2013:500, 2013.
- [30] Yosuke Yusa. The ARICH detector at Belle II experiment. *PoS*, EPS-HEP2013:556, 2013.
- [31] T. Abe et al. The Belle detector. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 550(2-3):329–402, 2005.
- [32] A. Kuzmin. Electromagnetic calorimeter of belle ii. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 958:162235, 2020. Proceedings of the Vienna Conference on Instrumentation 2019.

- [33] Hamamatsu Photonics. S2744 series silicon photodiode. https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/s2744-08_etc_kpin1049e.pdf, 2023. Datasheet, accessed March 24, 2026.
- [34] Leo Piilonen. Belle ii klm detector. jun 2022. Belle II Week, 13-17 June 2022.
- [35] T. Kuhr, C. Pulvermacher, M. Ritter, T. Hauth, and N. Braun. The belle II core software. *Computing and Software for Big Science*, 3(1), 11 2018.
- [36] Michael De Nuccio et al. *Belle II Starter Kit*, 2019.
- [37] A. Abashian et al. Belle ecl detector description. *Nucl. Instr. and Meth. A*, 479:117–119, 2002.
- [38] S. Nishida et al. The top counter of the belle ii detector: Commissioning and initial operation. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 1037:169693, 2026. Belle II collaboration.